

**Bangladesh University of Engineering and Technology**

Department of Computer Science & Engineering

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## Electronic Circuits Question Bank

EEE 263 · Electronic Circuits

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Year Coverage: 2011-12 to 2024-25

Topics: Diode Characteristics, Wave Shaping & Clipping/Clamping,  
BJT & MOSFET Amplifiers, Op-Amps (Filters, Schmitt Trigger, Analog Computing), Oscillators & Timers (555) & more

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BUET · EEE 263 · Electronic Circuits

# Electronic Circuits

Part I — Semiconductor Device Characteristics

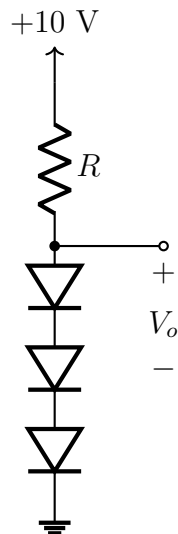
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Ideal device models, I-V characteristics, biasing, and circuit analysis for Diode, BJT, and MOSFET

## S1 Ideal Diode Characteristics

### Diode Circuit Analysis

**Q1.** For the circuit shown,  $I$  is the current flowing through each of the diodes and  $R$  is unknown. The diodes are identical with  $I_S = 10^{-16}$  A and  $n = 1$ . (i) Find the value of  $(I, R)$  such that  $V_o = 2.4$  V is maintained. (ii) If a current of 1 mA is drawn away from the output terminated by a load, what is the change in output voltage?



(15 marks)

[EEE 263 | L-2/T-1 | 2024–25]

### Answer

#### Theory & Principle

In analog electronic circuits, a series string of forward-biased diodes is frequently utilized to create a simple, cost-effective voltage regulator or reference source. The circuit operates on the principle that a forward-biased diode maintains a relatively stable voltage drop across its terminals over a wide range of currents.

The relationship between the current flowing through a forward-biased p-n junction diode ( $I$ ) and its terminal voltage ( $V_D$ ) is governed by the Shockley diode equation:

$$I = I_S \left( e^{\frac{V_D}{nV_T}} - 1 \right) \approx I_S e^{\frac{V_D}{nV_T}} \quad (1)$$

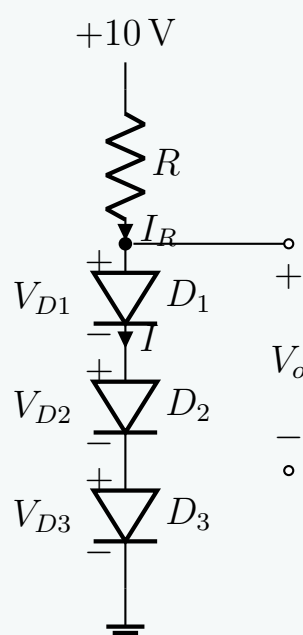
where:

- $I_S$  is the reverse saturation current.
- $n$  is the emission coefficient or ideality factor ( $n = 1$  for this problem).
- $V_T$  is the thermal voltage, given by  $V_T = \frac{kT}{q}$  ( $V_T \approx 25$  mV at room temperature 290 K or 25.85 mV at 300 K).

When small variations in current occur due to an external load, the corresponding change in voltage can be accurately analyzed using the **small-signal diode model**. The dynamic (incremental) resistance  $r_d$  of a single diode at its DC operating point is defined as:

$$r_d = \left( \frac{dI}{dV_D} \right)^{-1} = \frac{nV_T}{I} \quad (2)$$

#### Circuit Operation



The circuit consists of a constant supply voltage  $V_{CC} = +10\text{ V}$  supplying current through a current-limiting resistor  $R$  into three series-connected identical diodes. Since the diodes are connected in series and no load current is initially drawn, the same quiescent current  $I$  flows through each diode.

The total output voltage  $V_o$  is equal to the sum of the forward voltage drops across the three identical diodes:

$$V_o = V_{D1} + V_{D2} + V_{D3} = 3V_D \quad (3)$$

—

### Step-by-Step Solution

Because the exact value of thermal voltage  $V_T$  varies slightly depending on the assumed textbook reference temperature (25 mV vs. 25.85 mV), both standard engineering cases are detailed below to ensure complete accuracy.

#### Part (i): Find $(I, R)$ for $V_o = 2.4\text{ V}$

Given that the three diodes are identical and share the same current  $I$ , the voltage drop across each individual diode is:

$$V_D = \frac{V_o}{3} = \frac{2.4\text{ V}}{3} = 0.8\text{ V} \quad (4)$$

Using the simplified diode current equation, we isolate the current  $I$ :

$$I = I_S e^{\frac{V_D}{nV_T}} \quad (5)$$

By applying Ohm's law across the resistor  $R$ , the current through the resistor is identical to the diode string current ( $I_R = I$ ) under no-load conditions:

$$R = \frac{V_{CC} - V_o}{I} = \frac{10\text{ V} - 2.4\text{ V}}{I} = \frac{7.6\text{ V}}{I} \quad (6)$$

**Case A:** Assuming  $V_T = 25\text{ mV}$  (Standard textbook approximation)

$$\begin{aligned} I &= 10^{-16} \times e^{\frac{0.8}{1 \times 0.025}} = 10^{-16} \times e^{32} \\ I &= 10^{-16} \times (7.89637 \times 10^{13}) = \mathbf{7.896\text{ mA}} \end{aligned} \quad (7)$$

Now, calculating the value of  $R$ :

$$R = \frac{7.6\text{ V}}{7.89637 \times 10^{-3}\text{ A}} = \mathbf{962.46\ \Omega} \quad (8)$$

**Case B:** Assuming  $V_T = 25.85\text{ mV}$  (At  $T = 300\text{ K}$ )

$$\begin{aligned} I &= 10^{-16} \times e^{\frac{0.8}{1 \times 0.02585}} = 10^{-16} \times e^{30.9478} \\ I &= 10^{-16} \times (2.7481 \times 10^{13}) = \mathbf{2.748\text{ mA}} \end{aligned} \quad (9)$$

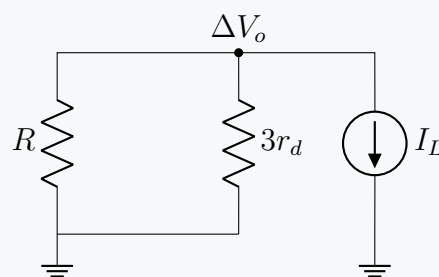
Now, calculating the value of  $R$ :

$$R = \frac{7.6\text{ V}}{2.7481 \times 10^{-3}\text{ A}} = \mathbf{2765.55\ \Omega = 2.766\text{ k}\Omega} \quad (10)$$

—

#### Part (ii): Impact of Drawing a 1 mA Load Current

When a load current  $I_L = 1\text{ mA}$  is drawn away from the output terminal, it changes the small-signal operating environment. We employ the **\*\*small-signal equivalent circuit\*\*** looking into the output node.



The small-signal resistance of the total diode string is the sum of the small-signal resistances of the three individual series diodes:

$$r_{\text{total}} = 3r_d = 3 \left( \frac{nV_T}{I} \right) = \frac{3V_T}{I} \quad (11)$$

The total incremental output resistance  $R_{\text{out}}$  observed at the output node is the parallel combination of the biasing resistor  $R$  and the incremental resistance of the diode chain ( $3r_d$ ):

$$R_{\text{out}} = R \parallel 3r_d = \frac{R \cdot 3r_d}{R + 3r_d} \quad (12)$$

When an incremental load current  $\Delta I_L = 1\text{ mA}$  is drawn out of the node, the change in output voltage  $\Delta V_o$  is determined via Ohm's law:

$$\Delta V_o = -I_L \cdot R_{\text{out}} \quad (13)$$

**Case A:** Evaluating with  $V_T = 25$  mV values ( $I = 7.896$  mA,  $R = 962.46$   $\Omega$ )

$$3r_d = \frac{3 \times 25 \text{ mV}}{7.89637 \text{ mA}} = 9.498 \text{ } \Omega \quad (14)$$

$$R_{\text{out}} = 962.46 \text{ } \Omega \parallel 9.498 \text{ } \Omega = \frac{962.46 \times 9.498}{962.46 + 9.498} = 9.405 \text{ } \Omega \quad (15)$$

$$\Delta V_o = -(1 \text{ mA}) \times 9.405 \text{ } \Omega = -\mathbf{9.41} \text{ mV} \quad (16)$$

**Case B:** Evaluating with  $V_T = 25.85$  mV values ( $I = 2.748$  mA,  $R = 2765.55$   $\Omega$ )

$$3r_d = \frac{3 \times 25.85 \text{ mV}}{2.7481 \text{ mA}} = 28.219 \text{ } \Omega \quad (17)$$

$$R_{\text{out}} = 2765.55 \text{ } \Omega \parallel 28.219 \text{ } \Omega = \frac{2765.55 \times 28.219}{2765.55 + 28.219} = 27.934 \text{ } \Omega \quad (18)$$

$$\Delta V_o = -(1 \text{ mA}) \times 27.934 \text{ } \Omega = -\mathbf{27.93} \text{ mV} \quad (19)$$

### Summary of Final Results

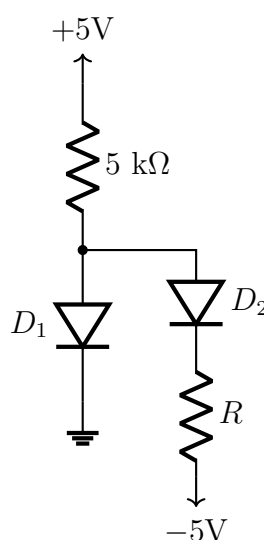
The values depend on the standard convention chosen for the thermal voltage  $V_T$ :

| Parameter                           | Case A ( $V_T = 25$ mV) | Case B ( $V_T = 25.85$ mV) |
|-------------------------------------|-------------------------|----------------------------|
| Quiescent Current, $I$              | 7.90 mA                 | 2.75 mA                    |
| Biasing Resistor, $R$               | 962.5 $\Omega$          | 2.766 k $\Omega$           |
| Incremental Resistance, $3r_d$      | 9.50 $\Omega$           | 28.22 $\Omega$             |
| Output Voltage Change, $\Delta V_o$ | −9.41 mV                | −27.93 mV                  |

### Important Notes & Exam Tips

- **Small-Signal Validity:** The small-signal model assumes  $\Delta V_D \ll V_T$ . In Case A,  $\Delta V_D = \frac{\Delta V_o}{3} \approx -3.14$  mV, which is significantly smaller than 25 mV. This justifies the highly accurate usage of linear incremental analysis.
- **Voltage Regulator Load Regulation:** The negative sign in  $\Delta V_o$  indicates that as the regulator circuit supplies current to an external network, the terminal voltage drops slightly. A lower output resistance ( $R_{\text{out}}$ ) guarantees superior voltage regulation performance.

**Q2.** For the circuit given, determine the range of values of resistor  $R$  such that  $D_1$  does not conduct and  $D_2$  conducts. Assume ideal diodes.



(15 marks)

[EEE 263 | L-2/T-1 | 2022–23]

### Answer

#### Theory & Principle

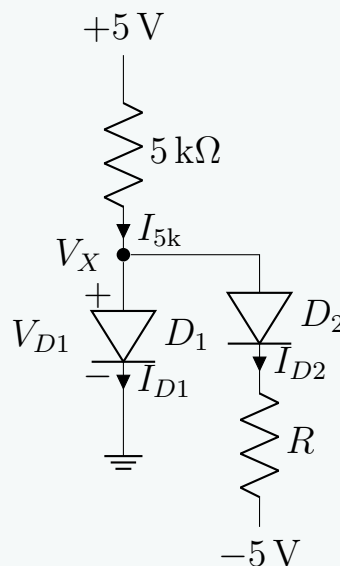
An ideal diode acts as a perfect electronic switch with two distinct states of operation:

- ON (Forward-Biased / Conducting State):** When forward-biased, an ideal diode behaves as a short circuit (zero voltage drop across its terminals,  $V_D = 0$  V) and permits a forward current  $I_D > 0$ .
- OFF (Reverse-Biased / Non-Conducting State):** When reverse-biased, an ideal diode behaves as an open circuit ( $I_D = 0$  A) and sustains a reverse terminal voltage  $V_D \leq 0$  V.

To determine the operating range of circuit parameters under specified diode conduction states, we substitute each diode with its corresponding ideal equivalent circuit component and solve the resulting circuit equations using Kirchhoff's laws.

### Circuit Operation & Schematic

The target condition states that diode  $D_1$  must remain **OFF** (non-conducting) while diode  $D_2$  must remain **ON** (conducting). We define the voltage at the central node between the  $5\text{ k}\Omega$  resistor, the anode of  $D_1$ , and the anode of  $D_2$  as  $V_X$ .



Under the specified conditions:

- Since  $D_1$  is **OFF**, it acts as an open circuit, meaning  $I_{D1} = 0\text{ A}$ .
- Since  $D_2$  is **ON**, it acts as a short circuit ( $V_{D2} = 0\text{ V}$ ), directly linking node  $V_X$  to the top terminal of resistor  $R$ .

### Mathematical Derivation

With  $D_1$  open-circuited and  $D_2$  short-circuited, the circuit reduces to a single continuous series branch connected between the  $+5\text{ V}$  and  $-5\text{ V}$  power rails.

#### 1. Expressing the Node Voltage $V_X$

The total series resistance of this active branch is given by:

$$R_{\text{total}} = 5\text{ k}\Omega + R \quad (1)$$

The total voltage drop across this series combination is:

$$V_{\text{total}} = (+5\text{ V}) - (-5\text{ V}) = 10\text{ V} \quad (2)$$

The current flowing through the branch ( $I_{5k} = I_{D2}$ ) is:

$$I_{D2} = \frac{V_{\text{total}}}{R_{\text{total}}} = \frac{10\text{ V}}{5\text{ k}\Omega + R} \quad (3)$$

We can now determine the node voltage  $V_X$  by subtracting the voltage drop across the  $5\text{ k}\Omega$  resistor from the  $+5\text{ V}$  supply rail:

$$V_X = 5\text{ V} - I_{5k} \cdot (5\text{ k}\Omega) \quad (4)$$

Substituting the expression for  $I_{5k}$  into the equation yields:

$$\begin{aligned} V_X &= 5 - \left( \frac{10}{5\text{ k}\Omega + R} \right) \cdot 5\text{ k}\Omega \\ V_X &= 5 \left[ 1 - \frac{10}{5\text{ k}\Omega + R} \right] \\ V_X &= 5 \left[ \frac{5\text{ k}\Omega + R - 10}{5\text{ k}\Omega + R} \right] \\ V_X &= 5 \left[ \frac{R - 5\text{ k}\Omega}{5\text{ k}\Omega + R} \right] \end{aligned} \quad (5)$$

## 2. Applying the Non-Conduction Condition for $D_1$

For diode  $D_1$  to remain **OFF** (non-conducting), its anode-to-cathode voltage  $V_{D1}$  must be less than or equal to the ideal threshold voltage (0 V). Because its cathode is connected directly to ground (0 V):

$$V_{D1} = V_X - 0 \text{ V} \leq 0 \text{ V} \implies V_X \leq 0 \text{ V} \quad (6)$$

Substituting our derived equation for  $V_X$  into this condition:

$$5 \left[ \frac{R - 5 \text{ k}\Omega}{5 \text{ k}\Omega + R} \right] \leq 0 \quad (7)$$

Since a physical resistance  $R$  cannot be negative, the denominator ( $5 \text{ k}\Omega + R$ ) is strictly positive ( $> 0$ ). Thus, the inequality depends entirely on the numerator:

$$R - 5 \text{ k}\Omega \leq 0 \implies R \leq 5 \text{ k}\Omega \quad (8)$$

## 3. Verifying the Conduction Condition for $D_2$

For diode  $D_2$  to remain **ON** (conducting), the forward current flowing through it must be strictly positive:

$$I_{D2} = \frac{10 \text{ V}}{5 \text{ k}\Omega + R} > 0 \quad (9)$$

Since  $10 \text{ V} > 0$ , this condition is satisfied for all valid physical resistances where  $R \geq 0 \Omega$ .

## Final Answer

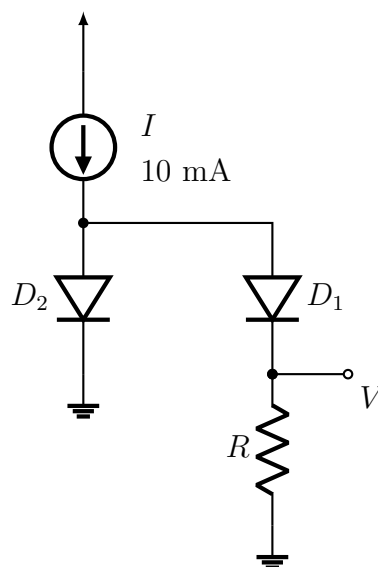
Combining the compliance boundaries for both states ( $R \geq 0 \Omega$  and  $R \leq 5 \text{ k}\Omega$ ), the complete range of values for resistor  $R$  is:

$$0 \Omega \leq R \leq 5 \text{ k}\Omega \quad (10)$$

## Exam Tips & Edge-Case Analysis

- **Boundary Testing ( $R = 5 \text{ k}\Omega$ ):** If  $R = 5 \text{ k}\Omega$ , the circuit forms a symmetric voltage divider between  $+5 \text{ V}$  and  $-5 \text{ V}$ , setting  $V_X = 0 \text{ V}$ . At this precise boundary,  $D_1$  experiences  $0 \text{ V}$  across its terminals but carries  $0 \text{ A}$  of current, which represents the critical threshold on the verge of conduction.
- **Behavior when  $R > 5 \text{ k}\Omega$ :** If  $R$  exceeds  $5 \text{ k}\Omega$ , the voltage  $V_X$  attempts to rise above  $0 \text{ V}$ . This immediately forward-biases the ideal diode  $D_1$ , clamping  $V_X$  firmly to  $0 \text{ V}$  and violating the condition that  $D_1$  must not conduct.

**Q3.** In the circuit shown, both diodes are identical. Find the value of  $R$  for which  $V = 50 \text{ mV}$ .



(16 marks)

[EEE 263 | L-2/T-1 | 2021–22]

## Answer

### Theory & Principle

In this circuit, a constant current source  $I$  feeds two parallel branches containing identical forward-biased diodes. The current distribution between these two branches is governed by the nonlinear voltage-current characteristic of the diodes, expressed by the Shockley diode equation:

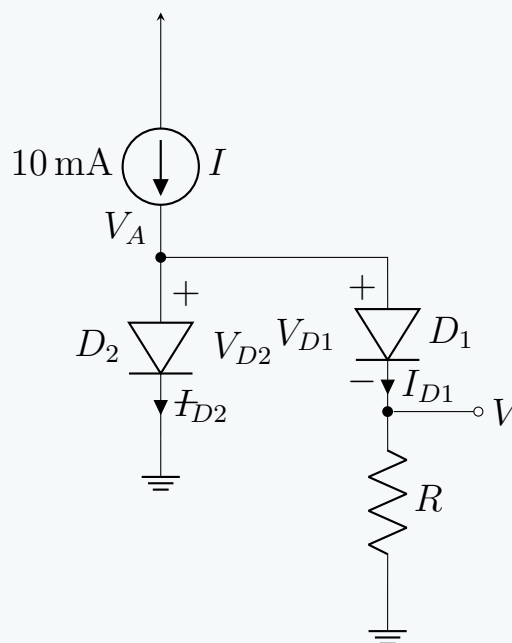
$$I_D = I_S \left( e^{\frac{V_D}{nV_T}} - 1 \right) \approx I_S e^{\frac{V_D}{nV_T}} \quad (1)$$

where:

- $I_S$  is the reverse saturation current (identical for both  $D_1$  and  $D_2$ ).
- $n$  is the ideality factor (assumed to be  $n = 1$ ).
- $V_T$  is the thermal voltage, standardly taken as 25 mV at room temperature.
- $V_D$  is the voltage drop across the diode.

By expressing the ratio of the currents in the two identical diodes, the common  $I_S$  term cancels out, allowing us to find the exact current division based strictly on the potential difference between their cathodes.

### Circuit Operation & Schematic



The total current  $I = 10$  mA enters the common anode node (let's call its voltage  $V_A$ ) and splits into two currents:  $I_{D2}$  flowing through diode  $D_2$  and  $I_{D1}$  flowing through diode  $D_1$ .

$$I_{D1} + I_{D2} = I = 10 \text{ mA} \quad (2)$$

### Mathematical Derivation & Step-by-Step Solution

#### 1. Determine Diode Voltages

The anode of both diodes is at potential  $V_A$ .

- The cathode of  $D_2$  is grounded (0 V). Therefore, the voltage across  $D_2$  is:

$$V_{D2} = V_A - 0 = V_A \quad (3)$$

- The cathode of  $D_1$  is at potential  $V = 50$  mV. Therefore, the voltage across  $D_1$  is:

$$V_{D1} = V_A - V \quad (4)$$

#### 2. Current Ratio

Using the approximate exponential diode equation, the currents through the diodes are:

$$I_{D2} = I_S e^{\frac{V_A}{nV_T}} \quad (5)$$

$$I_{D1} = I_S e^{\frac{V_A - V}{nV_T}} \quad (6)$$

Taking the ratio of  $I_{D2}$  to  $I_{D1}$  yields:

$$\frac{I_{D2}}{I_{D1}} = \frac{I_S e^{\frac{V_A}{nV_T}}}{I_S e^{\frac{V_A - V}{nV_T}}} = e^{\frac{V_A - (V_A - V)}{nV_T}} = e^{\frac{V}{nV_T}} \quad (7)$$

Assuming standard parameters of  $n = 1$  and thermal voltage  $V_T = 25$  mV, and given  $V = 50$  mV:

$$\frac{I_{D2}}{I_{D1}} = e^{\frac{50 \text{ mV}}{25 \text{ mV}}} = e^2 \approx 7.389 \quad (8)$$

### 3. Calculate Branch Currents

From Kirchhoff's Current Law (KCL):

$$I_{D1} + I_{D2} = 10 \text{ mA} \quad (9)$$

Substitute  $I_{D2} = I_{D1} \cdot e^2$  into the KCL equation:

$$\begin{aligned} I_{D1} + I_{D1}e^2 &= 10 \text{ mA} \\ I_{D1}(1 + e^2) &= 10 \text{ mA} \\ I_{D1} &= \frac{10 \text{ mA}}{1 + e^2} \end{aligned} \quad (10)$$

Calculating the numerical value:

$$I_{D1} = \frac{10 \text{ mA}}{1 + 7.389056} = \frac{10 \text{ mA}}{8.389056} \approx 1.192 \text{ mA} \quad (11)$$

### 4. Calculate Resistor $R$

The voltage across the resistor is  $V = 50 \text{ mV}$ , and the current flowing through it is  $I_{D1}$ . Using Ohm's Law:

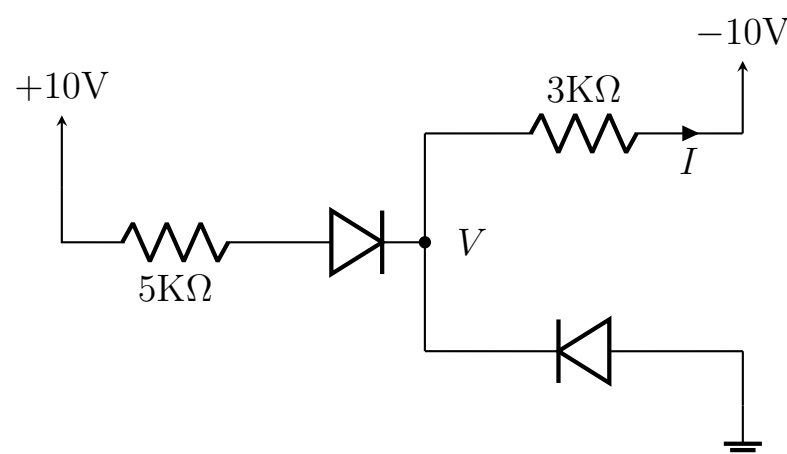
$$\begin{aligned} R &= \frac{V}{I_{D1}} \\ R &= \frac{50 \times 10^{-3} \text{ V}}{1.192 \times 10^{-3} \text{ A}} \\ R &\approx 41.946 \Omega \end{aligned} \quad (12)$$

### Final Answer

To maintain a voltage of  $V = 50 \text{ mV}$  across the resistor, assuming a thermal voltage of  $V_T = 25 \text{ mV}$ , the required value of  $R$  is:

$$\mathbf{R \approx 41.95 \Omega} \quad (13)$$

**Q4.** Find the values of  $I$  and  $V$  in the circuit shown. Use the  $0.7 \text{ V}$  drop diode model.



(10.67 marks)

[EEE 263 | L-2/T-1 | 2014–15]

### Answer

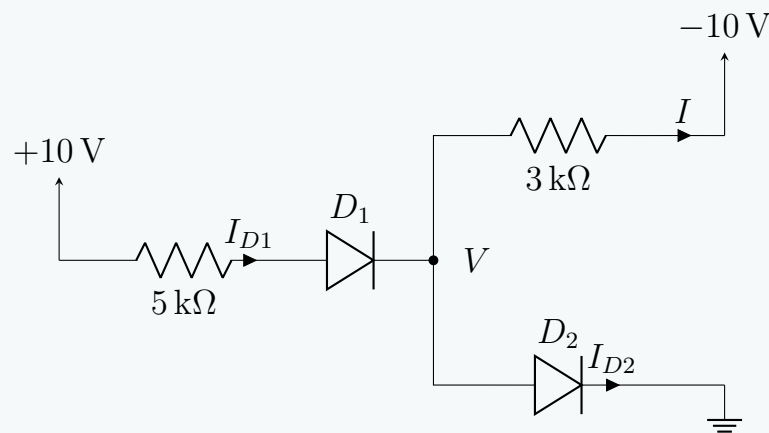
#### Theory & Principle

To analyze circuits containing nonlinear components like diodes, we systematically assume an operating state (ON or OFF) for each diode, solve the resulting linear circuit, and then verify if our assumptions hold true against the physical rules of the diode model. In the **\*\*Constant Voltage Drop (CVD) model\*\***:

- If a diode is assumed **ON** (forward-biased), it is replaced by a DC voltage source of  $V_D = 0.7 \text{ V}$  (for silicon). We must then verify that the current flowing through it is positive ( $I_D > 0$ ).
- If a diode is assumed **OFF** (reverse-biased), it is replaced by an open circuit ( $I_D = 0 \text{ A}$ ). We must then verify that the voltage across its terminals is less than the turn-on voltage ( $V_D < 0.7 \text{ V}$ ).

#### Circuit Operation & Schematic

For clarity in our analysis, let us designate the diode in the primary series path as  $D_1$  and the diode in the lower branch connecting to ground as  $D_2$ .



### Step-by-Step Solution

#### Hypothesis 1: Assume both $D_1$ and $D_2$ are ON

If  $D_2$  is ON, the voltage at node  $V$  is clamped by the forward voltage drop of  $D_2$  relative to ground:

$$V = 0\text{ V} + 0.7\text{ V} = 0.7\text{ V} \quad (1)$$

If  $D_1$  is also ON, its anode is at  $V + 0.7\text{ V} = 1.4\text{ V}$ . The current through  $D_1$  is:

$$I_{D1} = \frac{10\text{ V} - 1.4\text{ V}}{5\text{ k}\Omega} = \frac{8.6\text{ V}}{5\text{ k}\Omega} = 1.72\text{ mA} \quad (2)$$

The current  $I$  flowing through the  $3\text{ k}\Omega$  resistor to the  $-10\text{ V}$  source is:

$$I = \frac{V - (-10\text{ V})}{3\text{ k}\Omega} = \frac{0.7\text{ V} + 10\text{ V}}{3\text{ k}\Omega} = 3.567\text{ mA} \quad (3)$$

Applying Kirchhoff's Current Law (KCL) at node  $V$ :

$$\begin{aligned} I_{D1} &= I + I_{D2} \\ 1.72\text{ mA} &= 3.567\text{ mA} + I_{D2} \\ I_{D2} &= 1.72\text{ mA} - 3.567\text{ mA} = -1.847\text{ mA} \end{aligned} \quad (4)$$

**Conclusion for Hypothesis 1:** Since the derived diode current  $I_{D2}$  is negative, the assumption that  $D_2$  is ON is **incorrect**.

#### Hypothesis 2: Assume $D_1$ is ON and $D_2$ is OFF

Since  $D_2$  is OFF, it acts as an open circuit ( $I_{D2} = 0\text{ A}$ ). Consequently, all current flowing through  $D_1$  must travel through the  $3\text{ k}\Omega$  resistor. Therefore,  $I_{D1} = I$ .

We can apply Kirchhoff's Voltage Law (KVL) along the single continuous loop from the  $+10\text{ V}$  source to the  $-10\text{ V}$  source:

$$\begin{aligned} V_{\text{start}} - I(5\text{ k}\Omega) - V_{D1} - I(3\text{ k}\Omega) &= V_{\text{end}} \\ 10 - I(5) - 0.7 - I(3) &= -10 \\ 9.3 - 8I &= -10 \\ 8I &= 19.3 \end{aligned} \quad (5)$$

Solving for  $I$ :

$$I = \frac{19.3}{8}\text{ mA} = \mathbf{2.4125\text{ mA}} \quad (6)$$

Next, we calculate the voltage  $V$  at the node between the diode  $D_1$  and the  $3\text{ k}\Omega$  resistor. We can evaluate this from the right side of the circuit:

$$\begin{aligned} V - I(3\text{ k}\Omega) &= -10 \\ V &= -10 + 3(2.4125) \\ V &= -10 + 7.2375 \\ V &= \mathbf{-2.7625\text{ V}} \end{aligned} \quad (7)$$

(Self-check using the left side:  $V = 10 - I(5\text{ k}\Omega) - 0.7 = 9.3 - 2.4125(5) = 9.3 - 12.0625 = -2.7625\text{ V}$ . The calculation is consistent.)

### Verification of Assumptions

- **Verify  $D_1$  is ON:** The current through  $D_1$  is  $I = 2.4125\text{ mA} > 0$ . *Assumption holds.*
- **Verify  $D_2$  is OFF:** The voltage across  $D_2$  is  $V_{\text{anode}} - V_{\text{cathode}} = V - 0 = -2.7625\text{ V}$ . Since  $-2.7625\text{ V} < 0.7\text{ V}$ ,  $D_2$  remains reverse-biased. *Assumption holds.*

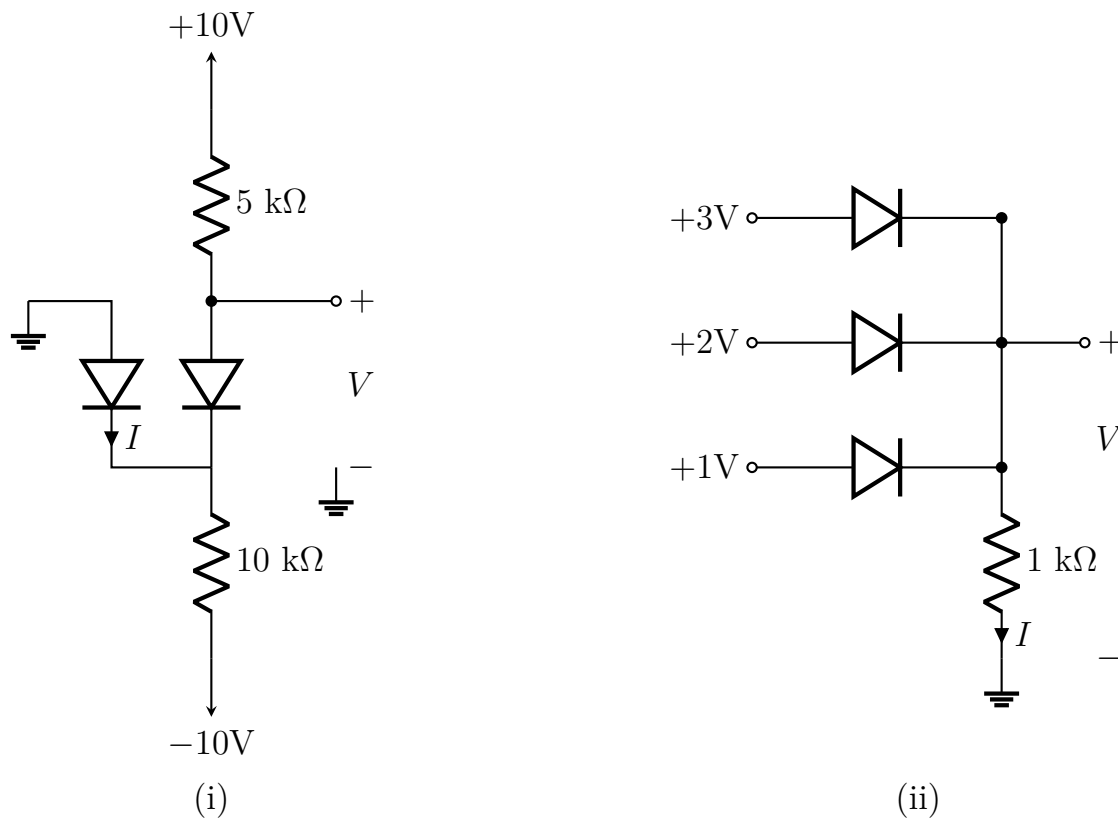
### Final Answer

Both initial assumptions are successfully verified. The exact values are:

$$\mathbf{I = 2.4125\text{ mA} \approx 2.41\text{ mA}} \quad (8)$$

$$\mathbf{V = -2.7625\text{ V} \approx -2.76\text{ V}} \quad (9)$$

**Q5.** Assuming the diodes to be ideal, find the values of  $I$  and  $V$  in the circuits shown.



(16 marks)

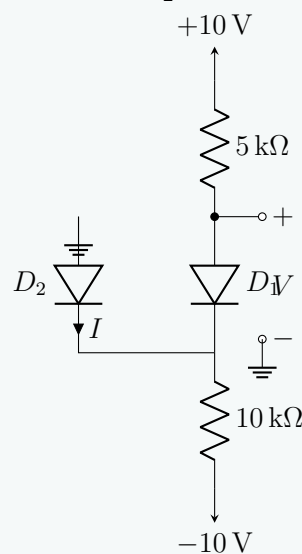
[EEE 263 | L-2/T-1 | 2011–12]

## Answer

### Part (i): Two-Diode Circuit Analysis

#### 1. Circuit Schematic

For clarity, we label the right diode as  $D_1$  and the left diode as  $D_2$ . The node voltage  $V$  is measured with respect to ground.



#### 2. Theory & Assumptions

To solve circuits containing ideal diodes, we assume an operating state (ON or OFF) for each diode, compute the resulting voltages and currents, and then verify if the physical conditions for our assumptions hold true:

- If **ON**, the ideal diode acts as a short circuit (0 V drop), and the forward current must be positive ( $I_D > 0$ ).
- If **OFF**, it acts as an open circuit (0 A current), and the reverse voltage must be negative ( $V_D < 0$ ).

**Hypothesis:** We assume that  $D_1$  is **ON** and  $D_2$  is **OFF**.

- Since  $D_2$  is OFF, it acts as an open circuit. Therefore, no current flows through its branch, which immediately gives us  $I = 0$  A.
- Since  $D_1$  is ON, it behaves as a short circuit. The  $5\text{ k}\Omega$  and  $10\text{ k}\Omega$  resistors are now connected in a single continuous series path between the  $+10\text{ V}$  and  $-10\text{ V}$  sources.

#### 3. Mathematical Derivation & Verification

With  $D_1$  shorted and  $D_2$  open, the total current  $I_{\text{series}}$  flowing through the right branch is determined by the total voltage difference divided by the total equivalent resistance:

$$I_{\text{series}} = \frac{10\text{ V} - (-10\text{ V})}{5\text{ k}\Omega + 10\text{ k}\Omega} = \frac{20\text{ V}}{15\text{ k}\Omega} = \frac{4}{3}\text{ mA} \approx 1.33\text{ mA} \quad (1)$$

Because this current flows downward through  $D_1$ , the forward current is  $I_{D1} = 1.33\text{ mA} > 0$ . The assumption that  $D_1$  is ON is verified.

Next, we calculate the node voltage  $V$ . Using Kirchhoff's Voltage Law (KVL) from the top  $+10\text{ V}$  source down to node  $V$ :

$$\begin{aligned} V &= 10\text{ V} - I_{\text{series}}(5\text{ k}\Omega) \\ V &= 10 - \left(\frac{4}{3}\text{ mA}\right)(5\text{ k}\Omega) \\ V &= 10 - \frac{20}{3} = \frac{10}{3}\text{ V} \approx 3.33\text{ V} \end{aligned} \quad (2)$$

Finally, we must verify the OFF state of  $D_2$ . The anode of  $D_2$  is connected to ground ( $0\text{ V}$ ). Because  $D_1$  is a short circuit, the cathode of  $D_2$  is at the same potential as  $V$  ( $3.33\text{ V}$ ).

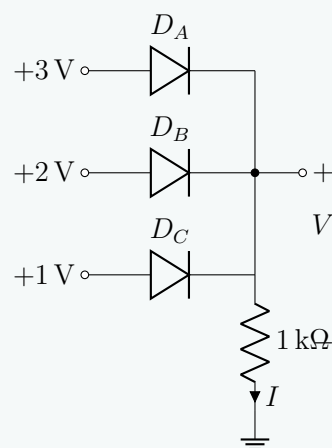
$$V_{D2} = V_{\text{anode}} - V_{\text{cathode}} = 0\text{ V} - 3.33\text{ V} = -3.33\text{ V} \quad (3)$$

Since  $V_{D2} < 0\text{ V}$ , diode  $D_2$  is reverse-biased. *The assumption that  $D_2$  is OFF is verified.*

## Part (ii): Three-Diode Circuit Analysis

### 1. Circuit Schematic

Let us label the three diodes  $D_A$ ,  $D_B$ , and  $D_C$  corresponding to their respective input voltages.



### 2. Theory & Principle

This configuration represents a classic analog **maximum-voltage selector** (a Diode OR Gate). In a common-cathode parallel array, the ideal diode connected to the highest positive potential will conduct, clamping the output node  $V$  to its input voltage. This clamping action simultaneously reverse-biases all other diodes in the circuit.

**Hypothesis:** We assume that  $D_A$  (connected to  $+3\text{ V}$ ) is **ON**, while  $D_B$  ( $+2\text{ V}$ ) and  $D_C$  ( $+1\text{ V}$ ) are **OFF**.

### 3. Mathematical Derivation & Verification

Since  $D_A$  is assumed ON and ideal, it acts as a short circuit, directly tying the common cathode node to its anode voltage:

$$V = 3\text{ V} \quad (4)$$

Now, we calculate the current  $I$  flowing through the  $1\text{ k}\Omega$  resistor to ground via Ohm's Law:

$$I = \frac{V - 0\text{ V}}{1\text{ k}\Omega} = \frac{3\text{ V}}{1\text{ k}\Omega} = 3\text{ mA} \quad (5)$$

Because  $D_B$  and  $D_C$  are open circuits, this entire  $3\text{ mA}$  current is supplied by  $D_A$ . Since  $I_{DA} = 3\text{ mA} > 0$ , *the assumption that  $D_A$  is ON is verified.*

Lastly, we check the voltage drops across the other two diodes to ensure they are safely OFF:

$$V_{DB} = V_{\text{anode},B} - V = 2\text{ V} - 3\text{ V} = -1\text{ V} \quad (6)$$

$$V_{DC} = V_{\text{anode},C} - V = 1\text{ V} - 3\text{ V} = -2\text{ V} \quad (7)$$

Because both  $V_{DB} < 0$  and  $V_{DC} < 0$ , both  $D_B$  and  $D_C$  are reverse-biased. *The assumption that  $D_B$  and  $D_C$  are OFF is verified.*

## Final Answers

For Circuit (i):

$$\mathbf{I = 0\text{ mA}, \quad V = 3.33\text{ V} \quad \left(\text{or } \frac{10}{3}\text{ V}\right)} \quad (8)$$

For Circuit (ii):

$$\mathbf{I = 3\text{ mA}, \quad V = 3\text{ V}} \quad (9)$$

### Diode Models (Ideal, CVD, Piecewise Linear)

**Q1.** Describe the different models for diode forward characteristics. (7 marks)

[EEE 263 | L-2/T-1 | 2022–23]

#### Answer

##### Introduction

In analog electronic circuit analysis and design, the semiconductor  $p$ - $n$  junction diode exhibits a highly non-linear forward conduction characteristic. To simplify circuit analysis, engineers use various mathematical and circuit approximations depending on the required accuracy and numerical complexity. These approximations are classified into four primary models: the **Exponential Model**, the **Ideal Model**, the **Constant-Voltage-Drop Model**, and the **Piecewise-Linear Model**.

##### 1. The Exponential (Exact) Model

###### Theory and Circuit Operation

The Exponential Model is the most accurate physical description of the diode's forward behavior. It is derived from solid-state physics and accounts for carrier diffusion across the  $p$ - $n$  junction under forward-bias conditions.

###### Mathematical Derivation

The current-voltage ( $i$ - $v$ ) relation is governed by the **Shockley Diode Equation**:

$$I_D = I_S \left( e^{\frac{V_D}{nV_T}} - 1 \right) \quad (1)$$

For a significant forward-bias voltage ( $V_D \gg nV_T$ ), the term  $-1$  becomes negligible, simplifies the relation to:

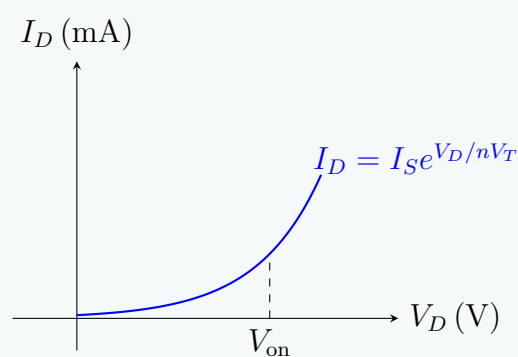
$$I_D \approx I_S e^{\frac{V_D}{nV_T}} \quad (2)$$

Where:

- $I_D$  = Diode forward current (A)
- $V_D$  = Diode forward voltage drop (V)
- $I_S$  = Reverse saturation current (typically  $10^{-15}$  A to  $10^{-12}$  A for small-signal silicon diodes)
- $n$  = Ideality factor (typically 1 to 2, depending on the material and fabrication structure)
- $V_T$  = Thermal voltage, expressed as:

$$V_T = \frac{kT}{q} \quad (3)$$

At room temperature ( $T \approx 300$  K),  $V_T \approx 25.3$  mV  $\approx 25$  mV or 26 mV.



##### Applications

Highly precise analog designs, log/antilog amplifiers, temperature sensors, and computer-aided design (SPICE) simulations.

##### 2. The Ideal Diode Model

###### Theory and Circuit Operation

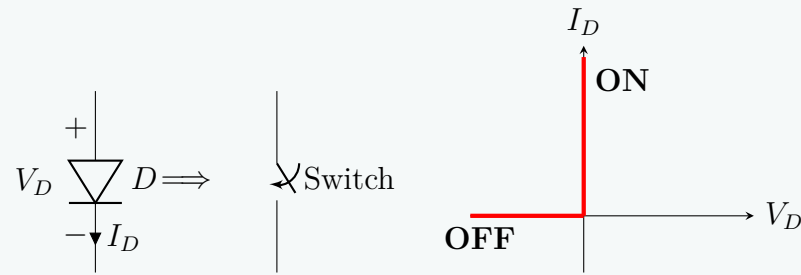
The Ideal Diode Model is the simplest approximation, treating the diode as a perfect electronic switch. It assumes zero forward resistance and an instantaneous transition from non-conduction to conduction at a zero threshold voltage.

###### Mathematical Derivation

The behavior is mathematically structured as a piecewise conditional state:

$$\begin{cases} V_D = 0, & \text{for } I_D > 0 \quad (\text{ON State / Short Circuit}) \\ I_D = 0, & \text{for } V_D < 0 \quad (\text{OFF State / Open Circuit}) \end{cases} \quad (4)$$

### Circuit Representation & $i$ - $v$ Curve



### Applications

Initial circuit inspection, high-voltage power supply rectification, and conceptual logic gate behavior analyses.

## 3. The Constant-Voltage-Drop (CVD) Model

### Theory and Circuit Operation

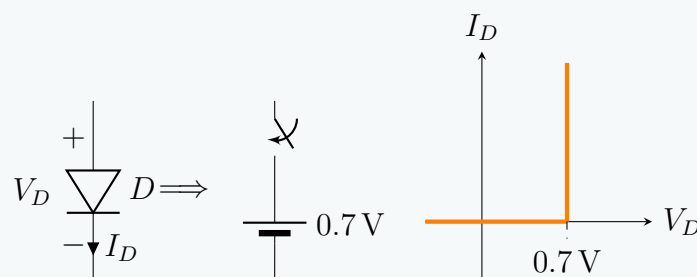
The Constant-Voltage-Drop Model acknowledges that a semiconductor  $p$ - $n$  junction requires a specific potential barrier clearance to initiate conduction. Once the threshold voltage is reached, the diode conducts current with a fixed voltage drop, completely independent of the magnitude of the forward current.

### Mathematical Derivation

For silicon diodes, this stable threshold drop ( $V_D$ ) is conventionally set to 0.7 V. The conditional state equation yields:

$$\begin{cases} V_D = 0.7 \text{ V,} & \text{for } I_D > 0 \quad (\text{ON State}) \\ I_D = 0, & \text{for } V_D < 0.7 \text{ V} \quad (\text{OFF State}) \end{cases} \quad (5)$$

### Circuit Representation & $i$ - $v$ Curve



### Applications

Standard analysis of typical analog circuits, power supply regulators, wave-shaping clippers, and clamping networks.

## 4. The Piecewise-Linear (PWL) Model

### Theory and Circuit Operation

The Piecewise-Linear model scales up the CVD model by embedding an internal series dynamic resistance ( $r_D$ ). This linearizes the slope of the forward exponential curve beyond the initial conduction threshold voltage ( $V_{D0}$ ), matching the behavior of a real diode under varying current levels.

### Mathematical Derivation

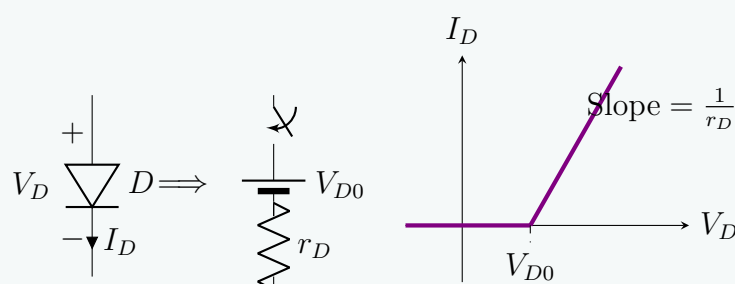
When forward-biased beyond  $V_{D0}$ , the total forward voltage drop across the diode becomes linear with respect to the forward current:

$$V_D = V_{D0} + I_D r_D \quad \text{for } V_D \geq V_{D0} \quad (6)$$

The dynamic forward resistance is calculated mathematically as the inverse slope of the operational  $i$ - $v$  linear trace:

$$r_D = \frac{\Delta V_D}{\Delta I_D} = \left. \frac{dv_D}{di_D} \right|_{Q\text{-point}} = \frac{nV_T}{I_Q} \quad (7)$$

### Circuit Representation & $i$ - $v$ Curve



### Applications

Precise small-signal analysis, power electronics thermal estimations, and small-signal incremental amplifier architectures.

### Summary and Comparison Model Table

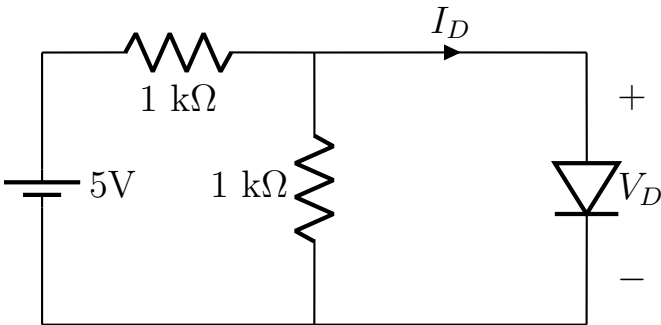
| Parameter              | Exponential Model           | Ideal Model              | Constant-Voltage Drop | Piecewise-Linear                             |
|------------------------|-----------------------------|--------------------------|-----------------------|----------------------------------------------|
| Threshold ( $V_{on}$ ) | Variable ( $I_D$ dependent) | 0 V                      | 0.7 V (Silicon)       | $V_{D0} \approx 0.6\text{ V} - 0.7\text{ V}$ |
| Internal Resistance    | Non-linear, dynamic         | 0 $\Omega$               | 0 $\Omega$            | Finite dynamic $r_D$                         |
| Complexity Level       | Very High (Transcendental)  | Extremely Low            | Low                   | Moderate                                     |
| Accuracy Class         | Exact Physics Reference     | First-Pass Approximation | Standard Standard     | Practical High Local Linearity               |

### Exam Tips

- When solving generic exam circuit problems, unless specified otherwise, default to the **Constant-Voltage-Drop Model** (0.7 V for Si) as it strikes the best balance between speed and practical accuracy.
- Always clarify your model assumption at the start of your calculation steps to secure full marks for method derivation.

**Q2.** Determine the current  $I_D$  and diode voltage  $V_D$  for the circuit shown using:

- (a) Piecewise linear model ( $V_{DO} = 0.7\text{ V}$ ,  $r_D = 20\text{ }\Omega$ )
- (b) Constant voltage drop model ( $V_{DO} = 0.7\text{ V}$ )
- (c) Ideal diode model



(18 marks)

[EEE 263 | L-2/T-1 | 2011–12]

### Answer

#### 1. Circuit Analysis and Thévenin Equivalent

To determine the current  $I_D$  and voltage  $V_D$  across the diode efficiently for multiple models, we first simplify the linear portion of the circuit connected to the diode using Thévenin's theorem.

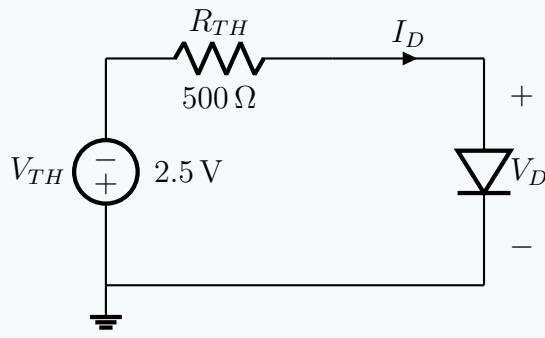
**Original Circuit**

**Thévenin Simplification**

We remove the diode from the circuit to find the open-circuit Thévenin voltage ( $V_{TH}$ ) and the Thévenin equivalent resistance ( $R_{TH}$ ) looking back into the terminals.

$$V_{TH} = 5\text{ V} \times \frac{1\text{ k}\Omega}{1\text{ k}\Omega + 1\text{ k}\Omega} = 5\text{ V} \times 0.5 = 2.5\text{ V} \tag{1}$$
$$R_{TH} = 1\text{ k}\Omega \parallel 1\text{ k}\Omega = \frac{1\text{ k}\Omega \times 1\text{ k}\Omega}{1\text{ k}\Omega + 1\text{ k}\Omega} = 0.5\text{ k}\Omega = 500\text{ }\Omega \tag{2}$$

Since  $V_{TH} = 2.5\text{ V}$  is strictly positive (and greater than typical diode turn-on voltages), we can safely assume the diode is **forward-biased (ON)** in all subsequent models.



## 2. Step-by-Step Solution for Diode Models

### (a) Piecewise Linear Model

**Theory:** In the piecewise linear model, a forward-biased diode is replaced by a DC voltage source  $V_{DO}$  in series with a dynamic forward resistance  $r_D$ .

Given:  $V_{DO} = 0.7 \text{ V}$ ,  $r_D = 20 \Omega$ .

Applying Kirchhoff's Voltage Law (KVL) to the Thévenin equivalent circuit:

$$V_{TH} - I_D R_{TH} - V_{DO} - I_D r_D = 0 \quad (3)$$

$$I_D (R_{TH} + r_D) = V_{TH} - V_{DO} \quad (4)$$

$$I_D = \frac{V_{TH} - V_{DO}}{R_{TH} + r_D} \quad (5)$$

Substituting the values:

$$I_D = \frac{2.5 \text{ V} - 0.7 \text{ V}}{500 \Omega + 20 \Omega} = \frac{1.8 \text{ V}}{520 \Omega} \approx 3.4615 \text{ mA} \quad (6)$$

The voltage across the diode  $V_D$  is the sum of the threshold voltage and the drop across its internal resistance:

$$V_D = V_{DO} + I_D r_D = 0.7 \text{ V} + (3.4615 \times 10^{-3} \text{ A})(20 \Omega) = 0.7 \text{ V} + 0.0692 \text{ V} = 0.7692 \text{ V} \quad (7)$$

### (b) Constant Voltage Drop Model

**Theory:** This model assumes that once the diode is forward-biased, it maintains a constant voltage drop  $V_{DO}$  regardless of the current flowing through it. It neglects the internal forward resistance ( $r_D = 0$ ).

Given:  $V_{DO} = 0.7 \text{ V}$ .

Applying KVL:

$$I_D = \frac{V_{TH} - V_{DO}}{R_{TH}} \quad (8)$$

Substituting the values:

$$I_D = \frac{2.5 \text{ V} - 0.7 \text{ V}}{500 \Omega} = \frac{1.8 \text{ V}}{500 \Omega} = 3.6 \text{ mA} \quad (9)$$

The voltage across the diode is simply the constant threshold voltage:

$$V_D = 0.7 \text{ V} \quad (10)$$

### (c) Ideal Diode Model

**Theory:** The ideal diode model treats the forward-biased diode as a perfect conductor (short circuit) with zero voltage drop.

Given:  $V_{DO} = 0 \text{ V}$ ,  $r_D = 0 \Omega$ .

Applying KVL:

$$I_D = \frac{V_{TH}}{R_{TH}} \quad (11)$$

Substituting the values:

$$I_D = \frac{2.5 \text{ V}}{500 \Omega} = 5.0 \text{ mA} \quad (12)$$

The voltage across the diode is zero:

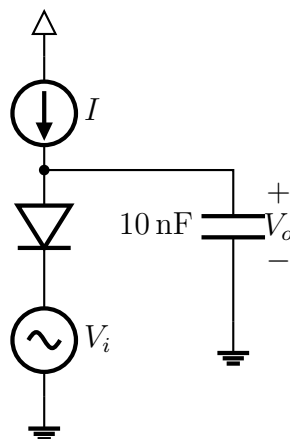
$$V_D = 0 \text{ V} \quad (13)$$

## 3. Final Summary

| Part | Diode Model                                                       | Diode Current ( $I_D$ ) | Diode Voltage ( $V_D$ ) |
|------|-------------------------------------------------------------------|-------------------------|-------------------------|
| (a)  | Piecewise Linear ( $V_{DO} = 0.7 \text{ V}$ , $r_D = 20 \Omega$ ) | 3.46 mA                 | 0.769 V                 |
| (b)  | Constant Voltage Drop ( $V_{DO} = 0.7 \text{ V}$ )                | 3.60 mA                 | 0.700 V                 |
| (c)  | Ideal Diode                                                       | 5.00 mA                 | 0.000 V                 |

## Small Signal Model

- Q1.** In the circuit shown, “ $I$ ” is a DC current source and “ $V_i$ ” is a sinusoidal signal with small amplitude ( $< 10\text{ mV}$ ) and a frequency of  $100\text{ kHz}$ . Representing the diode by its small-signal resistance  $r_d$  (a function of  $I$ ):
- sketch the small-signal equivalent circuit for determining the sinusoidal output  $V_o$  across the capacitor;
  - find the phase shift between  $V_i$  and  $V_o$ ;
  - find the value of  $I$  that will provide a phase shift of  $-45^\circ$ ;
  - find the range of phase shift achieved as  $I$  is varied from  $0.1$  to  $10$  times the value found in (iii). Assume  $n = 1$ .



(20 marks)

[EEE 263 | L-2/T-1 | 2023–24]

## Answer

## Circuit Operation and Small-Signal Modeling

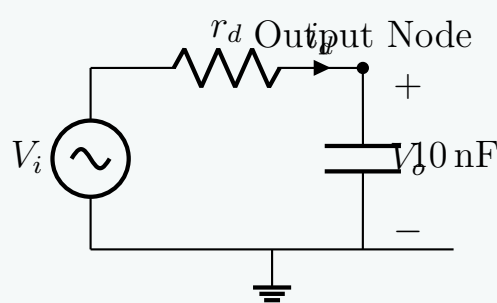
The given circuit operates with a combination of a DC bias and a small-signal AC variation.

- DC Analysis:** The ideal DC current source  $I$  forces a constant bias current through the diode, establishing the quiescent operating point (Q-point). Because  $V_i$  is a purely AC sinusoidal source, its DC voltage is  $0\text{ V}$  (acting as a short circuit to ground in DC analysis). Consequently, the entire DC current  $I$  flows through the diode to ground.
- AC Small-Signal Analysis:** By applying the superposition principle, the DC current source  $I$  is deactivated (replaced by an open circuit because it is an ideal current source). The diode is replaced by its small-signal dynamic resistance  $r_d$ , which is determined by the DC bias current  $I$ .

## (i) Small-Signal Equivalent Circuit

To construct the small-signal equivalent circuit for determining the AC output  $V_o$ :

- The ideal DC current source  $I$  becomes an open circuit.
- The diode is replaced by its small-signal resistance  $r_d$ .
- The rest of the AC components ( $V_i$  and the  $10\text{ nF}$  capacitor) remain active.



*Note: The circuit is drawn horizontally for standard signal flow. This electrically matches the vertical topology of the original circuit, forming a standard RC low-pass filter.*

 (ii) Phase Shift between  $V_i$  and  $V_o$ 

From the small-signal equivalent circuit, the system operates as an AC voltage divider. The output voltage  $V_o$  is taken across the capacitor. Using complex impedances, the transfer function  $H(j\omega)$  is:

$$\frac{V_o}{V_i} = \frac{Z_C}{r_d + Z_C} = \frac{\frac{1}{j\omega C}}{r_d + \frac{1}{j\omega C}} \quad (1)$$

$$\frac{V_o}{V_i} = \frac{1}{1 + j\omega r_d C} \quad (2)$$

The phase shift  $\phi$  between the output  $V_o$  and input  $V_i$  is the phase angle of the transfer function:

$$\phi = \angle \left( \frac{1}{1 + j\omega r_d C} \right) = \angle(1) - \angle(1 + j\omega r_d C) \quad (3)$$

$$\phi = 0^\circ - \tan^{-1} \left( \frac{\omega r_d C}{1} \right) \quad (4)$$

$$\phi = -\tan^{-1}(\omega r_d C) \quad (5)$$

### (iii) Value of $I$ for a $-45^\circ$ Phase Shift

We are given the required condition for the phase shift:

$$\phi = -45^\circ \quad (6)$$

Substituting this into our phase equation:

$$-\tan^{-1}(\omega r_d C) = -45^\circ \quad (7)$$

$$\omega r_d C = \tan(45^\circ) = 1 \quad (8)$$

Solving for the required small-signal resistance  $r_d$ :

$$r_d = \frac{1}{\omega C} \quad (9)$$

Using the given parameters ( $f = 100$  kHz and  $C = 10$  nF):

- Angular frequency  $\omega = 2\pi f = 2\pi \times 10^5$  rad/s
- Capacitance  $C = 10 \times 10^{-9}$  F =  $10^{-8}$  F

$$r_d = \frac{1}{(2\pi \times 10^5)(10^{-8})} = \frac{1}{2\pi \times 10^{-3}} = \frac{500}{\pi} \Omega \approx 159.15 \Omega \quad (10)$$

The small-signal resistance of a diode is inversely proportional to its DC bias current:

$$r_d = \frac{nV_T}{I} \implies I = \frac{nV_T}{r_d} \quad (11)$$

Assuming a standard room temperature thermal voltage  $V_T \approx 25$  mV and  $n = 1$  (given):

$$I = \frac{1 \times 25 \times 10^{-3} \text{ V}}{159.15 \Omega} \approx 1.571 \times 10^{-4} \text{ A} = \mathbf{157.1 \mu A} \quad (12)$$

### (iv) Range of Phase Shift as $I$ is Varied

Let  $I_{45}$  be the current found in part (iii) that yields  $\omega r_d C = 1$ . Because  $r_d = \frac{nV_T}{I}$ , the resistance  $r_d$  scales inversely with  $I$ . Consequently, the unitless product  $\omega r_d C$  also scales inversely with  $I$ :

$$\omega r_d C = \frac{I_{45}}{I} \quad (13)$$

We evaluate the phase shift  $\phi = -\tan^{-1}(\omega r_d C)$  at the two extremes of the given current range:

(a) **When**  $I = 0.1 \times I_{45}$ :

$$\omega r_d C = \frac{I_{45}}{0.1 I_{45}} = 10 \quad (14)$$

$$\phi_1 = -\tan^{-1}(10) \approx \mathbf{-84.29^\circ} \quad (15)$$

(b) **When**  $I = 10 \times I_{45}$ :

$$\omega r_d C = \frac{I_{45}}{10 I_{45}} = 0.1 \quad (16)$$

$$\phi_2 = -\tan^{-1}(0.1) \approx \mathbf{-5.71^\circ} \quad (17)$$

**Final Conclusion:** By varying the DC bias current  $I$  from 0.1 to 10 times the center  $-45^\circ$  condition value, the phase shift can be electronically tuned over a range from  $-5.71^\circ$  to  $-84.29^\circ$ . This highlights the circuit's fundamental utility as a highly tunable current-controlled phase shifter.

**Q2.** The small-signal model of a diode is said to be valid for voltage variations of about 10 mV. To what percentage current change does this correspond for  $n = 1$  (consider both positive and negative signals)? What is the maximum allowable voltage signal (positive or negative) if the current change is limited to 10%? **(10 marks)** [EEE 263 | L-2/T-1 | 2014–15]

## Answer

### Theory: Diode Large-Signal and Small-Signal Analysis

The total instantaneous current  $i_D$  flowing through a forward-biased diode is related to the total instantaneous voltage  $v_D$  across it by the exact exponential equation:

$$i_D = I_S e^{\frac{v_D}{nV_T}} \quad (1)$$

In a small-signal application, the diode operates at a DC quiescent point (Q-point) defined by a DC voltage  $V_D$  and a corresponding DC current  $I_D$ , upon which a small time-varying AC signal  $v_d$  is superimposed:

$$v_D(t) = V_D + v_d(t) \quad (2)$$

Substituting this into the diode equation yields the total instantaneous current:

$$i_D = I_S e^{\frac{V_D + v_d}{nV_T}} = \left( I_S e^{\frac{V_D}{nV_T}} \right) e^{\frac{v_d}{nV_T}} = I_D e^{\frac{v_d}{nV_T}} \quad (3)$$

The change in the diode current,  $\Delta i$ , due to the AC signal  $v_d$  is:

$$\Delta i = i_D - I_D = I_D \left( e^{\frac{v_d}{nV_T}} - 1 \right) \quad (4)$$

The **fractional (or percentage) current change** is thus defined as:

$$\frac{\Delta i}{I_D} = e^{\frac{v_d}{nV_T}} - 1 \quad (5)$$

*Assumption:* For typical small-signal analysis at room temperature, the thermal voltage is assumed to be  $V_T \approx 25$  mV. The ideality factor is given as  $n = 1$ .

#### Part 1: Percentage Current Change for $v_d = \pm 10$ mV

To evaluate the validity of the small-signal approximation for a  $\pm 10$  mV variation, we use Equation (5).

##### Case 1A: Positive Voltage Signal ( $v_d = +10$ mV)

$$\frac{\Delta i}{I_D} = e^{\frac{+10 \text{ mV}}{1 \times 25 \text{ mV}}} - 1 \quad (6)$$

$$= e^{0.4} - 1 \quad (7)$$

$$\approx 1.4918 - 1 = 0.4918 \quad (8)$$

Expressed as a percentage, a  $+10$  mV variation causes a  $+49.2\%$  increase in current.

##### Case 1B: Negative Voltage Signal ( $v_d = -10$ mV)

$$\frac{\Delta i}{I_D} = e^{\frac{-10 \text{ mV}}{1 \times 25 \text{ mV}}} - 1 \quad (9)$$

$$= e^{-0.4} - 1 \quad (10)$$

$$\approx 0.6703 - 1 = -0.3297 \quad (11)$$

Expressed as a percentage, a  $-10$  mV variation causes a  $-33.0\%$  decrease in current.

*Important Note:* The asymmetry ( $+49.2\%$  vs.  $-33.0\%$ ) arises directly from the highly non-linear (exponential) nature of the diode's  $i-v$  characteristic. The small-signal (linear) model predicts a symmetric  $\pm 40\%$  change ( $\frac{v_d}{V_T} = \frac{10}{25}$ ), illustrating the approximation error when  $v_d$  is as large as  $10$  mV.

#### Part 2: Maximum Voltage Signal for a 10% Current Change

To ensure high linearity, we often restrict the current change to  $\pm 10\%$ . We must find the maximum allowable voltage signal  $v_d$  that satisfies this constraint.

Rearranging Equation (5) to solve for  $v_d$ :

$$v_d = nV_T \ln \left( 1 + \frac{\Delta i}{I_D} \right) \quad (12)$$

##### Case 2A: Positive Current Limit ( $\frac{\Delta i}{I_D} = +0.10$ )

$$v_{d(\text{max}+)} = 1 \times 25 \text{ mV} \times \ln(1 + 0.10) \quad (13)$$

$$= 25 \text{ mV} \times \ln(1.10) \quad (14)$$

$$\approx 25 \text{ mV} \times 0.0953 \quad (15)$$

$$= +\mathbf{2.38 \text{ mV}} \quad (16)$$

##### Case 2B: Negative Current Limit ( $\frac{\Delta i}{I_D} = -0.10$ )

$$v_{d(\text{max}-)} = 1 \times 25 \text{ mV} \times \ln(1 - 0.10) \quad (17)$$

$$= 25 \text{ mV} \times \ln(0.90) \quad (18)$$

$$\approx 25 \text{ mV} \times (-0.1054) \quad (19)$$

$$= -\mathbf{2.63 \text{ mV}} \quad (20)$$

### Final Answer Summary

| Parameter                                               | Calculated Value |
|---------------------------------------------------------|------------------|
| Percentage Current Change for $v_d = +10 \text{ mV}$    | <b>+49.2%</b>    |
| Percentage Current Change for $v_d = -10 \text{ mV}$    | <b>-33.0%</b>    |
| Maximum Positive Voltage Signal for +10% Current Change | <b>+2.38 mV</b>  |
| Maximum Negative Voltage Signal for -10% Current Change | <b>-2.63 mV</b>  |

**Exam Tip:** Notice that to keep the small-signal model highly accurate (within 10% distortion), the AC voltage signal amplitude must be kept very small—generally strictly less than roughly 2.5 mV for  $n = 1$ .

### Peak Rectifier

**Q1.** Consider a half-wave rectifier circuit with  $R = 1 \text{ k}\Omega$  load operating from a 120 V (rms), 60 Hz household supply through a 10-to-1 step-down transformer, augmented with a filter capacitor  $C$  across the load. Assume  $V_D = 0.7 \text{ V}$  for any current.

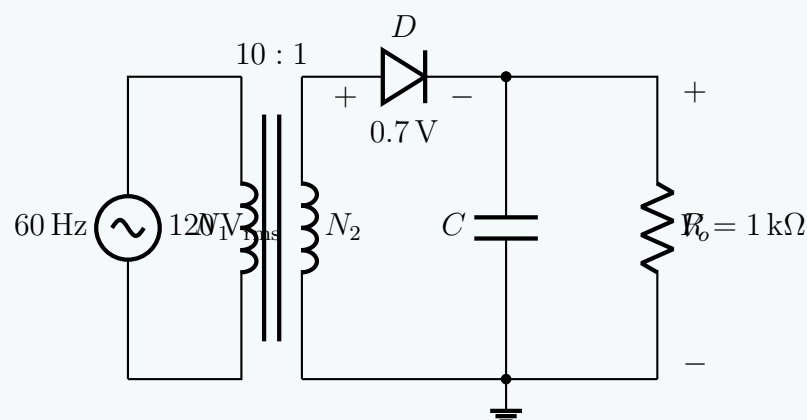
- Find  $C$  for a peak-to-peak ripple of 10% of peak output.
- Find  $C$  for a peak-to-peak ripple of 1% of peak output.
- Find the average output voltage for each of the case (i) & (ii) .
- Find the fraction of the cycle for which the diode conducts for each of the case (i) & (ii).
- Find the average and peak diode currents for each of the case (i) & (ii).

**(20 marks)**

[EEE 263 | L-2/T-1 | 2024–25]

### Answer

#### Circuit Diagram



#### Theory & Initial Calculations

**1. Secondary Voltage Calculation:** The step-down transformer reduces the primary AC voltage by the turns ratio  $N_1/N_2 = 10$ .

$$V_{s(\text{rms})} = \frac{V_{p(\text{rms})}}{10} = \frac{120 \text{ V}}{10} = 12 \text{ V}_{\text{rms}} \quad (1)$$

The peak voltage of the secondary transformer winding is:

$$V_{s(\text{peak})} = V_{s(\text{rms})} \times \sqrt{2} = 12\sqrt{2} \approx 16.97 \text{ V} \quad (2)$$

**2. Peak Output Voltage ( $V_p$ ):** For a half-wave rectifier, the diode conducts when forward-biased, causing a constant voltage drop  $V_D = 0.7 \text{ V}$ . The peak output voltage across the load and capacitor is:

$$V_p = V_{s(\text{peak})} - V_D = 16.97 \text{ V} - 0.7 \text{ V} = 16.27 \text{ V} \quad (3)$$

*Note: The ripple frequency  $f$  for a half-wave rectifier is equal to the source frequency (60 Hz).*

#### Step-by-Step Solution

We utilize the standard approximations for peak rectifier circuits (valid when  $V_r \ll V_p$ ):

- Ripple Voltage:**  $V_r \approx \frac{V_p}{fRC}$
- DC Output Voltage:**  $V_{\text{dc}} \approx V_p - \frac{V_r}{2}$
- Conduction Angle:**  $\theta_c = \omega\Delta t \approx \sqrt{\frac{2V_r}{V_p}}$  (in radians)
- Average Diode Current (Full Cycle):**  $I_{D,\text{avg}} = I_{\text{load}} = \frac{V_{\text{dc}}}{R}$
- Peak Diode Current:**  $i_{D,\text{max}} = I_{\text{load}} \left(1 + 2\pi\sqrt{\frac{2V_p}{V_r}}\right)$

**Case (i): Peak-to-Peak Ripple is 10% of Peak Output**

Given  $V_r = 0.10 \times V_p = 0.10 \times 16.27 \text{ V} = 1.627 \text{ V}$ .

(i) **Filter Capacitor  $C$ :**

$$C = \frac{V_p}{V_r f R} = \frac{V_p}{0.10 V_p f R} = \frac{1}{0.10 \times 60 \times 1000} = \frac{1}{6000} \approx \mathbf{166.7 \mu\text{F}} \quad (4)$$

(iii) **Average Output Voltage ( $V_{dc}$ ):**

$$V_{dc} = V_p - \frac{V_r}{2} = 16.27 - \frac{1.627}{2} = 16.27 - 0.8135 = \mathbf{15.46 \text{ V}} \quad (5)$$

(iv) **Fraction of Cycle for Conduction:** The conduction angle  $\theta_c = \sqrt{\frac{2 \times 0.10 V_p}{V_p}} = \sqrt{0.20} \approx 0.4472 \text{ rad}$ .

$$\text{Fraction} = \frac{\theta_c}{2\pi} = \frac{0.4472}{2\pi} \approx \mathbf{0.0712} \text{ (or 7.12\%)} \quad (6)$$

(v) **Average and Peak Diode Currents:** The average diode current over the entire cycle equals the DC load current:

$$I_{D,avg} = I_{load} = \frac{V_{dc}}{R} = \frac{15.46 \text{ V}}{1 \text{ k}\Omega} = \mathbf{15.46 \text{ mA}} \quad (7)$$

The peak diode current is calculated as:

$$i_{D,max} = I_{load} \left( 1 + 2\pi \sqrt{\frac{2V_p}{0.10V_p}} \right) = 15.46 \text{ mA} \left( 1 + 2\pi \sqrt{20} \right) \approx 15.46(1 + 28.10) = \mathbf{449.8 \text{ mA}} \quad (8)$$

**Case (ii): Peak-to-Peak Ripple is 1% of Peak Output**

Given  $V_r = 0.01 \times V_p = 0.01 \times 16.27 \text{ V} = 0.1627 \text{ V}$ .

(ii) **Filter Capacitor  $C$ :**

$$C = \frac{V_p}{V_r f R} = \frac{1}{0.01 \times 60 \times 1000} = \frac{1}{600} \approx \mathbf{1666.7 \mu\text{F} \text{ (or 1.67 mF)}} \quad (9)$$

(iii) **Average Output Voltage ( $V_{dc}$ ):**

$$V_{dc} = V_p - \frac{V_r}{2} = 16.27 - \frac{0.1627}{2} = 16.27 - 0.08135 = \mathbf{16.19 \text{ V}} \quad (10)$$

(iv) **Fraction of Cycle for Conduction:** The conduction angle  $\theta_c = \sqrt{\frac{2 \times 0.01 V_p}{V_p}} = \sqrt{0.02} \approx 0.1414 \text{ rad}$ .

$$\text{Fraction} = \frac{\theta_c}{2\pi} = \frac{0.1414}{2\pi} \approx \mathbf{0.0225} \text{ (or 2.25\%)} \quad (11)$$

(v) **Average and Peak Diode Currents:** The average diode current over the entire cycle:

$$I_{D,avg} = I_{load} = \frac{V_{dc}}{R} = \frac{16.19 \text{ V}}{1 \text{ k}\Omega} = \mathbf{16.19 \text{ mA}} \quad (12)$$

The peak diode current is:

$$i_{D,max} = 16.19 \text{ mA} \left( 1 + 2\pi \sqrt{\frac{2V_p}{0.01V_p}} \right) = 16.19 \text{ mA} \left( 1 + 2\pi \sqrt{200} \right) \approx 16.19(1 + 88.86) = \mathbf{1454.8 \text{ mA} \text{ (or 1.45 A)}} \quad (13)$$

**Final Answer Summary Table**

| Part      | Parameter                                   | Case (i) 10% Ripple | Case (ii) 1% Ripple  |
|-----------|---------------------------------------------|---------------------|----------------------|
| (i), (ii) | Filter Capacitor ( $C$ )                    | 166.7 $\mu\text{F}$ | 1666.7 $\mu\text{F}$ |
| (iii)     | Average Output Voltage ( $V_{dc}$ )         | 15.46 V             | 16.19 V              |
| (iv)      | Conduction Fraction                         | 0.0712 (7.12%)      | 0.0225 (2.25%)       |
| (v)       | Cycle-Average Diode Current ( $I_{D,avg}$ ) | 15.46 mA            | 16.19 mA             |
| (v)       | Peak Diode Current ( $i_{D,max}$ )          | 449.8 mA            | 1454.8 mA            |

**Important Note on Diode Current:** The  $I_{D,avg}$  computed above is the average over the *entire cycle*, corresponding to the DC load current  $I_{load}$ . If a question explicitly asks for the average current strictly *during the conduction interval*, it is calculated as  $I_{D(cond\_avg)} = I_{load} / \text{Fraction}$ .

**Q2.** Consider a peak rectifier fed by a 60 Hz sinusoid with a peak value  $V_p = 100 \text{ V}$ . Let the load resistance  $R = 10 \text{ k}\Omega$ . Find the value

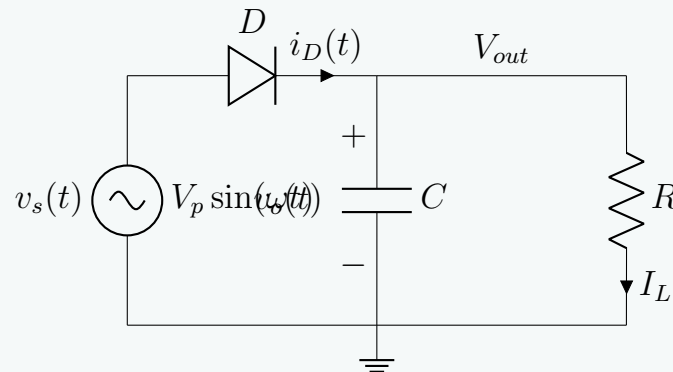
of capacitance  $C$  that will result in a peak-to-peak ripple of 2 V. Also calculate the fraction of the cycle during which the diode is conducting and the average and peak values of the diode current. **(10 marks)** [EEE 263 | L-2/T-1 | 2017–18]

### Answer

#### Assumptions:

For this analysis, we assume an ideal diode model ( $V_{D,on} = 0$  V) since the peak input voltage ( $V_p = 100$  V) is significantly larger than the typical forward voltage drop of a practical diode. Furthermore, since the specified ripple (2 V) is very small compared to the peak voltage ( $V_r \ll V_p$ ), the standard small-ripple textbook approximations hold true.

#### Circuit Diagram: Half-Wave Peak Rectifier



#### Given Parameters

- Source frequency,  $f = 60$  Hz
- Peak input voltage,  $V_p = 100$  V
- Load resistance,  $R = 10$  k $\Omega$
- Peak-to-peak ripple voltage,  $V_r = 2$  V

#### 1. Calculation of Filter Capacitance ( $C$ )

In a peak rectifier with a small ripple, the capacitor discharges linearly through the load resistor  $R$  over approximately the entire period  $T \approx \frac{1}{f}$ . The peak-to-peak ripple voltage  $V_r$  is given by:

$$V_r \approx \frac{V_p}{fRC}$$

Rearranging to solve for the capacitance  $C$ :

$$\begin{aligned} C &= \frac{V_p}{fRV_r} \\ &= \frac{100}{60 \times (10 \times 10^3) \times 2} \\ &= \frac{100}{1.2 \times 10^6} \approx 83.33 \times 10^{-6} \text{ F} = \mathbf{83.33 \mu\text{F}} \end{aligned}$$

#### 2. Conduction Angle and Fraction of the Cycle

The diode only conducts during a short interval  $\Delta t$  near the peak of the input voltage to replenish the charge lost by the capacitor. The conduction angle  $\theta_c = \omega\Delta t$  can be approximated by:

$$\omega\Delta t \approx \sqrt{\frac{2V_r}{V_p}}$$

Substituting the known values:

$$\omega\Delta t = \sqrt{\frac{2 \times 2}{100}} = \sqrt{0.04} = \mathbf{0.2 \text{ rad}}$$

To find the fraction of the cycle during which the diode conducts, we divide the conduction angle by a full period ( $2\pi$  radians):

$$\text{Fraction} = \frac{\omega\Delta t}{2\pi} = \frac{0.2}{2\pi} \approx \mathbf{0.0318 \text{ (or 3.18\%)}}$$

### 3. Average and Peak Diode Currents

During the non-conduction interval, the load current  $I_L$  is approximately constant and is supplied entirely by the capacitor:

$$I_L \approx \frac{V_p}{R} = \frac{100}{10 \text{ k}\Omega} = 10 \text{ mA}$$

#### Average Diode Current ( $i_{D,avg}$ ):

To maintain steady-state, the charge supplied by the diode during the short conduction interval must equal the charge drawn by the load over the full cycle. The average diode current during conduction is derived as:

$$\begin{aligned} i_{D,avg} &= I_L \left( 1 + \pi \sqrt{\frac{2V_p}{V_r}} \right) \\ &= 10 \text{ mA} \left( 1 + \pi \sqrt{\frac{2 \times 100}{2}} \right) \\ &= 10 \text{ mA} (1 + 10\pi) \\ &\approx 10 \text{ mA} \times (1 + 31.416) = \mathbf{324.16 \text{ mA}} \end{aligned}$$

#### Peak Diode Current ( $i_{D,max}$ ):

The maximum diode current occurs at the beginning of the conduction interval and is approximately twice the average charging current plus the load current:

$$\begin{aligned} i_{D,max} &= I_L \left( 1 + 2\pi \sqrt{\frac{2V_p}{V_r}} \right) \\ &= 10 \text{ mA} \left( 1 + 2\pi \sqrt{\frac{2 \times 100}{2}} \right) \\ &= 10 \text{ mA} (1 + 20\pi) \\ &\approx 10 \text{ mA} \times (1 + 62.832) = \mathbf{638.32 \text{ mA}} \end{aligned}$$

### Final Answer Summary

| Parameter                             | Calculated Value    |
|---------------------------------------|---------------------|
| Filter Capacitance ( $C$ )            | $83.33 \mu\text{F}$ |
| Conduction Angle ( $\omega\Delta t$ ) | $0.2 \text{ rad}$   |
| Conduction Fraction                   | $3.18\%$            |
| Average Diode Current ( $i_{D,avg}$ ) | $324.16 \text{ mA}$ |
| Peak Diode Current ( $i_{D,max}$ )    | $638.32 \text{ mA}$ |

**Q3.** Consider a half-wave peak rectifier fed with a voltage having a triangular waveform with 20 V peak-to-peak amplitude, zero average and 1 kHz frequency. Load resistance  $R = 100 \Omega$  and filter capacitor  $C = 100 \mu\text{F}$ . Assume the diode has a forward voltage drop of 0.7 V when conducting. Find: (i) average DC output voltage, (ii) time interval during which the diode conducts, (iii) average diode current during conduction, (iv) maximum diode current. **(20 marks)** [EEE 263 | L-2/T-1 | 2016–17]

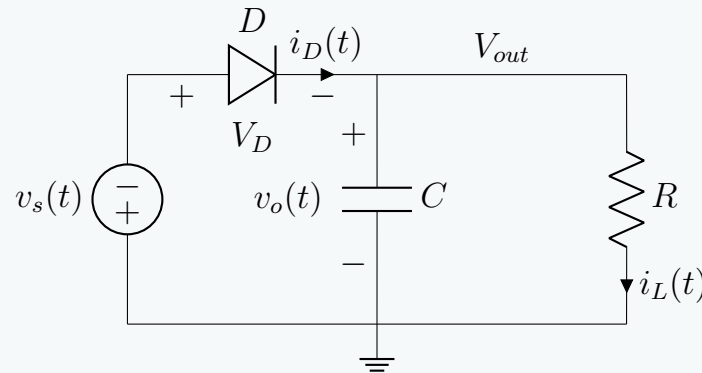
### Answer

#### Theory and Circuit Operation

A peak rectifier (or envelope detector) consists of a diode and a filter capacitor connected in parallel with a load resistance. When a zero-average triangular waveform is applied:

- Conduction Phase:** On the rising edge of the input voltage  $v_s(t)$ , once it exceeds the capacitor voltage plus the diode forward drop ( $v_s > v_o + V_D$ ), the diode turns ON. The capacitor quickly charges up, following the input waveform minus the diode drop:  $v_o(t) = v_s(t) - V_D$ .
- Non-Conduction Phase:** Immediately after the input passes its peak, the input slope decreases rapidly. The diode turns OFF because the capacitor cannot discharge fast enough to follow the sharp drop of the triangular input. The capacitor then discharges exponentially through the load resistor  $R$  with a time constant  $\tau = RC$ .

### Circuit Diagram



### Given Parameters

- Peak-to-peak input voltage:  $V_{pp} = 20 \text{ V}$
- Zero-average input implies: Peak input voltage  $V_p = \frac{V_{pp}}{2} = 10 \text{ V}$
- Input frequency:  $f = 1 \text{ kHz} \implies \text{Period } T = \frac{1}{f} = 1 \text{ ms}$
- Load resistance:  $R = 100 \Omega$
- Filter capacitance:  $C = 100 \mu\text{F}$
- Diode forward voltage drop:  $V_D = 0.7 \text{ V}$

The discharge time constant is:

$$\tau = RC = 100 \Omega \times 100 \mu\text{F} = 10 \text{ ms}$$

Since  $\tau = 10 \text{ ms} \gg T = 1 \text{ ms}$ , the small-ripple approximation and linear discharge assumption are highly accurate.

### Mathematical Derivation and Step-by-Step Solution

The maximum output voltage reached at the peak of the input is:

$$V_{omax} = V_p - V_D = 10 \text{ V} - 0.7 \text{ V} = 9.3 \text{ V}$$

**(i) Calculation of Peak-to-Peak Ripple ( $V_r$ ) and Average DC Output Voltage ( $V_{out,avg}$ )** During the non-conduction phase, which lasts for approximately the entire period ( $T - \Delta t \approx T$ ), the capacitor discharges linearly. The peak-to-peak ripple voltage  $V_r$  can be expressed as:

$$V_r \approx \frac{V_{omax}}{RC}(T - \Delta t) \quad (1)$$

On the rising edge of the triangular wave leading up to the peak (defined at  $t = 0$ ), the slope of the input voltage is constant:

$$\frac{dv_s}{dt} = \frac{V_{pp}}{T/2} = \frac{20 \text{ V}}{0.5 \text{ ms}} = 40,000 \text{ V/s} = 40 \text{ V/ms}$$

The diode turns ON at a time  $t = -\Delta t$  before the peak, where the input voltage rises enough to forward-bias the diode:

$$V_r = \left( \frac{dv_s}{dt} \right) \Delta t = 40,000 \cdot \Delta t$$

Equating the two relationships for  $V_r$  (using millisecond units for simplicity):

$$\begin{aligned} 40 \cdot \Delta t &= \frac{9.3}{10}(1 - \Delta t) \\ 40 \cdot \Delta t &= 0.93 - 0.93 \cdot \Delta t \\ 40.93 \cdot \Delta t &= 0.93 \implies \Delta t = \frac{0.93}{40.93} \approx \mathbf{0.02272 \text{ ms}} = \mathbf{22.72 \mu\text{s}} \end{aligned}$$

Now, substituting  $\Delta t$  back to find the precise ripple voltage  $V_r$ :

$$V_r = 40 \text{ V/ms} \times 0.02272 \text{ ms} = 0.9089 \text{ V}$$

The average DC output voltage is the midpoint of the linear ripple waveform:

$$\begin{aligned} V_{out,avg} &= V_{omax} - \frac{V_r}{2} \\ &= 9.3 \text{ V} - \frac{0.9089 \text{ V}}{2} = 9.3 \text{ V} - 0.4545 \text{ V} = \mathbf{8.8455 \text{ V}} \end{aligned}$$

**(ii) Time Interval During Which the Diode Conducts ( $\Delta t$ )** As derived above during the simultaneous solution for the ripple voltage:

$$\Delta t = \mathbf{22.72 \mu\text{s}}$$

**(iii) Average Diode Current During Conduction ( $i_{D,avg}$ )** During the conduction interval  $\Delta t$ , the diode current  $i_D(t)$  splits into the capacitor charging current  $i_C(t)$  and the load current  $i_L(t)$ :

$$i_D(t) = i_C(t) + i_L(t) = C \frac{dv_o}{dt} + \frac{v_o(t)}{R}$$

Since  $v_o(t) = v_s(t) - V_D$  during conduction, its slope matches the input slope:

$$i_C(t) = C \frac{dv_s}{dt} = 100 \mu\text{F} \times 40,000 \text{ V/s} = 4.0 \text{ A} \quad (\text{constant during conduction})$$

The average load current during this short conduction window is determined by the average output voltage during conduction, which matches  $V_{out,avg}$ :

$$i_{L,avg} = \frac{V_{out,avg}}{R} = \frac{8.8455 \text{ V}}{100 \Omega} = 0.08846 \text{ A} = 88.46 \text{ mA}$$

Thus, the total average current flowing through the diode \*during the conduction interval\* is:

$$\begin{aligned} i_{D,avg} &= i_C + i_{L,avg} \\ &= 4.0 \text{ A} + 0.08846 \text{ A} = \mathbf{4.0885 \text{ A}} \end{aligned}$$

**(iv) Maximum Diode Current ( $i_{D,max}$ )** The maximum diode current occurs exactly at the peak of the input ( $t = 0$ ), where the output voltage reaches its maximum value  $V_{omax} = 9.3 \text{ V}$ :

$$\begin{aligned} i_{D,max} &= i_C + i_{L,max} \\ &= C \frac{dv_o}{dt} + \frac{V_{omax}}{R} \\ &= 4.0 \text{ A} + \frac{9.3 \text{ V}}{100 \Omega} = 4.0 \text{ A} + 0.093 \text{ A} = \mathbf{4.093 \text{ A}} \end{aligned}$$

#### Final Answer Summary

| Part  | Parameter                                   | Formula / Relation                          | Value                                 |
|-------|---------------------------------------------|---------------------------------------------|---------------------------------------|
| (i)   | Average DC Output Voltage ( $V_{out,avg}$ ) | $V_{omax} - \frac{V_r}{2}$                  | <b>8.85 V</b>                         |
| (ii)  | Diode Conduction Interval ( $\Delta t$ )    | $\frac{V_r}{(dv_s/dt)}$                     | <b>22.72 <math>\mu\text{s}</math></b> |
| (iii) | Average Diode Current during Conduction     | $C \frac{dv_s}{dt} + \frac{V_{out,avg}}{R}$ | <b>4.09 A</b>                         |
| (iv)  | Maximum Diode Current ( $i_{D,max}$ )       | $C \frac{dv_s}{dt} + \frac{V_{omax}}{R}$    | <b>4.093 A</b>                        |

**Q4.** A peak rectifier circuit is operated with a  $1 \text{ k}\Omega$  load and a diode that can be modeled to have a  $0.7 \text{ V}$  drop at any current. It is fed by a  $50 \text{ Hz}$  sinusoid and the capacitor is chosen to provide a peak-to-peak ripple voltage of  $10\%$  of the peak output voltage. Consider the load current  $I_L = 15 \text{ mA}$ .

- What is the value of the capacitor?
- What fraction of the cycle does the diode conduct?
- What is the average diode current?
- What is the peak diode current?

**(16 marks)**

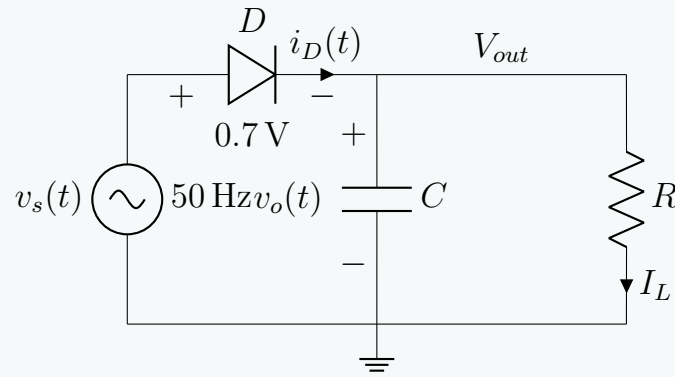
[EEE 263 | L-2/T-1 | 2014–15]

#### Answer

##### Assumptions & Initial Analysis:

We are analyzing a half-wave peak rectifier using the constant voltage drop (CVD) model for the diode ( $V_D = 0.7 \text{ V}$ ). Because the ripple voltage is small ( $10\%$  of the peak output voltage), the load current  $I_L$  is approximately constant during the discharge phase. Let  $V_{omax}$  be the peak output voltage across the load. Given  $I_L = 15 \text{ mA}$  and  $R = 1 \text{ k}\Omega$ , we can determine the peak output voltage:

$$V_{omax} = I_L R = 15 \text{ mA} \times 1 \text{ k}\Omega = 15 \text{ V}$$

**Circuit Diagram: Peak Rectifier**

**Given Parameters**

- Input frequency,  $f = 50 \text{ Hz}$
- Load resistance,  $R = 1 \text{ k}\Omega$
- Load current,  $I_L = 15 \text{ mA}$
- Peak output voltage,  $V_{omax} = 15 \text{ V}$
- Peak-to-peak ripple voltage,  $V_r = 10\% \text{ of } V_{omax} = 0.1 \times 15 \text{ V} = 1.5 \text{ V}$
- Diode forward drop,  $V_D = 0.7 \text{ V}$

**1. Value of the Filter Capacitor ( $C$ )**

For a small ripple, the capacitor discharges linearly through the load resistor  $R$  over approximately the entire period  $T \approx \frac{1}{f}$ . The peak-to-peak ripple voltage  $V_r$  is related to the load current by:

$$V_r \approx \frac{V_{omax}}{fRC} = \frac{I_L}{fC}$$

Rearranging to solve for the capacitance  $C$ :

$$\begin{aligned} C &= \frac{I_L}{fV_r} \\ &= \frac{15 \times 10^{-3} \text{ A}}{50 \text{ Hz} \times 1.5 \text{ V}} \\ &= \frac{15 \times 10^{-3}}{75} = 0.2 \times 10^{-3} \text{ F} = \mathbf{200 \mu\text{F}} \end{aligned}$$

**2. Fraction of the Cycle the Diode Conducts**

The diode conducts only during a brief interval  $\Delta t$  to replenish the charge on the capacitor. The conduction angle  $\theta_c = \omega\Delta t$  is approximated by:

$$\omega\Delta t \approx \sqrt{\frac{2V_r}{V_{omax}}}$$

Substituting the values:

$$\omega\Delta t = \sqrt{\frac{2 \times 1.5}{15}} = \sqrt{\frac{3}{15}} = \sqrt{0.2} \approx \mathbf{0.4472 \text{ rad}}$$

The fraction of the cycle is the conduction angle divided by the full period ( $2\pi$ ):

$$\text{Fraction} = \frac{\omega\Delta t}{2\pi} = \frac{0.4472}{2\pi} \approx \mathbf{0.0712} \text{ (or } \mathbf{7.12\%})$$

**3. Average Diode Current**

The average diode current during the conduction interval is required to replace the charge lost to the load during the entire cycle. It is given by:

$$\begin{aligned} i_{D,avg} &= I_L \left( 1 + \pi \sqrt{\frac{2V_{omax}}{V_r}} \right) \\ &= 15 \text{ mA} \left( 1 + \pi \sqrt{\frac{2 \times 15}{1.5}} \right) \\ &= 15 \text{ mA} (1 + \pi\sqrt{20}) \\ &\approx 15 \text{ mA} \times (1 + 14.0496) = \mathbf{225.74 \text{ mA}} \end{aligned}$$

### 4. Peak Diode Current

The maximum diode current occurs at the very beginning of the conduction interval when the charging rate is highest. It is approximated as:

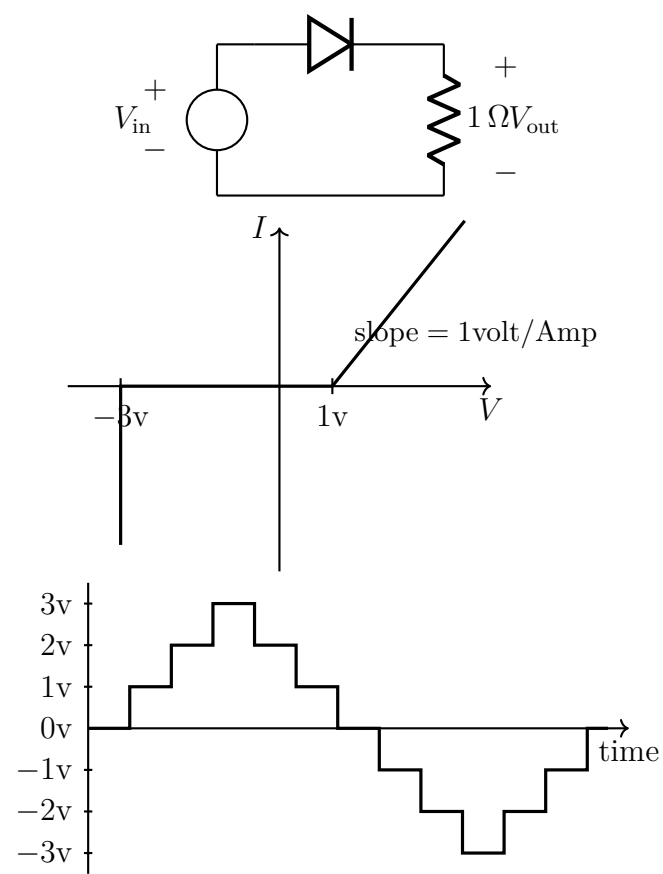
$$\begin{aligned} i_{D,max} &= I_L \left( 1 + 2\pi \sqrt{\frac{2V_{omax}}{V_r}} \right) \\ &= 15 \text{ mA} \left( 1 + 2\pi \sqrt{\frac{2 \times 15}{1.5}} \right) \\ &= 15 \text{ mA} \left( 1 + 2\pi \sqrt{20} \right) \\ &\approx 15 \text{ mA} \times (1 + 28.099) = \mathbf{436.49 \text{ mA}} \end{aligned}$$

#### Final Answer Summary

| Parameter                                | Calculated Value    |
|------------------------------------------|---------------------|
| 1. Filter Capacitance ( $C$ )            | $200 \mu\text{F}$   |
| 2. Conduction Fraction                   | $7.12\%$            |
| 3. Average Diode Current ( $i_{D,avg}$ ) | $225.74 \text{ mA}$ |
| 4. Peak Diode Current ( $i_{D,max}$ )    | $436.49 \text{ mA}$ |

I-V Characteristics & Waveform Analysis

**Q1.** (a) What would be the output voltage if the diode and the resistor were connected in parallel? **(10.67 marks)** [EEE 263 | L-2/T-1 | 2013–14]

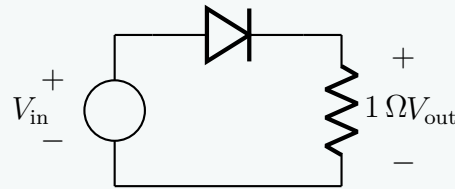


- (b) The I-V characteristics of a diode and a sinusoidal input voltage signal  $V_{in}$  are given in the figure. Draw the output voltage  $V_{out}$  across the resistor. **(16 marks)**
- (c) What would be the output voltage  $V_{out}$  across the resistor if the diode had a constant 0.7 V voltage drop? **(10 marks)**

Answer

**Preliminary Note on the Input Signal:** Although the problem text mentions a “sinusoidal input voltage signal”, the provided figure explicitly illustrates a staircase waveform. In engineering analysis, when there is a discrepancy between a descriptive text and a provided graphical signal, the explicit graphical waveform is used. The solution below is derived using the staircase waveform.

### Reference Circuit Diagram



### Part (a): Diode and Resistor in Parallel

If the diode and the  $1\Omega$  resistor are connected in parallel, and this combination is directly fed by the input voltage source  $V_{in}$  (assuming an ideal voltage source with zero internal resistance), the output voltage  $V_{out}$  across the parallel combination is completely dictated by the source.

By the definition of parallel components connected to an ideal voltage source:

$$V_{out} = V_{in}$$

The waveform for  $V_{out}$  will be exactly identical to the input staircase waveform. (Note: The diode would draw current according to its I-V characteristics, but it would not alter the node voltage unless a series source resistance is present, which is not shown).

### Part (b): $V_{out}$ using the given I-V Characteristics

#### 1. Device Modeling from I-V Curve:

Based on the provided  $V - I$  graph, we can extract the piecewise linear model of the diode:

- **Forward Conduction Region:** For  $V_D \geq 1\text{ V}$ , the curve has a slope of  $1\text{ A/V}$ . Thus, the dynamic forward resistance is  $r_d = \frac{1}{\text{slope}} = 1\Omega$ . The diode behaves as a  $1\text{ V}$  DC source in series with a  $1\Omega$  resistor.
- **Cutoff Region:** For  $-3\text{ V} < V_D < 1\text{ V}$ , the current is  $I_D = 0\text{ A}$ .
- **Breakdown Region:** The diode drops vertically at  $V_D = -3\text{ V}$ , indicating a Zener breakdown voltage of  $3\text{ V}$ .

#### 2. Circuit Analysis:

Applying Kirchhoff's Voltage Law (KVL) to the series circuit:

$$V_{in} = V_D + V_{out} \quad \text{where} \quad V_{out} = I_D \cdot R = I_D(1\Omega)$$

Case 1: Diode is ON ( $V_{in} > 1\text{ V}$ )

Substituting the forward region model  $V_D = 1 + I_D(1)$ :

$$\begin{aligned} V_{in} &= (1 + I_D \cdot 1) + I_D \cdot 1 \\ V_{in} &= 1 + 2I_D \\ I_D &= \frac{V_{in} - 1}{2} \end{aligned}$$

Since  $V_{out} = I_D \cdot 1\Omega$ , we get:

$$V_{out} = \frac{V_{in} - 1}{2} \quad (\text{for } V_{in} > 1\text{ V})$$

Case 2: Diode is OFF ( $-3\text{ V} \leq V_{in} \leq 1\text{ V}$ )

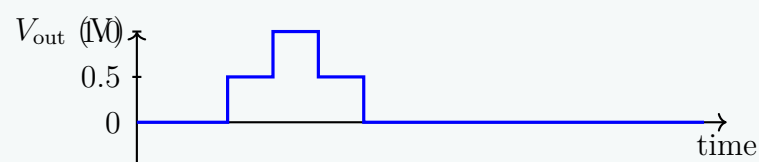
$I_D = 0\text{ A}$ , therefore  $V_{out} = 0\text{ V}$ . (Since  $V_{in}$  minimum is  $-3\text{ V}$ , it never strictly exceeds the breakdown threshold to cause reverse current).

#### 3. Waveform Calculation:

Mapping the discrete levels of the input staircase:

- $V_{in} = 0\text{ V to } 1\text{ V} \implies V_{out} = 0\text{ V}$
- $V_{in} = 2\text{ V} \implies V_{out} = \frac{2-1}{2} = 0.5\text{ V}$
- $V_{in} = 3\text{ V} \implies V_{out} = \frac{3-1}{2} = 1.0\text{ V}$
- $V_{in} \leq 0\text{ V} \implies V_{out} = 0\text{ V}$

### Output Waveform for Part (b):



**Part (c):  $V_{out}$  with Constant Voltage Drop (CVD) Model**
**1. Device Modeling:**

Assume the diode has a Constant Voltage Drop (CVD) of 0.7 V when forward biased ( $r_d = 0 \Omega$ ).

- When ON ( $V_{in} > 0.7 \text{ V}$ ):  $V_D = 0.7 \text{ V}$ .
- When OFF ( $V_{in} < 0.7 \text{ V}$ ):  $I_D = 0 \text{ A}$ .

**2. Circuit Analysis:**

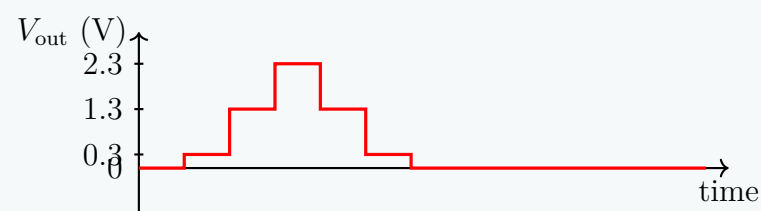
Applying KVL:

$$V_{out} = \begin{cases} V_{in} - 0.7 \text{ V}, & V_{in} > 0.7 \text{ V} \\ 0 \text{ V}, & V_{in} \leq 0.7 \text{ V} \end{cases}$$

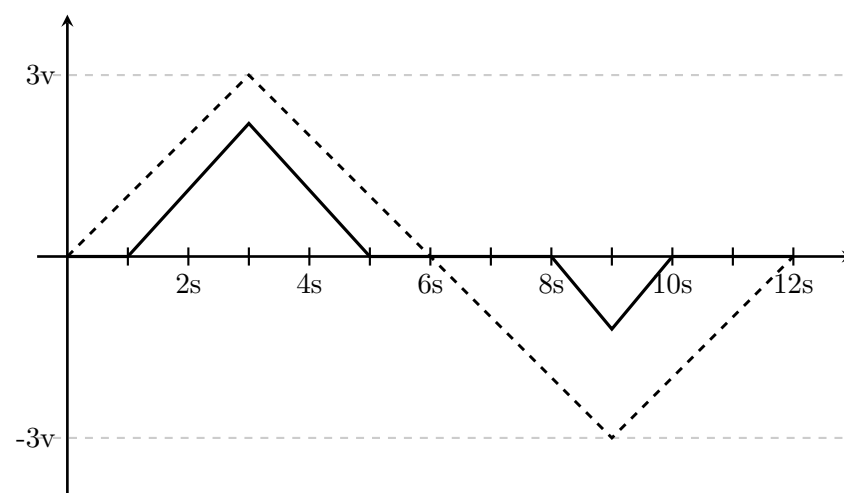
**3. Waveform Calculation:**

Mapping the discrete levels of the input staircase:

- $V_{in} = 0 \text{ V} \Rightarrow V_{out} = 0 \text{ V}$
- $V_{in} = 1 \text{ V} \Rightarrow V_{out} = 1 - 0.7 = 0.3 \text{ V}$
- $V_{in} = 2 \text{ V} \Rightarrow V_{out} = 2 - 0.7 = 1.3 \text{ V}$
- $V_{in} = 3 \text{ V} \Rightarrow V_{out} = 3 - 0.7 = 2.3 \text{ V}$
- $V_{in} \leq 0 \text{ V} \Rightarrow V_{out} = 0 \text{ V}$

**Output Waveform for Part (c):**


- Q2.** (a) An AC voltage source is applied to a series combination of a diode and a resistance. The dotted line in the figure is the input voltage waveform ( $V_{ac}$ ) and the solid line is the output voltage waveform ( $V_R$ ) across the resistance. Copy  $V_R$  from the figure to your answer script exactly to scale and draw the voltage waveform across the diode over the same graph. Given that  $\frac{dV_{ac}}{dt} = 1 \text{ V/sec}$  when  $V_{ac}$  is rising and  $\frac{dV_{ac}}{dt} = -1 \text{ V/sec}$  when  $V_{ac}$  is falling.



(20 marks)

[EEE 263 | L-2/T-1 | 2012-13]

- (b) Draw the I-V characteristics curve of the above diode exactly to scale (mention the voltage(s) at which the diode begins conduction). (16 marks) [EEE 263 | L-2/T-1 | 2012-13]
- (c) Draw the voltage waveform across the diode if it had been connected in parallel to the resistance. (10 marks) [EEE 263 | L-2/T-1 | 2012-13]

**Answer**
**Part (a): Graphical Derivation and Diode Voltage Waveform ( $V_D$ )**

By Kirchhoff's Voltage Law (KVL) applied to the series circuit loop:

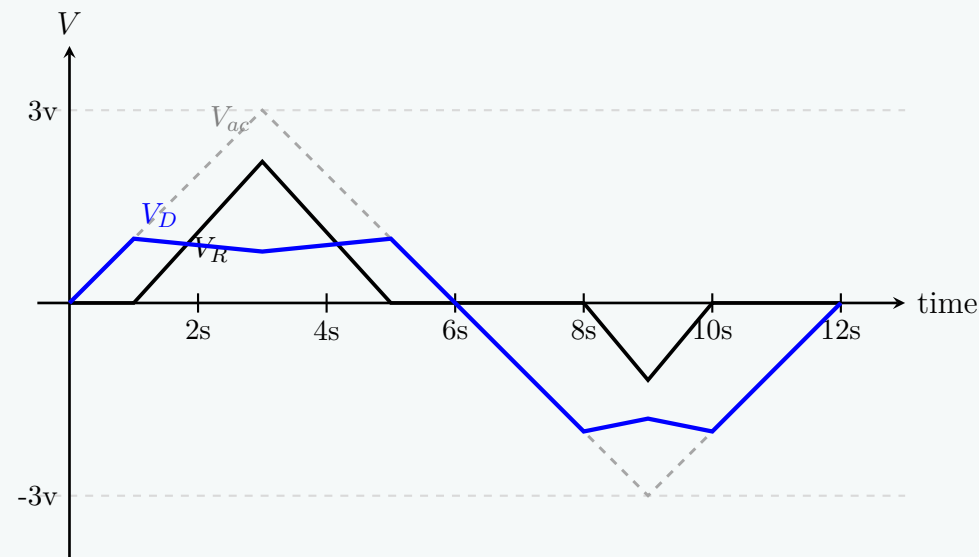
$$V_{ac}(t) = V_D(t) + V_R(t) \Rightarrow V_D(t) = V_{ac}(t) - V_R(t)$$

The diode voltage at any instant is found by subtracting the coordinates of the resistor voltage ( $V_R$ ) from the input AC voltage ( $V_{ac}$ ). Computing the key boundary transitions:

- **From 0 s to 1 s:**  $V_{ac}$  rises from 0 V to 1 V while  $V_R = 0 \text{ V} \Rightarrow V_D = V_{ac}$ . At  $t = 1 \text{ s}$ ,  $V_D = 1 \text{ V}$ .
- **At  $t = 3 \text{ s}$  (Positive Peak):**  $V_{ac} = 3 \text{ V}$ ,  $V_R = 2.2 \text{ V} \Rightarrow V_D = 3 - 2.2 = 0.8 \text{ V}$ .
- **At  $t = 5 \text{ s}$ :**  $V_{ac} = 1 \text{ V}$ ,  $V_R = 0 \text{ V} \Rightarrow V_D = 1 \text{ V}$ .

- At  $t = 6\text{ s}$ :  $V_{ac} = 0\text{ V}$ ,  $V_R = 0\text{ V} \Rightarrow V_D = 0\text{ V}$ .
- At  $t = 8\text{ s}$ :  $V_{ac} = -2\text{ V}$ ,  $V_R = 0\text{ V} \Rightarrow V_D = -2\text{ V}$ .
- At  $t = 9\text{ s}$  (**Negative Peak**):  $V_{ac} = -3\text{ V}$ ,  $V_R = -1.2\text{ V} \Rightarrow V_D = -3 - (-1.2) = -1.8\text{ V}$ .
- At  $t = 10\text{ s}$ :  $V_{ac} = -2\text{ V}$ ,  $V_R = 0\text{ V} \Rightarrow V_D = -2\text{ V}$ .
- At  $t = 12\text{ s}$ :  $V_{ac} = 0\text{ V}$ ,  $V_R = 0\text{ V} \Rightarrow V_D = 0\text{ V}$ .

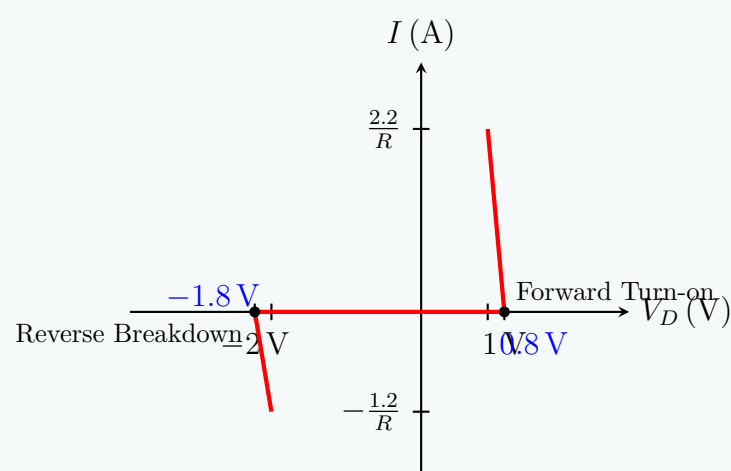
Below is the complete waveform graph containing  $V_{ac}$  (dashed gray),  $V_R$  (solid black), and the derived  $V_D$  (**solid blue line**):



### Part (b): I-V Characteristics Curve of the Diode

By tracking the relationship between the diode voltage  $V_D$  and the loop current  $I = \frac{V_R}{R}$ , we deduce the behavioral model of this unique device:

- Forward Conduction Threshold:** Forward conduction begins precisely at  $V_D = 1\text{ V}$  (corresponding to  $t = 1\text{ s}$  where  $V_R$  begins rising from  $0\text{ V}$ ). As current increases up to a peak value of  $I_{max} = \frac{2.2}{R}$ , the voltage across the diode drops slightly down to  $0.8\text{ V}$ , revealing a negative dynamic differential resistance ( $r_d < 0$ ).
- Reverse Conduction (Breakdown) Threshold:** Reverse conduction begins at  $V_D = -2\text{ V}$  (corresponding to  $t = 8\text{ s}$ ). As reverse current increases in magnitude to a peak value of  $I = -\frac{1.2}{R}$ , the voltage shifts to  $-1.8\text{ V}$ .
- Non-Conducting (Off) State:** For  $-2\text{ V} \leq V_D \leq 1\text{ V}$ , the current through the device is  $I = 0\text{ A}$ .



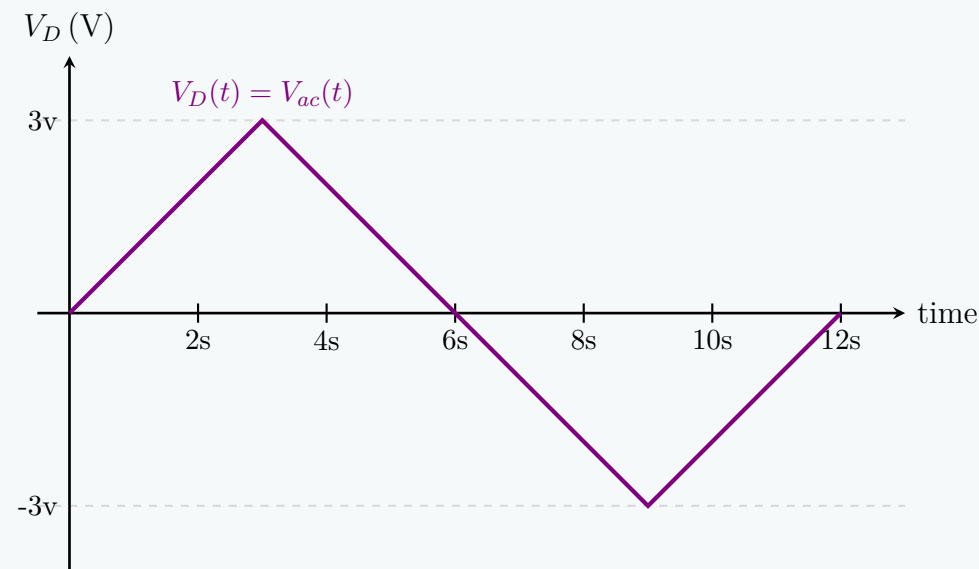
### Part (c): Diode Connected in Parallel with the Resistance

When the diode is connected in parallel with the resistance across an ideal independent AC voltage source, the voltage at the output nodes is entirely governed by the source itself.

By the fundamental definition of parallel connections across an ideal source:

$$V_D(t) = V_R(t) = V_{ac}(t)$$

Consequently, the diode does not split the source voltage with another series element. The voltage waveform across it will perfectly duplicate the input triangular waveform  $V_{ac}$ :



### Diode Thermal Characteristics

**Q1.** When a 15 A current is applied to a particular diode, the junction voltage immediately becomes 0.7 V. As power dissipation raises its temperature, the voltage eventually reaches 0.6 V. (i) What is the apparent rise in junction temperature? (ii) What is the temperature rise per watt of power dissipation? (iii) What is the power dissipated in the diode in its final state? **(15 marks)** [EEE 263 | L-2/T-1 | 2024–25]

### Answer

#### Theory and Assumptions

For a standard silicon pn-junction diode, the forward voltage drop  $V_D$  exhibits a negative temperature coefficient when operated at a constant forward current. As the temperature of the junction increases, the forward voltage required to maintain the same current decreases linearly.

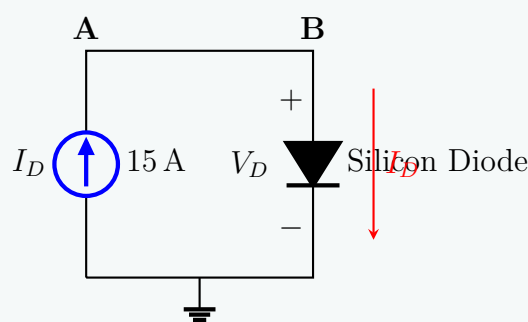
A widely accepted standard approximation in analog circuit design for the temperature coefficient of a silicon diode is:

$$k = \frac{\Delta V_D}{\Delta T} \approx -2.0 \text{ mV}/^\circ\text{C} \quad (1)$$

This means that for every  $1^\circ\text{C}$  rise in junction temperature, the diode voltage drops by approximately 2 mV. We will use this standard assumption to determine the temperature rise.

#### Test Circuit Equivalent Diagram

The following schematic models the testing condition described in the problem, where an ideal constant current source drives the diode, and the resulting voltage is measured.



#### Given Data

- Constant forward current,  $I_D = 15 \text{ A}$
- Initial junction voltage,  $V_{D1} = 0.7 \text{ V}$
- Final steady-state junction voltage,  $V_{D2} = 0.6 \text{ V}$

#### Step-by-Step Solution

##### (i) Apparent Rise in Junction Temperature

First, we determine the total change in the junction voltage:

$$\begin{aligned} \Delta V_D &= V_{D2} - V_{D1} \\ &= 0.6 \text{ V} - 0.7 \text{ V} \\ &= -0.1 \text{ V} \\ &= -100 \text{ mV} \end{aligned}$$

Using the standard temperature coefficient  $k = -2.0\text{ mV}/^\circ\text{C}$ , the apparent rise in junction temperature ( $\Delta T$ ) can be calculated as:

$$\begin{aligned}\Delta T &= \frac{\Delta V_D}{k} \\ &= \frac{-100\text{ mV}}{-2.0\text{ mV}/^\circ\text{C}} \\ &= 50^\circ\text{C}\end{aligned}\tag{2}$$

The junction temperature of the diode rises by  $50^\circ\text{C}$  relative to the ambient or initial state.

**(iii) Power Dissipated in the Diode in its Final State**

*Note:* We compute the final power first, as it represents the steady-state condition necessary to evaluate the thermal resistance required in part (ii).

The power dissipated by the diode at any given moment is the product of the forward voltage and the constant forward current. In the final state (thermal equilibrium), the voltage has settled at  $V_{D2} = 0.6\text{ V}$ .

$$\begin{aligned}P_{\text{final}} &= V_{D2} \times I_D \\ &= 0.6\text{ V} \times 15\text{ A} \\ &= 9.0\text{ W}\end{aligned}\tag{3}$$

**(ii) Temperature Rise per Watt of Power Dissipation**

The temperature rise per watt of power dissipation characterizes the thermal resistance ( $\theta_{JA}$  or thermal resistance from junction to ambient) of the diode assembly. Since the diode temperature reaches a stable plateau, the relevant power causing this steady-state temperature rise is the final power dissipation ( $P_{\text{final}}$ ).

$$\begin{aligned}\theta &= \frac{\Delta T}{P_{\text{final}}} \\ &= \frac{50^\circ\text{C}}{9.0\text{ W}} \\ &\approx 5.556^\circ\text{C/W}\end{aligned}\tag{4}$$

| Final Answers Summary                                             |                        |
|-------------------------------------------------------------------|------------------------|
| (i) Apparent Rise in Junction Temperature ( $\Delta T$ )          | $50^\circ\text{C}$     |
| (ii) Temperature Rise per Watt ( $\theta$ )                       | $5.56^\circ\text{C/W}$ |
| (iii) Final Steady-State Power Dissipation ( $P_{\text{final}}$ ) | $9.0\text{ W}$         |

- Q2.** When a  $15\text{ A}$  current is applied to a particular diode, the junction voltage immediately becomes  $0.7\text{ V}$ . As power dissipation raises its temperature, the voltage eventually reaches  $0.58\text{ V}$ .
- (i) What is the apparent rise in junction temperature?
  - (ii) What is the temperature rise per watt of power dissipation?
  - (iii) What is the power dissipated in the diode in its final state?
- (10 marks)**

[EEE 263 | L-2/T-1 | 2023–24]

Answer

**Theory and Assumptions**

For standard silicon pn-junction diodes, the forward voltage drop  $V_D$  decreases linearly as the junction temperature increases, provided the forward current remains constant.

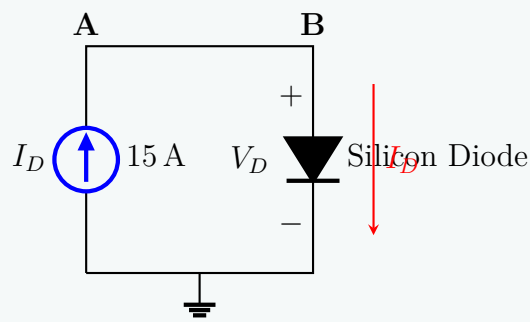
A standard textbook approximation for the temperature coefficient of a silicon diode’s forward voltage is:

$$k = \frac{\Delta V_D}{\Delta T} \approx -2.0\text{ mV}/^\circ\text{C}\tag{1}$$

This indicates that the junction voltage decreases by approximately  $2\text{ mV}$  for every  $1^\circ\text{C}$  rise in the diode’s junction temperature.

**Test Circuit Equivalent Diagram**

The circuit diagram below models the test condition. An ideal constant current source drives the diode, and the resulting voltage is measured across its terminals.



Given Data

- Constant forward current,  $I_D = 15\text{ A}$
- Initial junction voltage,  $V_{D1} = 0.7\text{ V}$
- Final steady-state junction voltage,  $V_{D2} = 0.58\text{ V}$

Step-by-Step Solution

(i) Apparent Rise in Junction Temperature

First, we calculate the total change in the diode’s forward voltage:

$$\begin{aligned}\Delta V_D &= V_{D2} - V_{D1} \\ &= 0.58\text{ V} - 0.70\text{ V} \\ &= -0.12\text{ V} \\ &= -120\text{ mV}\end{aligned}$$

Using the standard temperature coefficient  $k = -2.0\text{ mV}/^\circ\text{C}$ , the apparent rise in junction temperature ( $\Delta T$ ) is:

$$\begin{aligned}\Delta T &= \frac{\Delta V_D}{k} \\ &= \frac{-120\text{ mV}}{-2.0\text{ mV}/^\circ\text{C}} \\ &= 60^\circ\text{C}\end{aligned}\tag{2}$$

The junction temperature of the diode rises by  $60^\circ\text{C}$  relative to its initial state.

(iii) Power Dissipated in the Diode in its Final State

*Note: Calculating the final power state first provides the steady-state thermal conditions needed to evaluate the thermal resistance required in part (ii).*

The power dissipated by the diode at steady-state is the product of the final forward voltage and the constant forward current.

$$\begin{aligned}P_{\text{final}} &= V_{D2} \times I_D \\ &= 0.58\text{ V} \times 15\text{ A} \\ &= 8.7\text{ W}\end{aligned}\tag{3}$$

(ii) Temperature Rise per Watt of Power Dissipation

The temperature rise per watt of power dissipation represents the effective thermal resistance ( $\theta_{JA}$ ) of the system. Since the diode has reached a stable equilibrium, we use the final steady-state power dissipation ( $P_{\text{final}}$ ).

$$\begin{aligned}\theta &= \frac{\Delta T}{P_{\text{final}}} \\ &= \frac{60^\circ\text{C}}{8.7\text{ W}} \\ &\approx 6.897^\circ\text{C/W}\end{aligned}\tag{4}$$

| Final Answers Summary                                             |                        |
|-------------------------------------------------------------------|------------------------|
| (i) Apparent Rise in Junction Temperature ( $\Delta T$ )          | $60^\circ\text{C}$     |
| (ii) Temperature Rise per Watt ( $\theta$ )                       | $6.90^\circ\text{C/W}$ |
| (iii) Final Steady-State Power Dissipation ( $P_{\text{final}}$ ) | $8.7\text{ W}$         |

- Q1.** Consider a bridge rectifier circuit with a filter capacitor  $C$  placed across the load resistor  $R = 100\ \Omega$ . The transformer secondary delivers a sinusoid of  $12\text{ V}$  (rms) at  $60\text{ Hz}$ . Assume  $V_D = 0.8\text{ V}$ .
- Find the value of  $C$  such that the ripple voltage is no larger than  $1\text{ V}$  peak-to-peak.
  - What is the DC voltage at the output?
  - Find the load current.
  - Find the average and peak diode currents.
  - What is the peak reverse voltage across each diode?
- (15 marks)

[EEE 263 | L-2/T-1 | 2023–24]

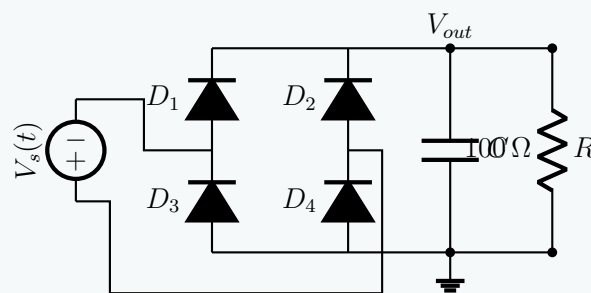
### Answer

#### Theory and Circuit Operation

A bridge rectifier utilizes four diodes arranged in a bridge configuration to convert an alternating current (AC) input into a pulsating direct current (DC) output. By adding a filter capacitor  $C$  in parallel with the load resistor  $R$ , the output voltage is smoothed out into a DC voltage with a small ripple component.

- Positive Half-Cycle:** When the secondary voltage is positive, diodes  $D_1$  and  $D_4$  are forward-biased (ON), while  $D_2$  and  $D_3$  are reverse-biased (OFF). Current flows from the transformer through  $D_1$ , through the load network ( $R \parallel C$ ), and returns through  $D_4$ .
- Negative Half-Cycle:** When the secondary voltage reverses, diodes  $D_2$  and  $D_3$  become forward-biased (ON), while  $D_1$  and  $D_4$  turn OFF. Current flows through  $D_2$ , through the load network in the same direction as before, and returns through  $D_3$ .
- Filter Operation:** The capacitor charges to the peak value of the rectified voltage ( $V_p$ ) during the conduction intervals. When the input voltage falls below the capacitor voltage, the diodes turn off, and the capacitor discharges through the load resistor  $R$  with a time constant  $\tau = RC$ . Because this occurs twice per line cycle, the ripple frequency is  $f_r = 2f$ .

#### Publication-Quality Circuit Schematic



#### Mathematical Derivations

The peak value of the secondary voltage ( $V_{sm}$ ) is given by:

$$V_{sm} = V_{s,\text{rms}} \times \sqrt{2} \quad (1)$$

Since two diodes conduct simultaneously during each half-cycle in a bridge rectifier, the peak rectified output voltage ( $V_p$ ) across the capacitor is reduced by two diode drops:

$$V_p = V_{sm} - 2V_D \quad (2)$$

For a full-wave rectifier with a heavy filter capacitor ( $RC \gg T$ ), the peak-to-peak ripple voltage ( $V_r$ ) can be approximated using the linear discharge model:

$$V_r = \frac{V_p}{2fRC} = \frac{I_L}{2fC} \quad (3)$$

where  $f$  is the input line frequency ( $60\text{ Hz}$ ).

#### Step-by-Step Solution

Given Parameters:

$$V_{s,\text{rms}} = 12\text{ V}, \quad f = 60\text{ Hz}, \quad R = 100\ \Omega, \quad V_D = 0.8\text{ V}, \quad V_r \leq 1\text{ V}$$

First, compute the input peak voltage:

$$V_{sm} = 12 \times \sqrt{2} \approx 16.971\text{ V} \quad (4)$$

Next, compute the peak output voltage:

$$V_p = 16.971 - 2(0.8) = 15.371\text{ V} \quad (5)$$

**(i) Find the value of  $C$  for  $V_r \leq 1\text{ V}$**

Rearranging the ripple voltage formula to solve for  $C$ :

$$\begin{aligned} C &= \frac{V_p}{2fRV_r} \\ &= \frac{15.371}{2 \times 60 \times 100 \times 1} \\ &= \frac{15.371}{12000} \approx 1.2809 \times 10^{-3}\text{ F} = 1281\ \mu\text{F} \end{aligned} \quad (6)$$

**(ii) Determine the DC output voltage**

The average or DC voltage ( $V_{DC}$ ) at the output is the peak value minus half of the peak-to-peak ripple voltage:

$$\begin{aligned} V_{DC} &= V_p - \frac{1}{2}V_r \\ &= 15.371 - \frac{1}{2}(1) = 14.871 \text{ V} \end{aligned} \quad (7)$$

**(iii) Find the load current**

The average load current ( $I_L$ ) is calculated via Ohm's law using the DC output voltage:

$$\begin{aligned} I_L &= \frac{V_{DC}}{R} \\ &= \frac{14.871}{100} = 0.1487 \text{ A} = 148.71 \text{ mA} \end{aligned} \quad (8)$$

**(iv) Find the average and peak diode currents**

Using the standard textbook equations for diode currents in full-wave capacitor-filter circuits:

**(a) Peak Diode Current ( $I_{D,\text{peak}}$ ):**

$$\begin{aligned} I_{D,\text{peak}} &= I_L \left( 1 + 2\pi\sqrt{\frac{V_p}{2V_r}} \right) \\ &= 0.14871 \times \left( 1 + 2\pi\sqrt{\frac{15.371}{2 \times 1}} \right) \\ &= 0.14871 \times \left( 1 + 2\pi\sqrt{7.6855} \right) \\ &= 0.14871 \times (1 + 2\pi \times 2.7723) \\ &= 0.14871 \times (1 + 17.419) = 0.14871 \times 18.419 \approx 2.739 \text{ A} \end{aligned} \quad (9)$$

**(b) Average Diode Current ( $I_{D,\text{avg}}$ ):**

- *Over a full cycle:* Each diode conducts for only one half-cycle, sharing the total load current equally over time.

$$I_{D,\text{avg}} = \frac{I_L}{2} = \frac{148.71 \text{ mA}}{2} \approx 74.36 \text{ mA} \quad (10)$$

- *During conduction interval ( $\Delta t$ ):* If required by interpretation, the average current during the brief interval when the diode conducts is:

$$\begin{aligned} I_{D,\text{avg, cond}} &= I_L \left( 1 + \pi\sqrt{\frac{V_p}{2V_r}} \right) \\ &= 0.14871 \times (1 + \pi \times 2.7723) \approx 1.444 \text{ A} \end{aligned} \quad (11)$$

**(v) Find the peak reverse voltage (PIV) across each diode**

In a bridge rectifier configuration, when a diode is reverse-biased, the maximum reverse voltage across it equals the peak transformer secondary voltage minus one diode drop:

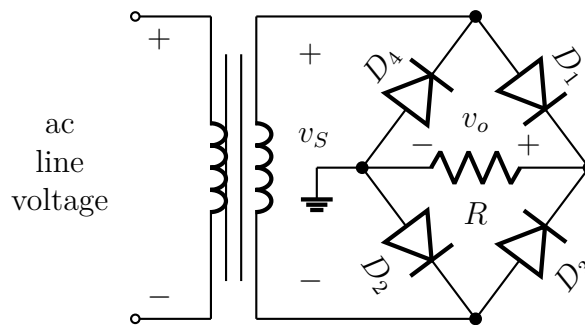
$$\begin{aligned} \text{PIV} &= V_{sm} - V_D \\ &= 16.971 - 0.8 = 16.171 \text{ V} \end{aligned} \quad (12)$$

| Final Answers Summary                             |                    |
|---------------------------------------------------|--------------------|
| (i) Required Filter Capacitance ( $C$ )           | 1281 $\mu\text{F}$ |
| (ii) DC Output Voltage ( $V_{DC}$ )               | 14.87 V            |
| (iii) Average Load Current ( $I_L$ )              | 148.71 mA          |
| (iv) Peak Diode Current ( $I_{D,\text{peak}}$ )   | 2.74 A             |
| - Average Diode Current (Full Cycle / Conduction) | 74.36 mA / 1.44 A  |
| (v) Peak Inverse Voltage (PIV)                    | 16.17 V            |

**Q2.** A full-wave bridge rectifier circuit with 1 k $\Omega$  load operates from a 120 V (rms), 60 Hz household supply through a 10:1 step-down transformer. Each diode has a voltage drop of 0.7 V at any current. (i) What is the peak value of the rectified voltage across the load? (ii) For what fraction of a cycle does each diode conduct? (iii) What are the average voltage across and average current through the load? **(10 marks)** [EEE 263 | L-2/T-1 | 2023–24]

**Q3.** A full-wave bridge rectifier circuit with a  $1\text{-k}\Omega$  load operates from a  $120\text{-V}$  (rms),  $60\text{-Hz}$  household supply through a  $10\text{-to-1}$  step-down transformer having a single secondary winding. Each diode can be modeled to have a  $0.7\text{-V}$  drop in forward bias. Find: the peak value of the rectified voltage across the load; the fraction of a cycle for which each diode conducts; the average voltage and average current through the load. **(16 marks)** [EEE 263 | L-2/T-1 | 2022–23]

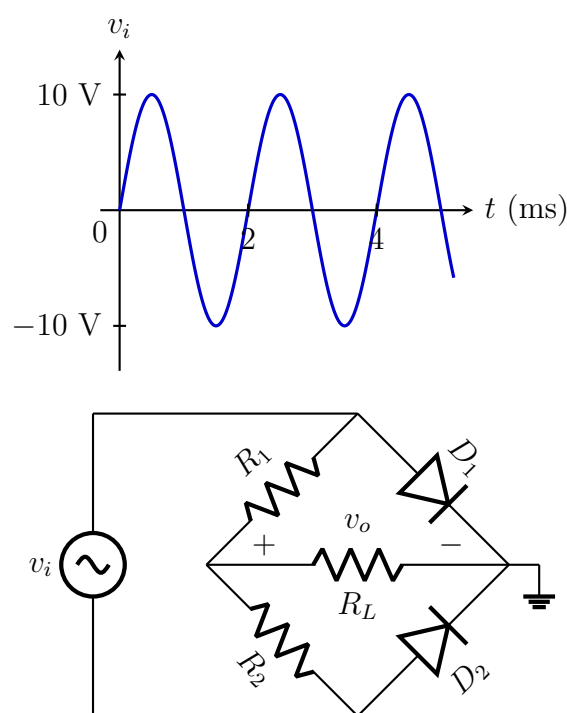
**Q4.** For the bridge rectifier circuit shown,  $v_s$  is a  $12\text{ V}$  (rms) sinusoid,  $V_D = 0.7\text{ V}$ , and  $R = 100\ \Omega$ . Find the average of the output voltage  $v_o$ , the peak diode current, and the PIV.



**(15 marks)**

[EEE 263 | L-2/T-1 | 2021–22]

**Q5.** Draw the transfer curve of the circuit shown, assuming all diodes are ideal,  $R_1 = R_2 = 20\text{ k}\Omega$ , and  $R_L = 10\text{ k}\Omega$ . (i) What is the maximum peak voltage of the output signal? (ii) What is the peak-inverse voltage? (iii) What is the average current through the load?



**(23 marks)**

[EEE 263 | L-2/T-1 | 2020–21]

**Q6.** Draw the circuit of a full-wave bridge rectifier and show the output wave shape for a sinusoidal input. **(16 marks)** [EEE 263 | L-2/T-1 | 2018–19]

**Q7.** For a full-wave bridge rectifier, find the value of peak current through the diode and PIV for the diode. Given:  $V_s = 12\text{ V}$  (rms),  $V_{DO} = 0.7\text{ V}$ ,  $R = 100\ \Omega$ . **(10 marks)** [EEE 263 | L-2/T-1 | 2018–19]

**Q8.** Comment on the value of RC time constant for a rectifier circuit. **(10 marks)** [EEE 263 | L-2/T-1 | 2018–19]

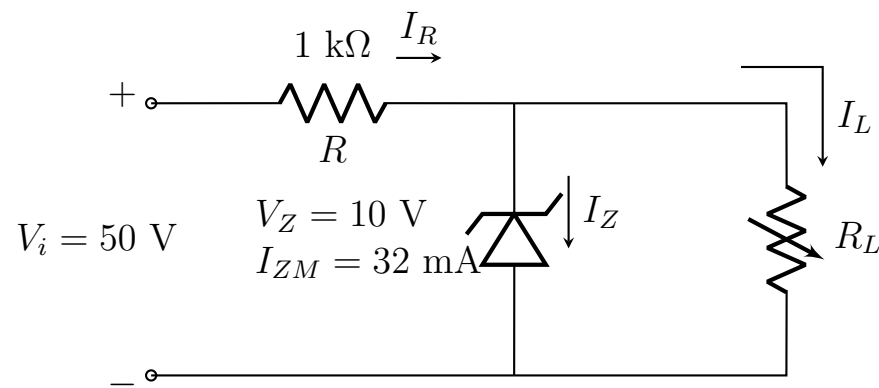
**Q9.** Show that the average output voltage of a bridge rectifier circuit is  $V_o = \frac{2}{\pi} V_s - 2V_{DO}$ . If the input voltage is a  $12\text{ V}$  (rms) sinusoid and the load resistance is  $100\ \Omega$ , calculate: (i) the average output voltage, (ii) the peak diode current, and (iii) the PIV of the diodes. Assume  $V_{DO} = 0.7\text{ V}$ . **(16 marks)** [EEE 263 | L-2/T-1 | 2011–12]

### Zener Diode Regulator

**Q1.** A  $6.8\text{ V}$  Zener diode exhibits its nominal voltage at a test current of  $5\text{ mA}$ . At this current the incremental resistance is  $r_z = 20\ \Omega$  and knee current  $I_{ZK} = 0.2\text{ mA}$ . The negative terminal is connected to  $V^+ = 10\text{ V}$  ( $\pm 1\text{ V}$ ) through a  $0.5\text{ k}\Omega$  resistor; positive terminal is grounded. (i) Find  $V_o$  with no load at nominal supply. (ii) Find the change in  $V_o$  as supply varies by  $\pm 1\text{ V}$ . (iii) Find the change in  $V_o$  when  $R_L = 2\text{ k}\Omega$  is connected; what happens if  $R_L = 0.5\text{ k}\Omega$ ? (iv) Find the minimum  $R_L$  for which the diode still operates in breakdown. **(20 marks)** [EEE 263 | L-2/T-1 | 2024–25]

**Q2.** A  $9.1\text{ V}$  Zener diode exhibits its nominal voltage at a test current of  $28\text{ mA}$ . At this current the incremental resistance is specified to be  $5\ \Omega$ . (i) Find  $V_{z0}$  in the Zener model. (ii) Find the Zener voltage at currents of  $10\text{ mA}$  and  $100\text{ mA}$ , respectively. (iii) Comment on the advantages/disadvantages of running the Zener diode at current values other than the nominal value. **(10 marks)** [EEE 263 | L-2/T-1 | 2023–24]

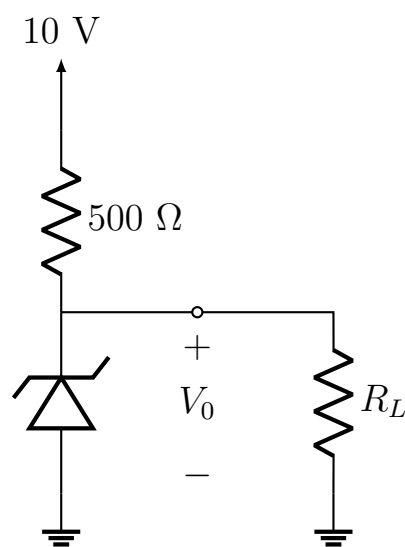
- Q3.** For the circuit given, determine the range of  $R_L$  and  $I_L$  that will result in  $V_{R_L}$  being maintained at 10 V. Also determine the maximum wattage rating of the diode. The Zener diode is a 10 V Zener with maximum current rating of 32 mA.



(18 marks)

[EEE 263 | L-2/T-1 | 2022–23]

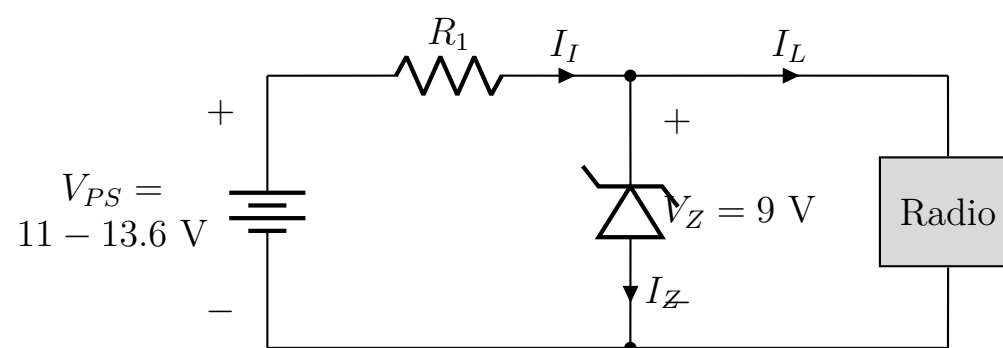
- Q4.** The zener diode in the circuit shown is specified to have  $V_Z = 6.8$  V at  $I_Z = 5$  mA,  $r_z = 20$   $\Omega$  and  $I_{ZK} = 0.2$  mA. (i) Find  $V_0$  at no load conditions. (ii) What is the minimum value of  $R_L$  for which the diode will still operate in the breakdown region?



(23 marks)

[EEE 263 | L-2/T-1 | 2020–21]

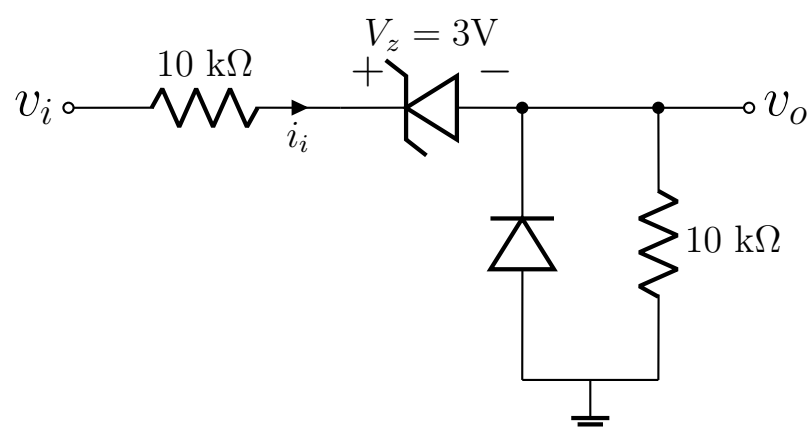
- Q5.** A voltage regulator circuit is to power a car radio at  $V_L = 9$  V from an automobile battery whose voltage may vary between 11 and 13.6 V. The current in the radio will vary between 0 (off) and 100 mA (full volume). Determine the value of the current-limiting resistor  $R_i$  and the maximum power dissipated in the Zener diode.



(18 marks)

[EEE 263 | L-2/T-1 | 2017–18]

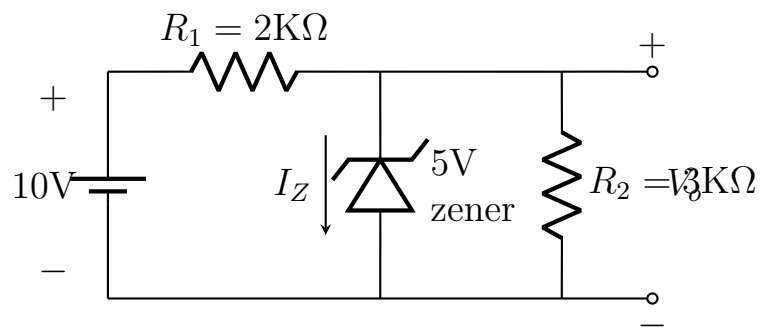
- Q6.** For the circuit shown in Figure , plot  $v_0$  versus  $v_i$  and  $i_1$  versus  $v_i$  over the range  $-10 \leq v_i \leq +10$  V. Assume the diodes to be ideal.



(13.67 marks)

[EEE 263 | L-2/T-1 | 2016–17]

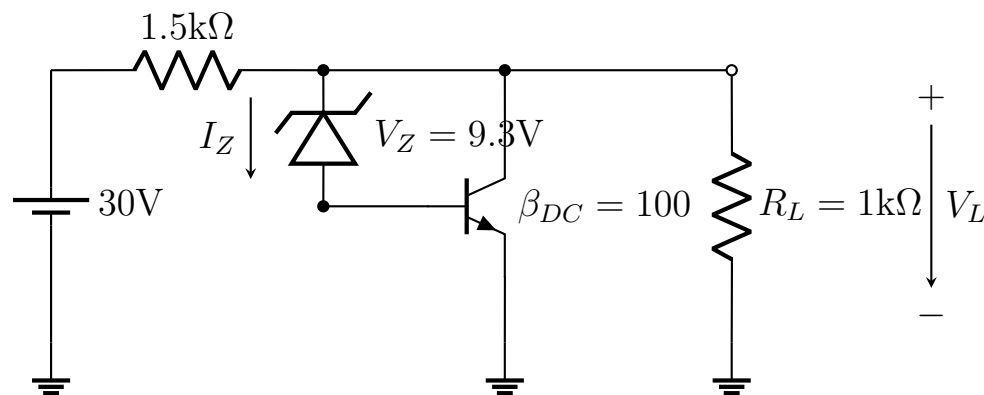
- Q7.** For the circuit shown, find  $V_a$  and  $I_z$ . Now if  $R_1$  and  $R_2$  are swapped, what will be the value of  $V_a$  and  $I_z$ ?



(15 marks)

[EEE 263 | L-2/T-1 | 2015–16]

Q8. Calculate the regulated load voltage  $V_L$  and the current  $I_Z$  flowing through the zener diode in the voltage regulator circuit shown.

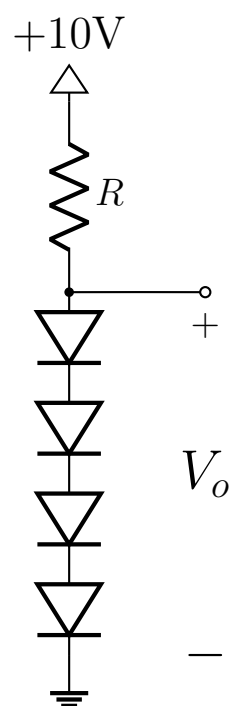


(16 marks)

[EEE 263 | L-2/T-1 | 2014–15]

### Diode Circuit Design

Q1. Design the circuit shown to provide an output voltage  $v_a = 3\text{ V}$ . Assume that the diodes available have  $0.7\text{ V}$  drop at  $1\text{ mA}$ , thermal voltage  $= 25\text{ mV}$ , and parameter  $n = 1$ .



(15 marks)

[EEE 263 | L-2/T-1 | 2015–16]

### Full Wave Rectifier

Q1. Draw the circuit of a full-wave rectifier with center-tapped secondary winding and draw the input and output voltage waveshapes. Comment on the values of RC time constant for a rectifier circuit. (20 marks)

[EEE 263 | L-2/T-1 | Jan–2021]

Q2. Draw the circuit diagram of a full wave rectifier with a filter capacitor. Assume the input is a sinusoidal signal with  $10\text{ kHz}$  frequency and  $7.07\text{ V}$  rms voltage, the load is  $1\text{ k}\Omega$  and capacitor used is  $1\text{ }\mu\text{F}$ . Diode has a forward voltage drop of  $0.7\text{ V}$ . Find:

- Ripple voltage.
- PIV and duty cycle.
- Peak output voltage.

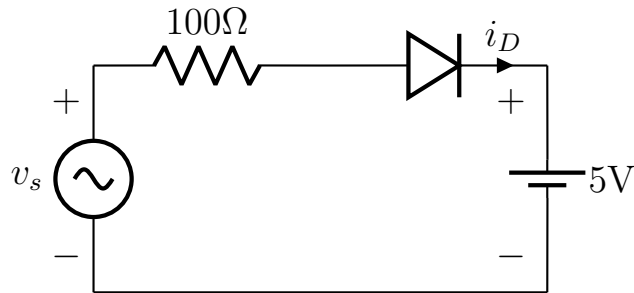
(16 marks)

[EEE 263 | L-2/T-1 | 2015–16]

## Battery Charger Circuit

**Q1.** The circuit shown is for charging a 5 V battery with  $V_s = 10 \sin(100\pi t)$ . Find:

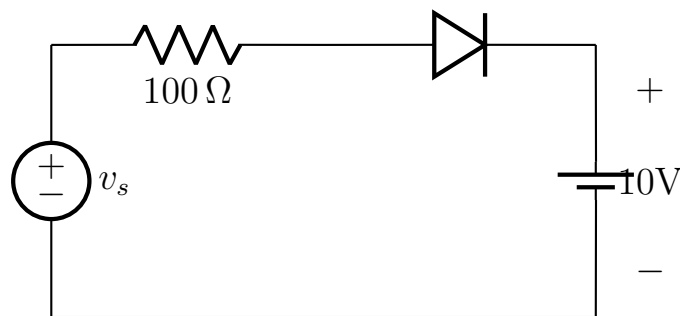
- Duty cycle (fraction of each cycle during which the diode conducts).
- Draw the diode current waveform indicating peak value.
- Maximum reverse bias voltage that appears across the diode.



(15 marks)

[EEE 263 | L-2/T-1 | 2015–16]

**Q2.** A circuit for charging a 10 V battery is shown in the figure. If  $V_s$  is a sinusoid with 20 V peak amplitude, find the fraction of each cycle during which the diode conducts. Assume the diode is ideal. Also find the peak value of the diode current and the maximum reverse bias voltage that appears across the diode.

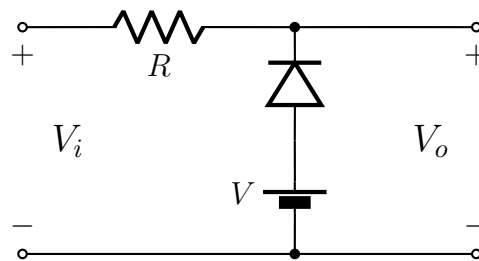


(15 marks)

[EEE 263 | L-2/T-1 | 2011–12]

## Diode Wave Shaping

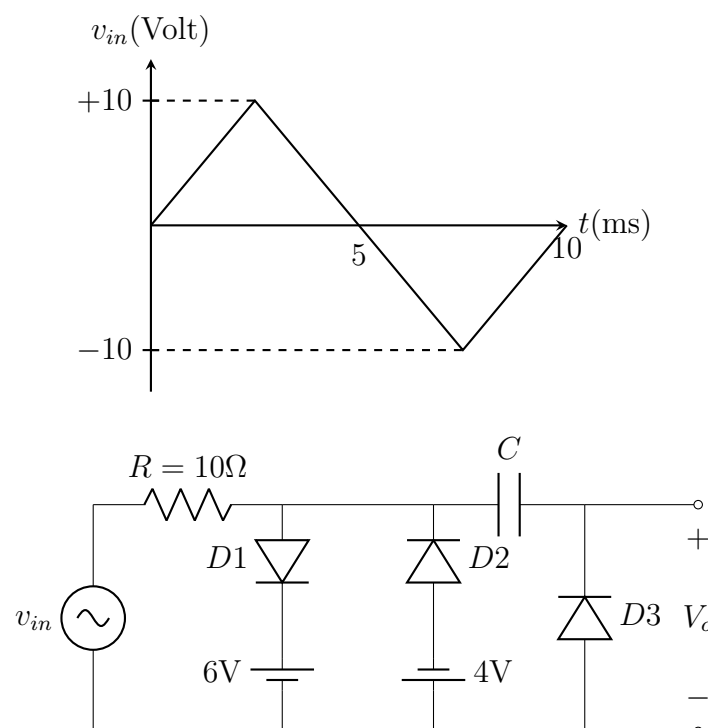
**Q1.** For a sinusoidal input, draw the output waveshape for the circuit shown.



(10 marks)

[EEE 263 | L-2/T-1 | 2018–19]

**Q2.** Assuming that the diodes are ideal, find the output voltage  $V_o$  for the circuit given. Draw the wave shape clearly.



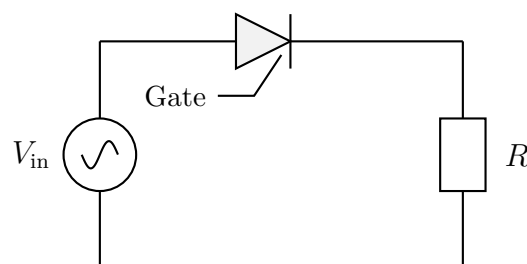
(20 marks)

[EEE 263 | L-2/T-1 | 2014–15]

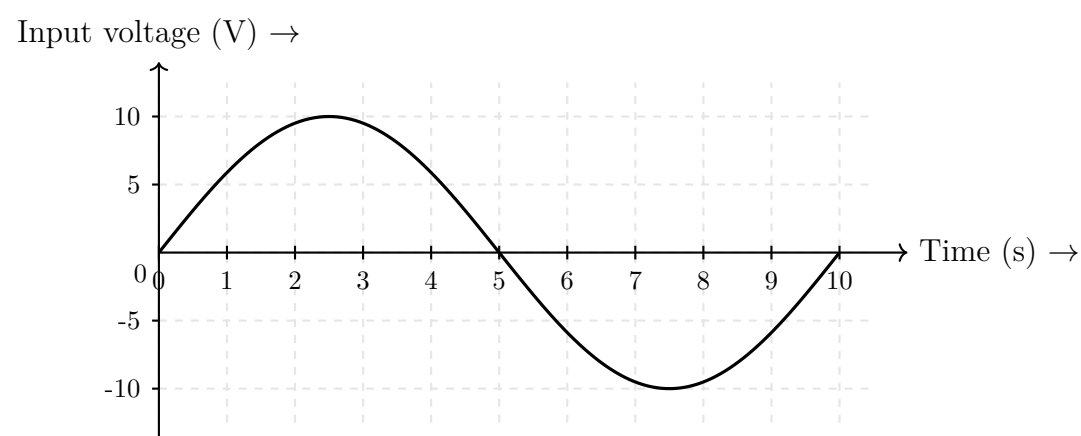
## Special Purpose Diodes/Thyristors

- Q1.** Draw the circuit diagram of controlled full wave rectifier using SCR. Draw the waveforms of sinusoidal input, gate pulses and rectified output. (11.67 marks) [EEE 263 | L-2/T-1 | 2014–15]
- Q2.** Explain SCR and TRIAC operation with suitable examples. (15 marks) [EEE 263 | L-2/T-1 | 2013–14]
- Q3.** Sketch the output wave form of the following circuit given in Fig. for Q. 1(c)(i) assuming output wave is obtained across the resistance. The input waveform and gate pulse sequence is given in Fig. for Q. 1(c)(ii).

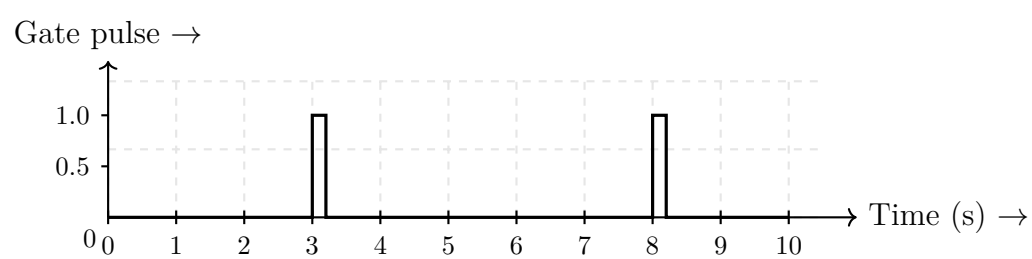
## 1. Circuit Diagram [Fig. 1(c)(i)]



## 2. Input Voltage Waveform [Fig. 1(c)(ii)]



## 3. Gate Pulse Sequence [Fig. 1(c)(ii)]



(20 marks)

[EEE 263 | L-2/T-1 | 2011–12]

- Q4.** Draw the I-V characteristics of an SCR and a TRIAC. (10 marks) [EEE 263 | L-2/T-1 | 2011–12]
- Q5.** Explain the operating principle of a Silicon controlled Rectifier with the help of 'Two transistor model'. (20 marks) [EEE 263 | L-2/T-1 | 2011–12]

BUET · EEE 263 · Electronic Circuits

# Electronic Circuits

S2 — BJT Characteristics

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Energy band profiles, DC analysis, biasing stability, circuit design

S2 BJT Characteristics

Energy Band Profile

Q1. Draw the energy band profile of a pnp transistor under zero bias and forward-active mode bias conditions. (16/23 marks) [EEE 263 | L-2/T-1 | 2020–21, 2018–19]

Device Structure and Operating Principle

Q1. Why is the third terminal (Base or Gate) used in a transistor? (10 marks) [EEE 263 | L-2/T-1 | Jan–2021]

Early Effect

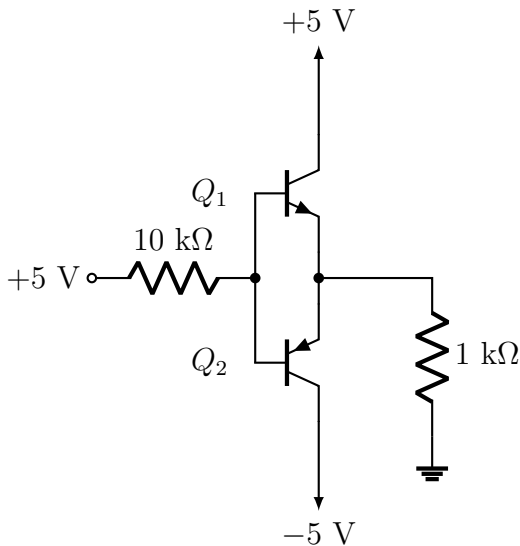
Q1. What is the Early-effect in a BJT? Draw the  $I$ – $V$  curve of a practical BJT showing the Early effect. (10 marks) [EEE 263 | L-2/T-1 | 2023–24]

Q2. What is the early effect for BJT? What are the consequences of early effect on BJT behavior? (16 marks) [EEE 263 | L-2/T-1 | Jan–2021]

Q3. Why biasing is needed for an amplifier using BJT? Draw the output characteristics of a BJT connected in CE configuration. Define Early effect. (10.667 marks) [EEE 263 | L-2/T-1 | 2011–12]

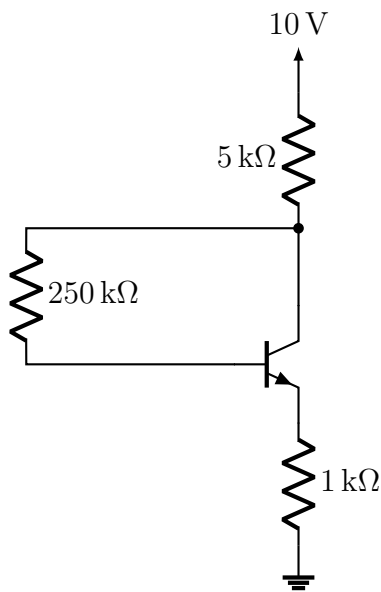
DC Analysis

Q1. For the circuit shown, find the voltages at all nodes and the currents through all branches. Assume  $\beta = 100$ .



(20 marks) [EEE 263 | L-2/T-1 | 2020–21]

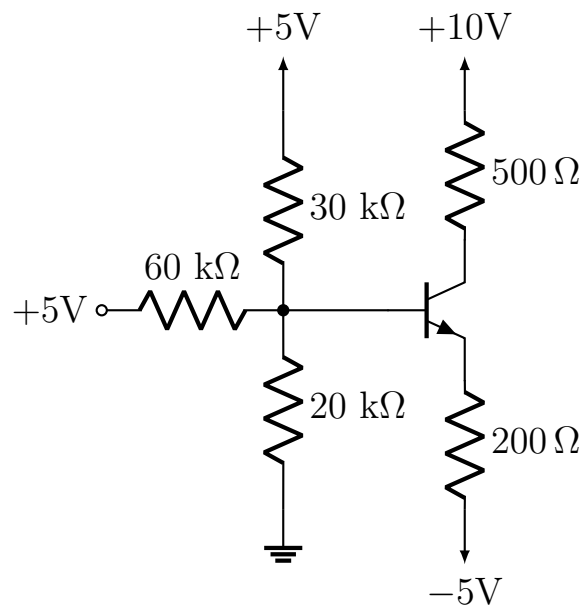
Q2. Calculate the collector current, emitter current and base current for the circuit shown in Figure . Also verify the operating mode of the transistor. Assume  $\beta = 100$ . (16)



(16 marks)

[EEE 263 | L-2/T-1 | 2017–18]

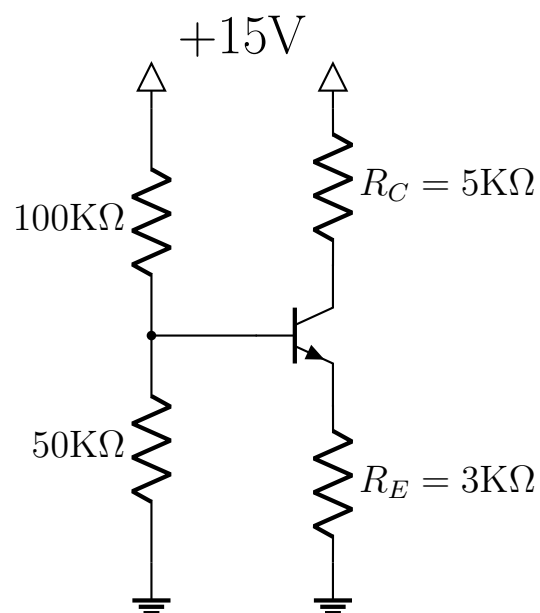
Q3. Find  $I_C$  and  $V_{CE}$  for the circuit shown. Assume  $\beta = 100$ .



(16 marks)

[EEE 263 | L-2/T-1 | 2016–17]

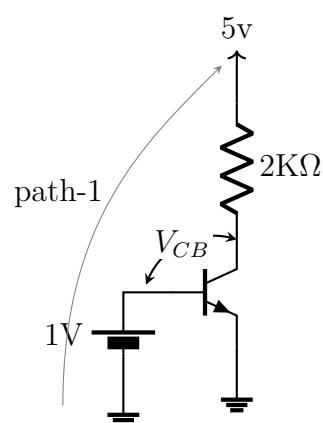
Q4. For the circuit shown, determine the voltages of all nodes and current through all branches. Assume  $\beta = 100$ .



(20 marks)

[EEE 263 | L-2/T-1 | 2015–16]

- Q5. (a) Express the collector current  $I_c$  as a function of  $V_{CB}$  that involves the 2 kΩ resistance (it will be a straight line equation). (16 marks) [EEE 263 | L-2/T-1 | 2012–13]
- (b) Find out the mathematical equation describing the  $I_c$ – $V_{CB}$  characteristic graph of the BJT (it will be a piecewise linear graph). (10 marks) [EEE 263 | L-2/T-1 | 2012–13]
- (c) Draw the two  $I_c$ – $V_{CB}$  graphs over the same plot exactly to scale and determine their intersection point. This intersection point gives the value of  $I_c$  and  $V_{CB}$  at the bias point. (10 marks) [EEE 263 | L-2/T-1 | 2012–13]
- (d) From the value of  $V_{CB}$ , can you determine the value of  $V_{CE}$  at the bias point? (10 marks) [EEE 263 | L-2/T-1 | 2012–13]



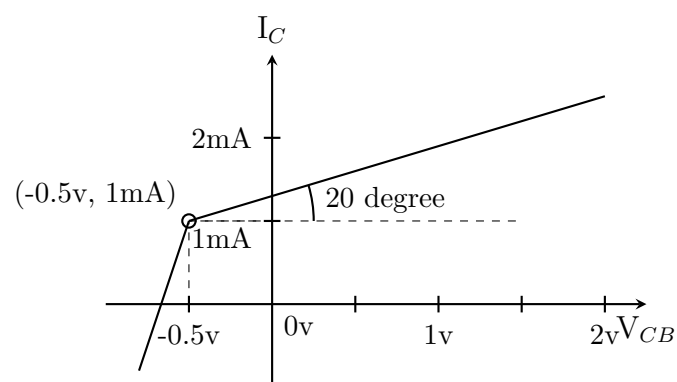
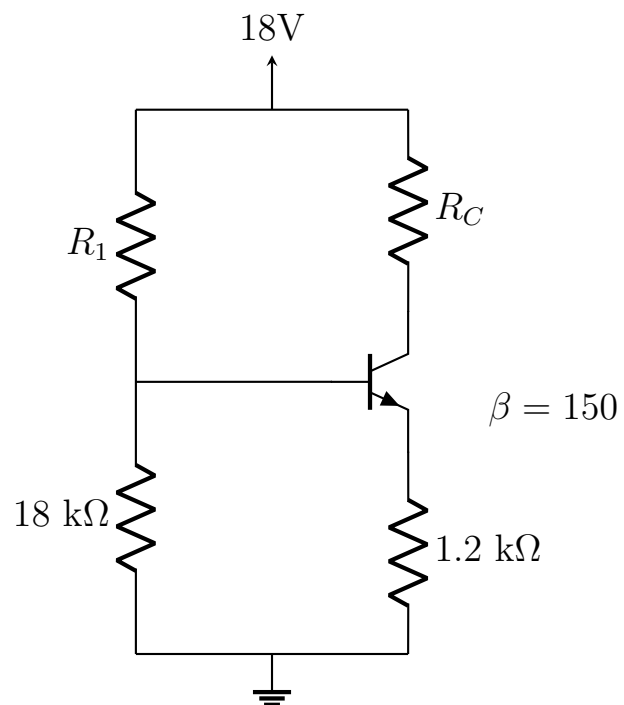


Figure-2

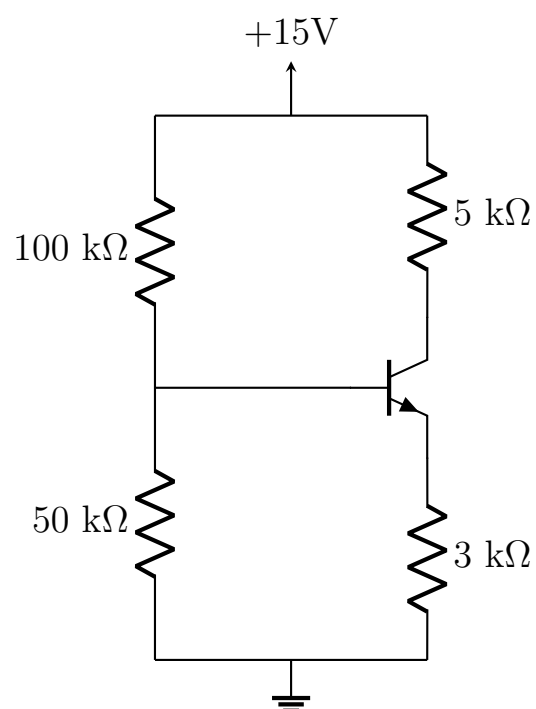
**Q6.** For the circuit shown, determine  $R_1$  and  $R_C$  if  $I_B = 0.02 \text{ mA}$  and  $V_{CE} = 10 \text{ V}$ .



(16 marks)

[EEE 263 | L-2/T-1 | 2011–12]

**Q7.** Determine all node voltages and branch currents for the circuit shown. Assume  $\beta = 100$  and  $V_{BE} = 0.7 \text{ V}$ .



(18 marks)

[EEE 263 | L-2/T-1 | 2011–12]

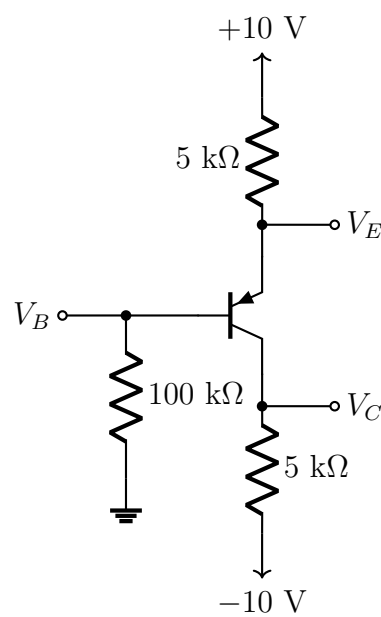
### BJT Saturation Parameters

**Q1.** A BJT for which  $I_B = 0.5 \text{ mA}$  has  $v_{CEsat} = 140 \text{ mV}$  at  $I_C = 10 \text{ mA}$  and  $v_{CEsat} = 170 \text{ mV}$  at  $I_C = 20 \text{ mA}$ . Estimate the values of: (i) saturation resistance  $R_{CEsat}$ ; (ii) offset voltage  $V_{CEoff}$ ; (iii)  $\beta_f$ ; (iv)  $\beta_r$ . (10 marks)

[EEE 263 | L-2/T-1 | 2023–24]

### Multi-BJT Circuit Analysis

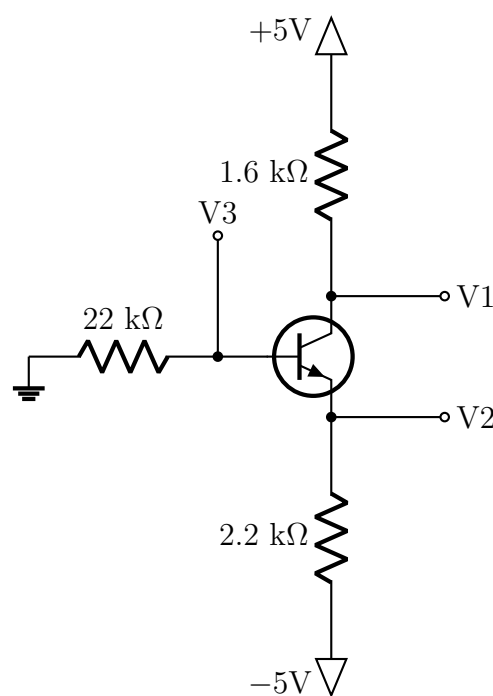
- Q1.** For the circuit shown, the emitter voltage was measured to be  $V_E = +1.7 \text{ V}$ . Assume  $|V_{BE}| = 0.7 \text{ V}$ . Find the values of the labeled node voltages and branch currents. What are the values of  $\alpha$  and  $\beta$  for the pnp transistor?



(15 marks)

[EEE 263 | L-2/T-1 | 2024–25]

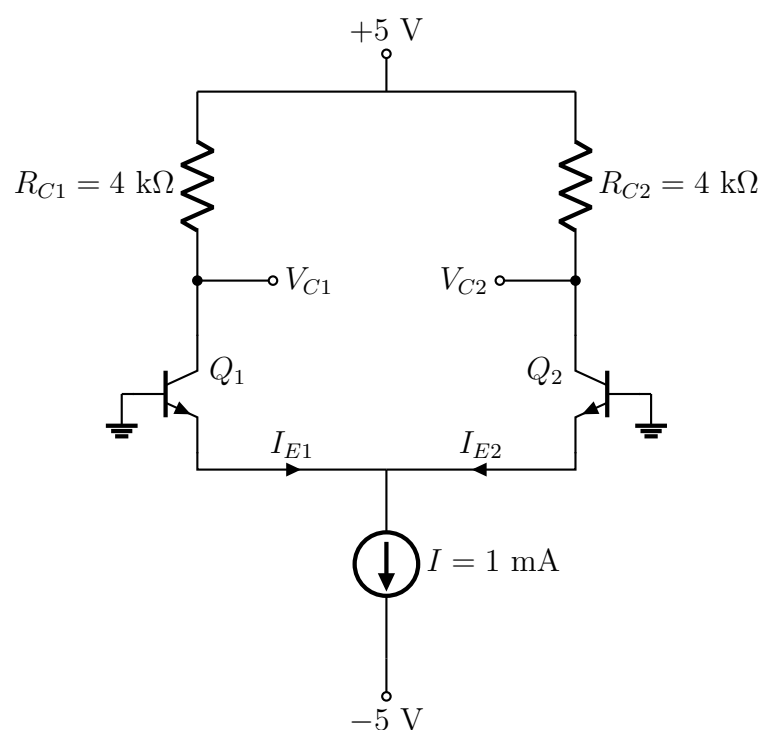
- Q2.** For the circuit shown, find the values of the labeled node voltages and branch currents. Assume  $\beta$  to be very high and  $|V_{BE}| = 0.7 \text{ V}$ . What will happen if  $\beta$  is reduced to a value of 100?



(15 marks)

[EEE 263 | L-2/T-1 | 2023–24]

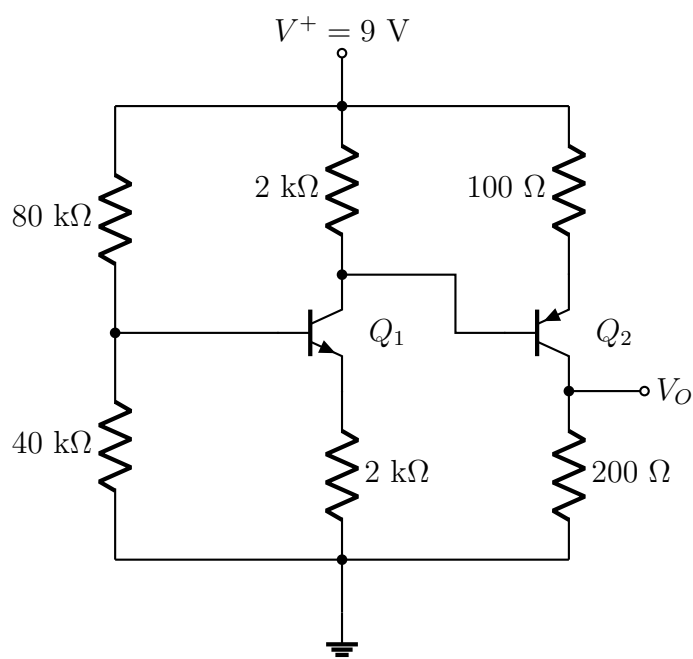
- Q3.** For the circuit shown with  $\beta = 200$  for each transistor, determine: (i)  $I_{E1}$ , (ii)  $I_{E2}$ , (iii)  $V_{C1}$ , and (iv)  $V_{C2}$ .



(15 marks)

[EEE 263 | L-2/T-1 | 2022–23]

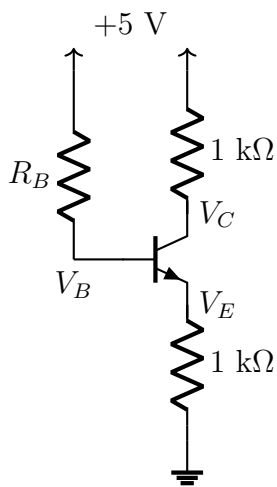
- Q4.** For the circuit shown,  $\beta_n = 120$  (npn) and  $\beta_p = 80$  (pnp). Determine  $I_{B1}$ ,  $I_{C1}$ ,  $I_{B2}$ ,  $I_{C2}$ ,  $V_{CE1}$ , and  $V_{EC2}$ .



(20 marks)

[EEE 263 | L-2/T-1 | 2022–23]

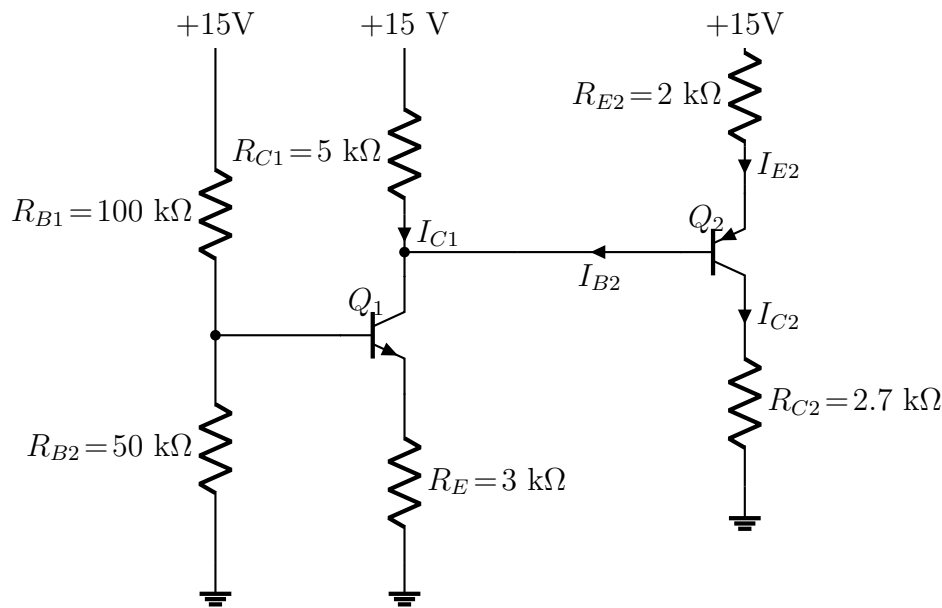
Q5. For the circuit shown, find  $V_B$ ,  $V_E$ , and  $V_C$  for  $R_B = 10\text{ k}\Omega$ . Let  $\beta = 100$ .



(16 marks)

[EEE 263 | L-2/T-1 | 2021–22]

Q6. Determine the currents  $I_{C1}$  and  $I_{C2}$  for the multi-BJT circuit shown.

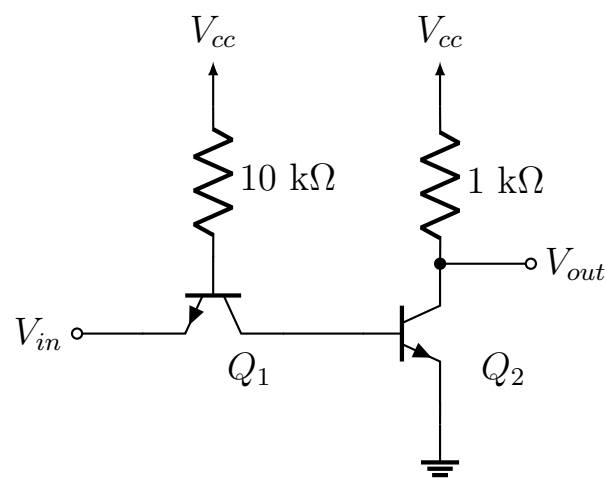


(30 marks)

[EEE 263 | L-2/T-1 | 2017–18]

Voltage Transfer Characteristics

Q1. Draw the voltage transfer characteristics by qualitatively analyzing the circuit shown.

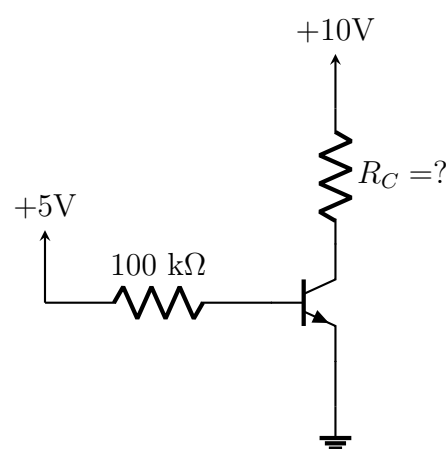


(15 marks)

[EEE 263 | L-2/T-1 | 2016–17]

### Operating Mode Design

- Q1.** The circuit shown is to be fabricated using a transistor type whose  $\beta$  can vary from 50 to 150. Design the circuit (find  $R_c$ ) so that all fabricated circuits are guaranteed to be in the active mode. What is the range of collector voltages that the fabricated circuits may exhibit?

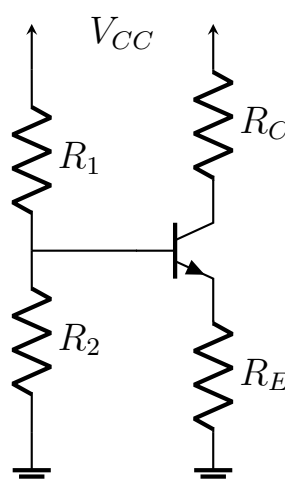


(20 marks)

[EEE 263 | L-2/T-1 | 2018–19]

### Biasing and Stability

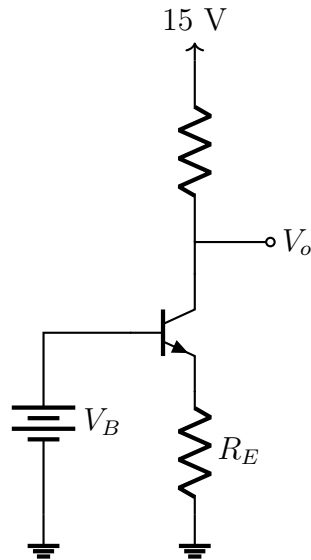
- Q1.** What is transistor biasing? Why do we need transistor biasing? How can you achieve biasing stability? (10 marks) [EEE 263 | L-2/T-1 | 2020–21]
- Q2.** Comment on the desired values of input impedance, output impedance, voltage gain, and current gain of a BJT amplifier. Comment on the values of these four parameters for CB, CE, and CC configurations. (16 marks) [EEE 263 | L-2/T-1 | Jan–2021]
- Q3.** (a) Find the expression for the emitter current in the circuit shown. What are the conditions that should be satisfied? How does the feedback operation of  $R_E$  work? (16 marks)



- (b) In the circuit of the previous question, if  $R_1 = 40 \text{ k}\Omega$ ,  $R_2 = 10 \text{ k}\Omega$ ,  $R_c = 1.5 \text{ k}\Omega$ ,  $R_E = 1 \text{ k}\Omega$ ,  $V_{cc} = 12 \text{ V}$ , and  $\beta = 1000$ , calculate the biasing point. How can the values of  $R_1$  and  $R_2$  be changed for a good biasing design? (10 marks) [EEE 263 | L-2/T-1 | 2014–15]

- Q4.** Why do we need to bias a BJT at a proper Q-point before using it as an amplifier? (5 marks) [EEE 263 | L-2/T-1 | 2012–13]

**Q5.** Graphically explain the benefit of having  $R_E > 0$  in the biasing scheme shown.

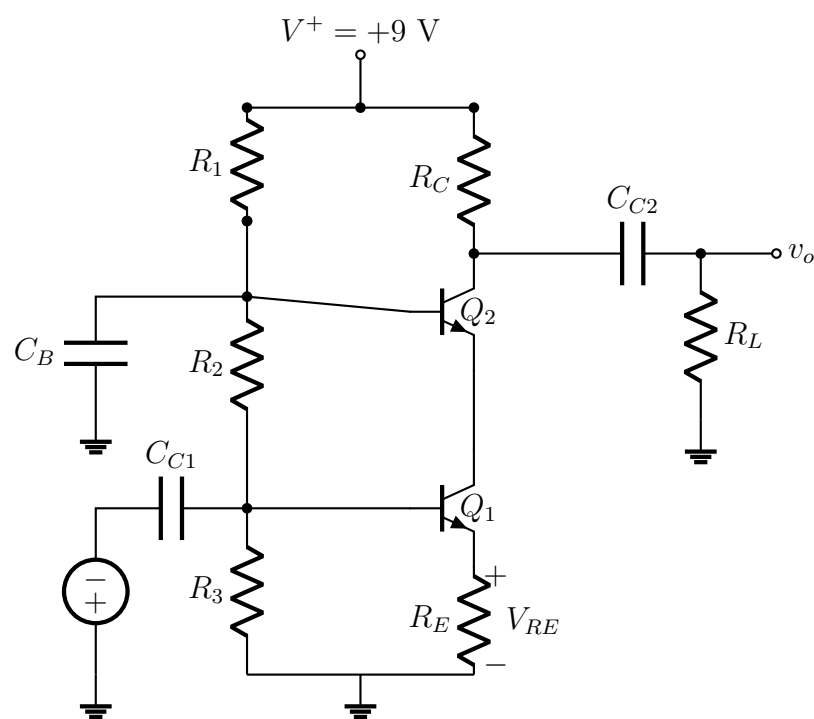


(16 marks)

[EEE 263 | L-2/T-1 | 2012–13]

### BJT Circuit Design

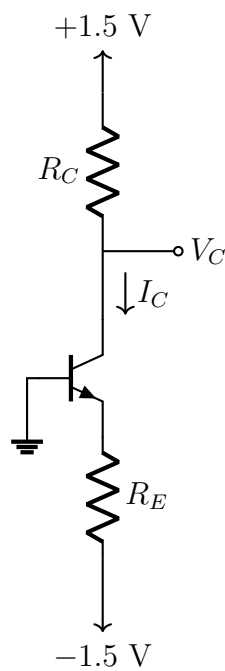
**Q1.** Design the circuit shown to meet the following specifications:  $V_{CE1} = V_{CE2} = 2.5\text{ V}$ ,  $V_{RE} = 0.7\text{ V}$ ,  $I_{C1} \cong I_{C2} \cong 1\text{ mA}$ , and  $I_{R1} \cong I_{R2} \cong I_{R3} = 0.10\text{ mA}$ .



(15 marks)

[EEE 263 | L-2/T-1 | 2022–23]

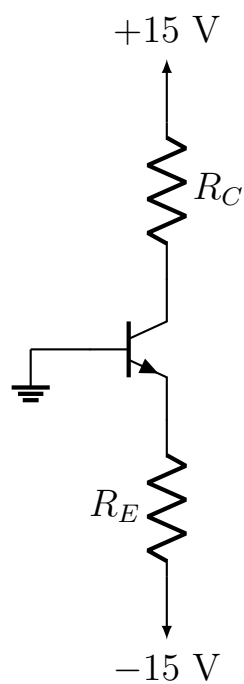
**Q2.** Design the circuit shown to establish  $I_C = 0.2\text{ mA}$  and  $V_C = 0.5\text{ V}$ . The transistor exhibits  $V_{BE} = 0.8\text{ V}$  at  $i_c = 1\text{ mA}$ , and  $\beta = 100$ .



(20 marks)

[EEE 263 | L-2/T-1 | 2021–22]

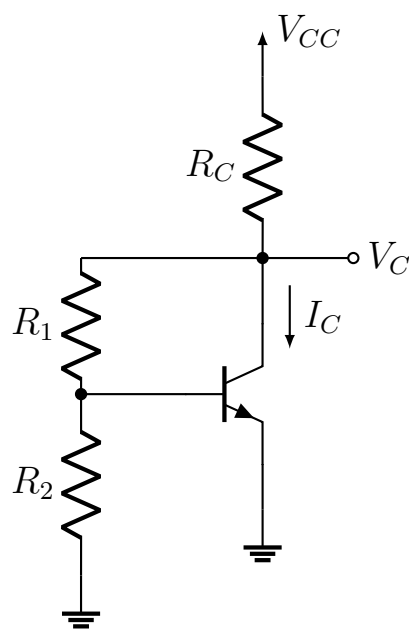
- Q3.** The transistor in the circuit has  $\beta = 100$  and exhibits  $V_{BE} = 0.7\text{ V}$  at  $i_c = 1\text{ mA}$ . Design the circuit so that a current of  $2\text{ mA}$  flows through the collector and a voltage of  $+5\text{ V}$  appears at the collector.



(20 marks)

[EEE 263 | L-2/T-1 | 2020–21]

- Q4.** Design the circuit shown to provide  $I_C = 3\text{ mA}$  and  $V_{CE} = 1.5\text{ V}$  for  $\beta = 90$ . Use  $V_{CC} = 3\text{ V}$  and a current through  $R_2$  equal to the base current.

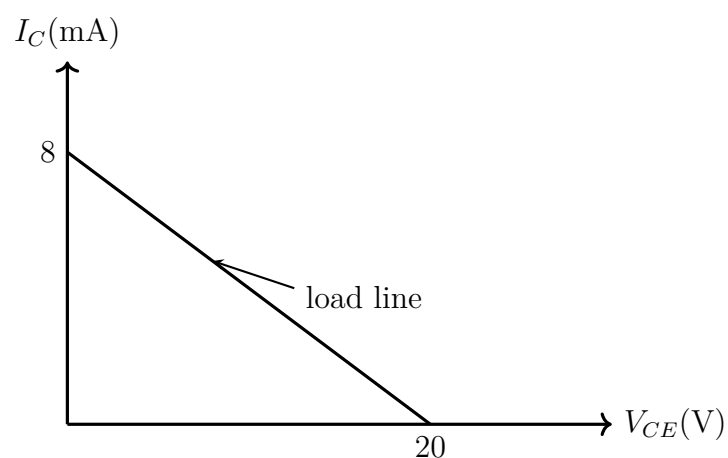


(15 marks)

[EEE 263 | L-2/T-1 | 2016–17]

### Linear Amplifier Constraints

- Q1.** For the figure shown, find out the maximum collector current that the device can support. If the operating point is set for that maximum current, what kind of problem will we face if we want to use the device as a linear amplifier?



(12 marks)

[EEE 263 | L-2/T-1 | 2011–12]

BUET · EEE 263 · Electronic Circuits

# Electronic Circuits

S3 — MOSFET Characteristics

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Structure, threshold voltage, DC analysis, biasing stability, CMOS logic

### S3 MOSFET Characteristics

#### Device Characteristics

- Q1.** Explain the different operation regions of an enhancement PMOS transistor with properly labeled  $I_D$ - $V_{SD}$  plots. Derive the threshold conditions for operating in those regions. **(8 marks)** [EEE 263 | L-2/T-1 | 2024–25]
- Q2.** For a particular MOSFET operating in the saturation region at a constant  $v_{gs}$ ,  $i_D = 2\text{ mA}$  at  $v_{ds} = 4\text{ V}$ . The  $i_D$  increases by 10% when  $v_{ds}$  is doubled. Find the corresponding values of  $r_0$ ,  $V_A$ , and  $\lambda$ . **(10 marks)** [EEE 263 | L-2/T-1 | 2023–24]
- Q3.** Briefly explain the I-V characteristics of an n-channel enhancement MOSFET with necessary plots. **(7 marks)** [EEE 263 | L-2/T-1 | 2022–23]
- Q4.** What is the difference between enhancement-type and depletion-type MOSFETs? Draw  $I_D$  vs.  $V_{GS}$  and  $I_D$  vs.  $V_{DS}$  characteristics for different values of  $V_{GS}$  for both enhancement-type and depletion-type MOSFETs. **(26 marks)** [EEE 263 | L-2/T-1 | 2018–19]
- Q5.** Showing different operating regions, describe the output characteristics of a MOSFET. Why does current saturation occur in a MOSFET as  $V_{DS}$  is increased? **(18 marks)** [EEE 263 | L-2/T-1 | Jan–2021]

#### Comparison with BJT

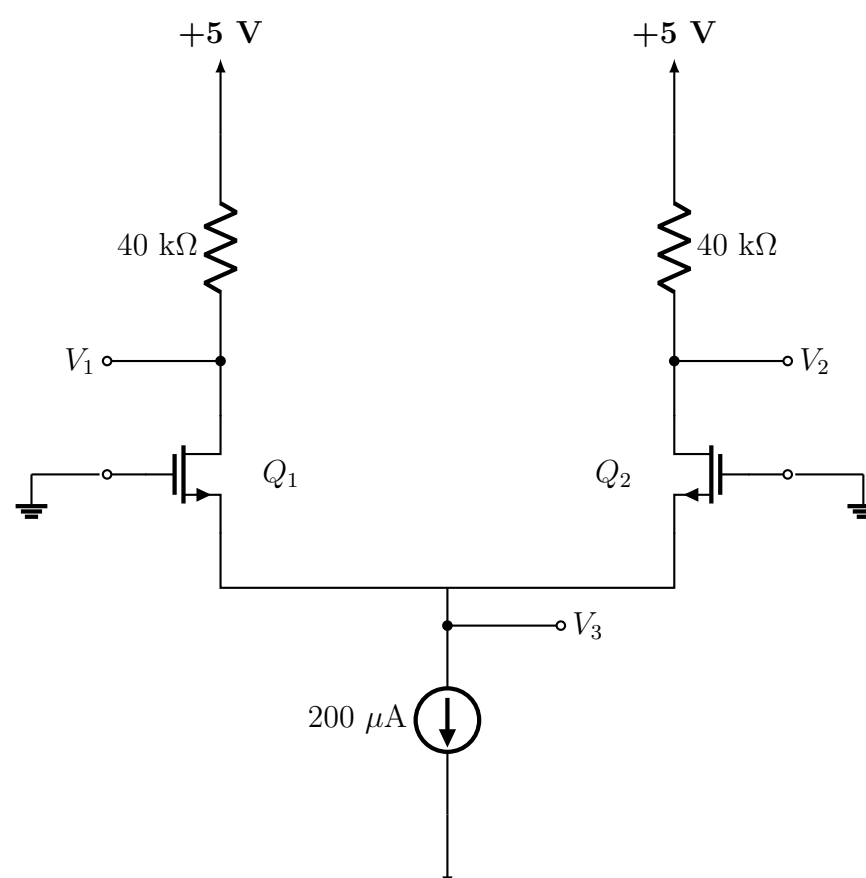
- Q1.** Compare MOSFET and BJT in terms of their operating principles. Mention each of their advantages and disadvantages. **(10 marks)** [EEE 263 | L-2/T-1 | 2013–14]
- Q2.** What are the mutual advantages and disadvantages of BJT and MOSFET? **(12 marks)** [EEE 263 | L-2/T-1 | 2012–13]

#### Threshold Voltage

- Q1.** Explain the concept of threshold voltage ( $V_t$ ) of a MOSFET. What happens to  $V_t$  and oxide capacitance ( $C_{ox}$ ) if: (i) the dielectric thickness is increased/decreased? (ii) the gate insulator is replaced with a more high- $k$  dielectric of the same thickness? **(20 marks)** [EEE 263 | L-2/T-1 | 2012–13]

#### DC Analysis & Transistor Sizing

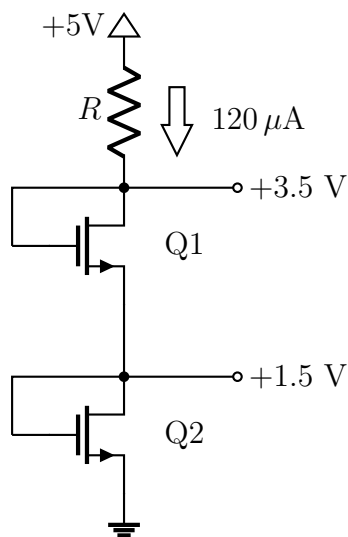
- Q1.** In the circuit shown, the aspect ratios of  $Q_1$  and  $Q_2$  are  $(W/L)_1 = 2(W/L)_2 = 20$ . Both transistors have  $V_t = 1\text{ V}$  and  $\mu_n C_{ox} = 100\text{ }\mu\text{A/V}^2$ . Find the node voltages  $V_1$ ,  $V_2$ , and  $V_3$ .



**(17 marks)**

[EEE 263 | L-2/T-1 | 2024–25]

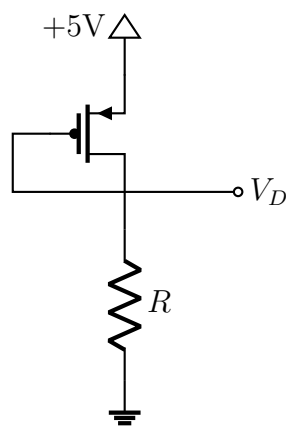
- Q2.** The NMOS transistors in the circuit have  $V_t = 1\text{ V}$ ,  $\mu_n C_{ox} = 120\text{ }\mu\text{A/V}^2$ ,  $\lambda = 0$ , and  $L_1 = L_2 = 1\text{ }\mu\text{m}$ . Find the required values of gate width  $W_1$  and  $W_2$  for  $Q_1$  and  $Q_2$ , and the value of  $R$ , for obtaining the voltage values indicated. The current in  $R$  will be  $120\text{ }\mu\text{A}$ .



(15 marks)

[EEE 263 | L-2/T-1 | 2023–24]

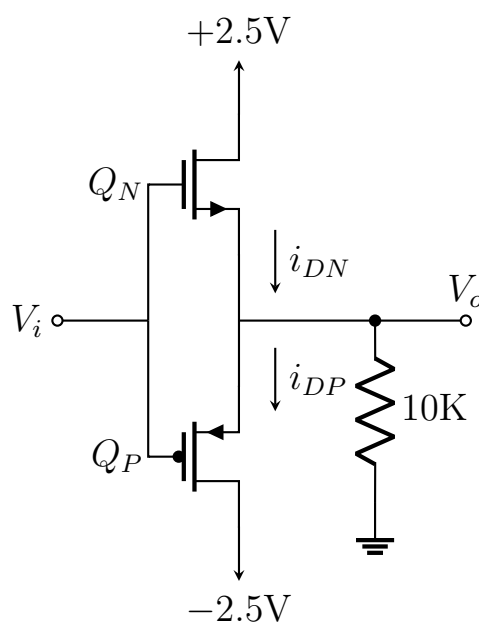
- Q3.** The PMOS transistor in the circuit shown has  $V_t = -0.7\text{ V}$ ,  $\mu_p C_{ox} = 60\text{ }\mu\text{A/V}^2$ ,  $L = 0.8\text{ }\mu\text{m}$ , and  $\lambda = 0$ . Find the value of  $W$  and  $R$  in order to establish a drain current of  $115\text{ }\mu\text{A}$  and a voltage  $V_D = 3.5\text{ V}$ .



(15 marks)

[EEE 263 | L-2/T-1 | 2023–24]

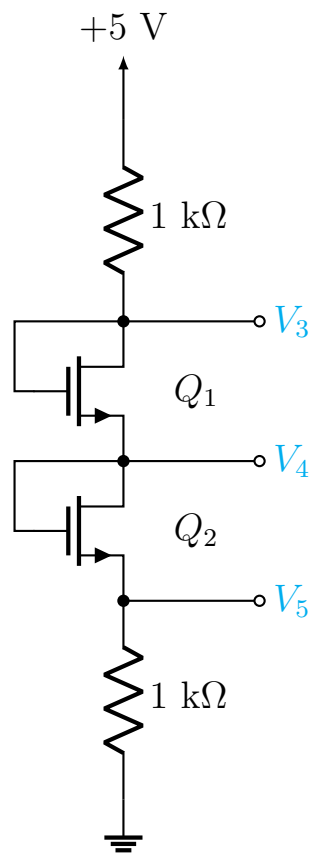
- Q4.** The NMOS and PMOS transistors in the circuit are matched with  $k_n = k_p = 1\text{ mA/V}^2$  and  $V_{tn} = -V_{tp} = 1\text{ V}$ . Assuming  $\lambda = 0$  for both MOSFETs, find the drain currents  $i_{DN}$  and  $i_{DP}$  and the voltage  $v_0$  for  $v_1 = 0\text{ V}$  and  $v_1 = +2.5\text{ V}$ .



(20 marks)

[EEE 263 | L-2/T-1 | 2022–23]

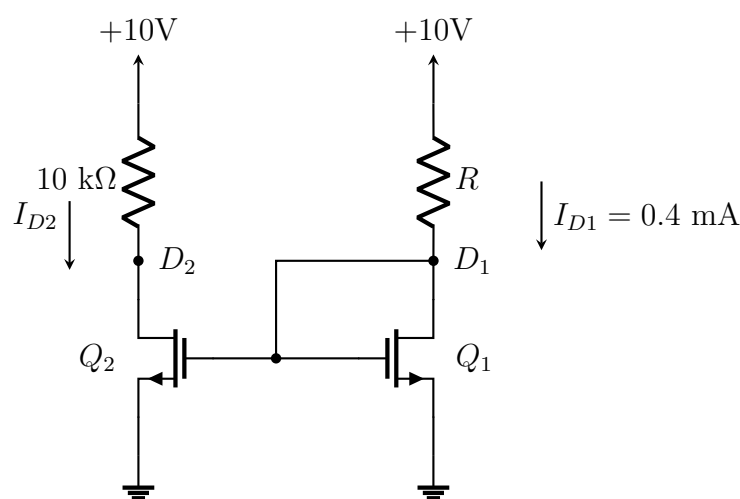
- Q5.** For the circuit shown, find the node voltages  $V_3$ ,  $V_4$ , and  $V_5$ . The NMOS transistors have  $V_t = 0.9\text{ V}$  and  $k'_n(W/L) = 1.5\text{ mA/V}^2$ .



(28 marks)

[EEE 263 | L-2/T-1 | Jan-2021]

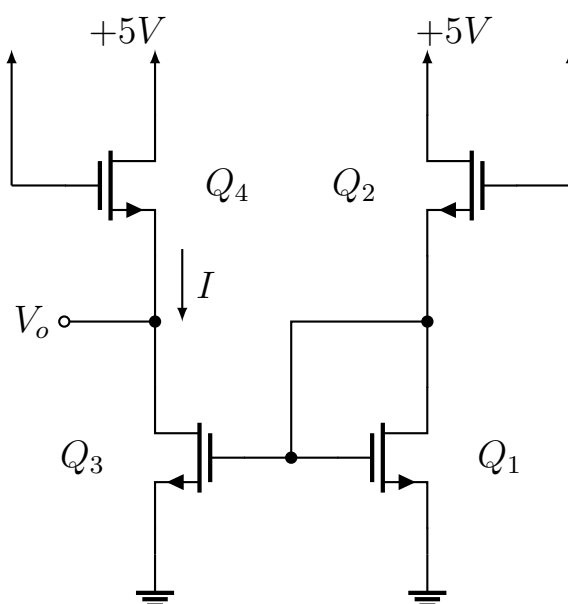
- Q6.** For the circuit shown, find  $R$  so that  $I_{D1} = 0.4 \text{ mA}$ . Also find  $V_{D1}$ ,  $V_{D2}$ , and  $I_{D2}$ .  $Q_1$  and  $Q_2$  are identical. Given:  $V_t = 2 \text{ V}$ ,  $\mu_n C_{ox} = 20 \mu\text{A}/\text{V}^2$ ,  $L = 10 \mu\text{m}$ ,  $W = 100 \mu\text{m}$ .



(20 marks)

[EEE 263 | L-2/T-1 | 2018-19]

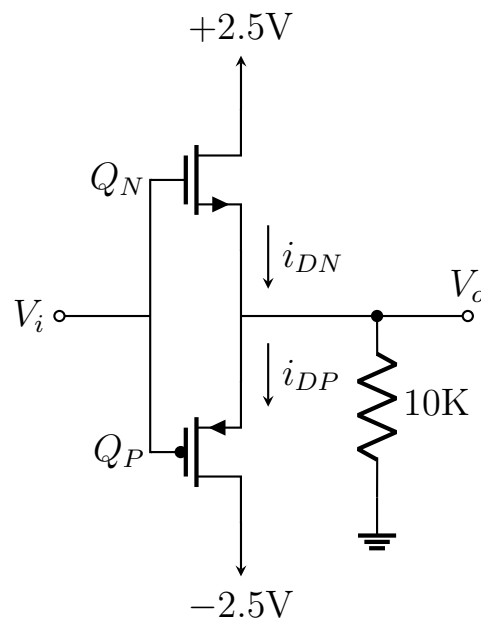
- Q7.** For the circuit shown, for all MOSFETs  $|V_t| = 1 \text{ V}$ ,  $\mu_n C_{ox} = 50 \mu\text{A}/\text{V}^2$ ,  $L = 1 \mu\text{m}$ ,  $W = 10 \mu\text{m}$ . Find  $V$  and  $I$ . If  $Q_3$  and  $Q_4$  are made to have  $W = 100 \mu\text{m}$ , find  $V$  and  $I$  in that case.



(26 marks)

[EEE 263 | L-2/T-1 | 2016-17]

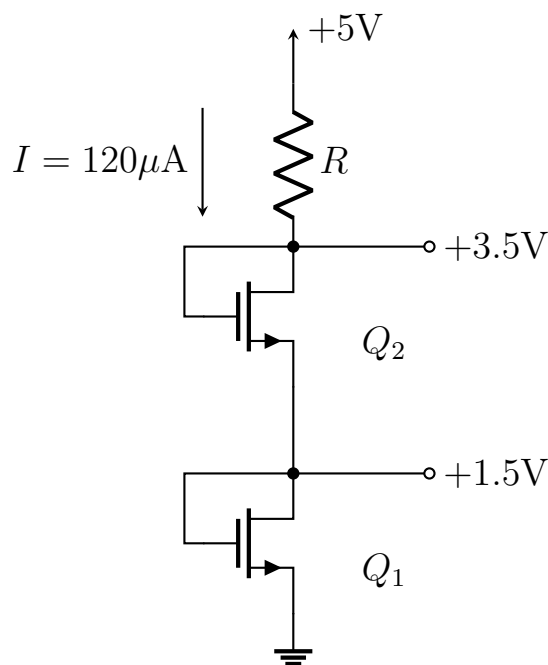
- Q8.** The NMOS and PMOS transistors in the circuit shown have  $k_n(W/L) = k_p(W/L) = 1 \text{ mA}/\text{V}^2$  and  $V_{tn} = -V_{tp} = 1 \text{ V}$ . Assume  $\lambda = 0$ . Find  $i_{DN}$ ,  $i_{DP}$ , and  $v_o$  for  $V_i = +2.5 \text{ V}$  and  $V_i = -2.5 \text{ V}$ .



(20 marks)

[EEE 263 | L-2/T-1 | 2014–15]

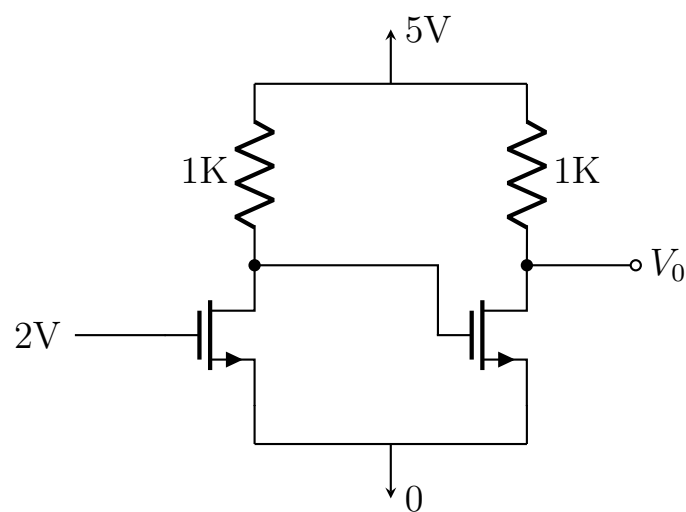
- Q9.** The NMOS transistors in the circuit have  $V_t = 1\text{ V}$ ,  $\mu_n C_{ox} = 120\text{ }\mu\text{A/V}^2$ ,  $L_1 = L_2 = 1\text{ }\mu\text{m}$ . Find  $W_1$ ,  $W_2$ , and  $R$ , where  $W_1$  and  $W_2$  are the gate widths of Q1 and Q2 respectively.



(16 marks)

[EEE 263 | L-2/T-1 | 2014–15]

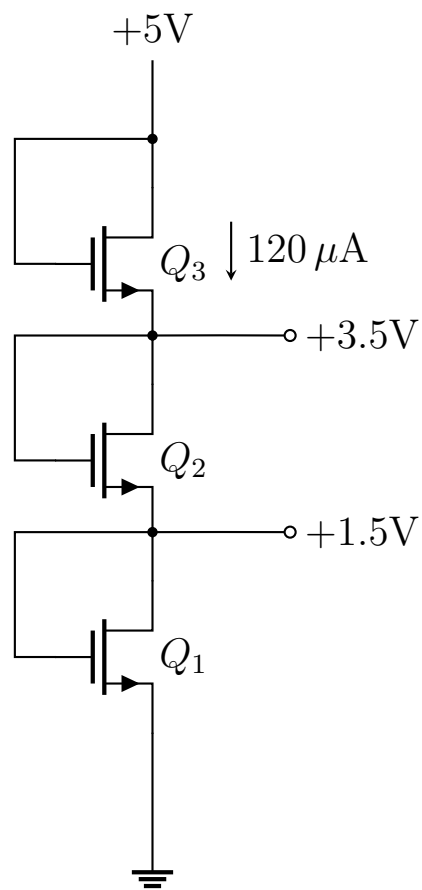
- Q10.** For the circuit shown, find the value of  $V_0$ . Both NMOS transistors have  $V_t = 1\text{ V}$ ,  $\lambda = 0$ , and  $\mu_n C_{ox}(W/L) = 1\text{ mA/V}^2$ . Supply voltage is  $5\text{ V}$ .



(16 marks)

[EEE 263 | L-2/T-1 | 2013–14]

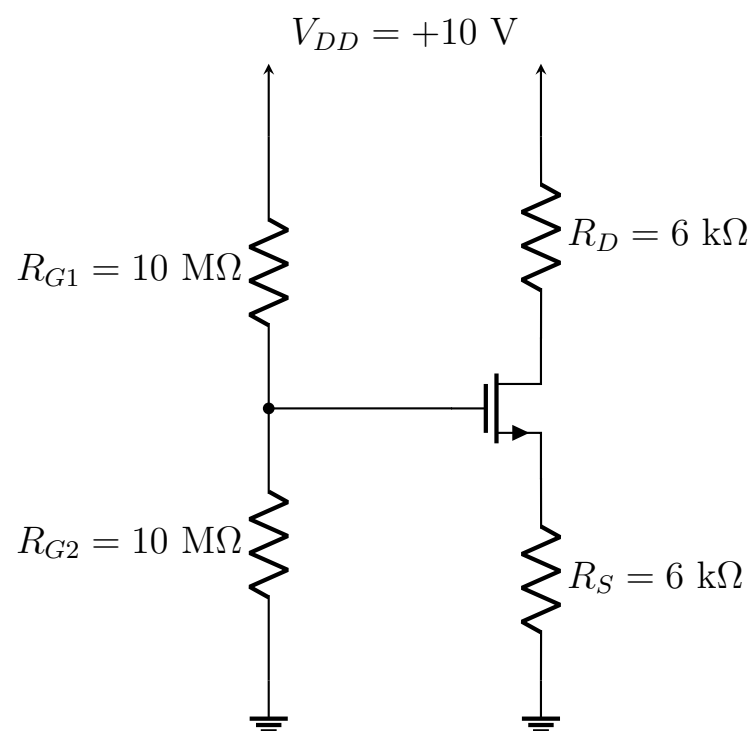
- Q11.** The NMOS transistors in the circuit have  $V_t = 1\text{ V}$ ,  $\mu_n C_{ox} = 120\text{ }\mu\text{A/V}^2$ , and  $L_1 = L_2 = L_3 = 1\text{ }\mu\text{m}$ . Find the required values of gate width for each of Q1, Q2, and Q3 to obtain the voltage and current values indicated in the figure.



(14 marks)

[EEE 263 | L-2/T-1 | 2012–13]

**Q12.** Determine all node voltages and drain current of the circuit. Assume  $V_t = 1\text{ V}$ ,  $K'_n W/L = 1\text{ mA/V}^2$ ,  $\lambda = 0$ ,  $V_{DD} = +10\text{ V}$ ,  $R_D = 6\text{ k}\Omega$ .

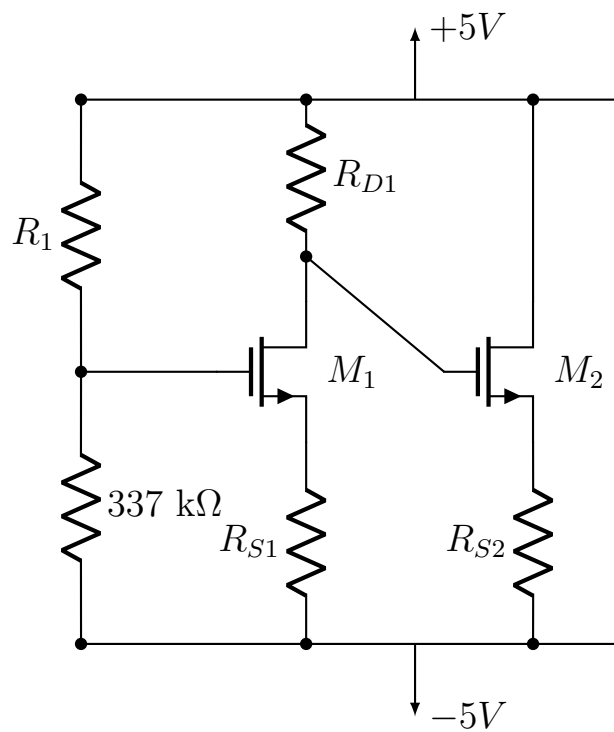


(20 marks)

[EEE 263 | L-2/T-1 | 2011–12]

### Multi-stage MOSFET Circuit Design

**Q1.** For the circuit shown with transistor parameters  $K_{n1} = 500\text{ }\mu\text{A/V}^2$ ,  $K_{n2} = 200\text{ }\mu\text{A/V}^2$ ,  $V_{T1} = V_{T2} = 1.2\text{ V}$ , design the circuit such that  $I_{D1} = 0.1\text{ mA}$ ,  $I_{D2} = 0.3\text{ mA}$ ,  $V_{DS1} = V_{DS2} = 5\text{ V}$ .



(20 marks)

[EEE 263 | L-2/T-1 | 2016–17]

### MOSFET Transconductance

**Q1.** Derive the condition of small signal operation considering a simple CS nMOS amplifier circuit. Find the expression of MOSFET transconductance  $g_m$  from there as well. (8 marks) [EEE 263 | L-2/T-1 | 2023–24]

**Q2.** How can the MOSFET transconductance  $g_m$  be varied with the design parameters  $W/L$  ratio and overdrive voltage  $V_{ov}$ ? Explain. (10 marks) [EEE 263 | L-2/T-1 | 2014–15]

### Voltage Transfer Characteristics

**Q1.** Draw the voltage transfer characteristic of a MOSFET. In which part of this graph is a MOSFET used as an amplifier, and why? What are the benefits and drawbacks of increasing bias voltage? (19 marks) [EEE 263 | L-2/T-1 | 2012–13]

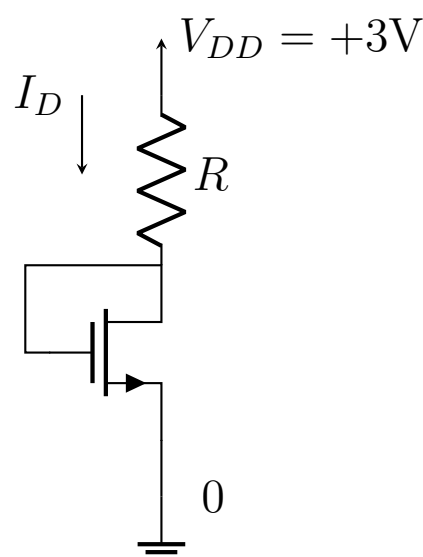
### MOSFET as Small-Signal Amplifier

**Q1.** Briefly explain the operation of a MOSFET as a small-signal amplifier and deduce its VTC from the  $I_D$ - $V_{DS}$  graph. (15 marks) [EEE 263 | L-2/T-1 | 2017–18]

### Biasing Stability

**Q1.** What is the reason for biasing a MOSFET? Write a short note on the biasing technique of a discrete MOSFET using a fixed voltage at the gate and a resistance  $R_s$  in the source. (11 marks) [EEE 263 | L-2/T-1 | 2017–18]

**Q2.** Design the circuit shown to obtain a current  $I_D = 80 \mu\text{A}$ . Given:  $V_t = 0.6 \text{ V}$ ,  $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$ ,  $L = 0.8 \mu\text{m}$ ,  $W = 4 \mu\text{m}$ , and  $V_A = 50 \text{ V}$ .



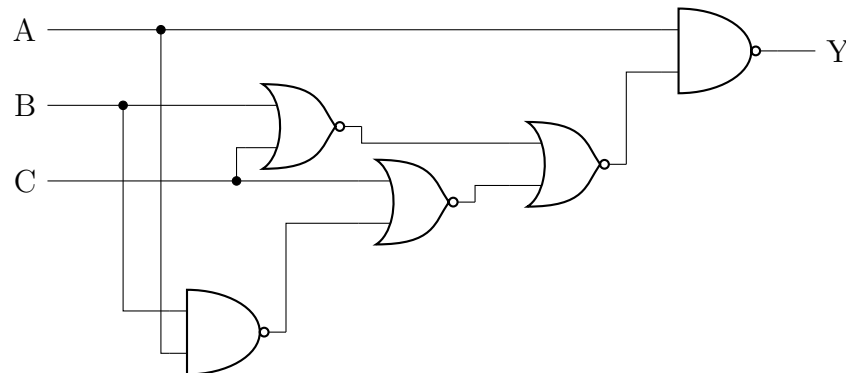
(16 marks)

[EEE 263 | L-2/T-1 | 2013–14]

**Q3.** Explain graphically the benefit of using a resistance at the MOSFET source terminal. How does it counteract the problem of unpredictable variation in process parameters such as  $V_t$ ,  $C_{ox}$ ,  $\mu_n$ ? **(20 marks)** [EEE 263 | L-2/T-1 | 2012–13]

## CMOS Logic Implementation

**Q1.** The diagram shows a Boolean function  $Y$  being implemented as a combinational circuit. Design a CMOS transistor-level implementation of the Boolean function using the minimum number of transistors.

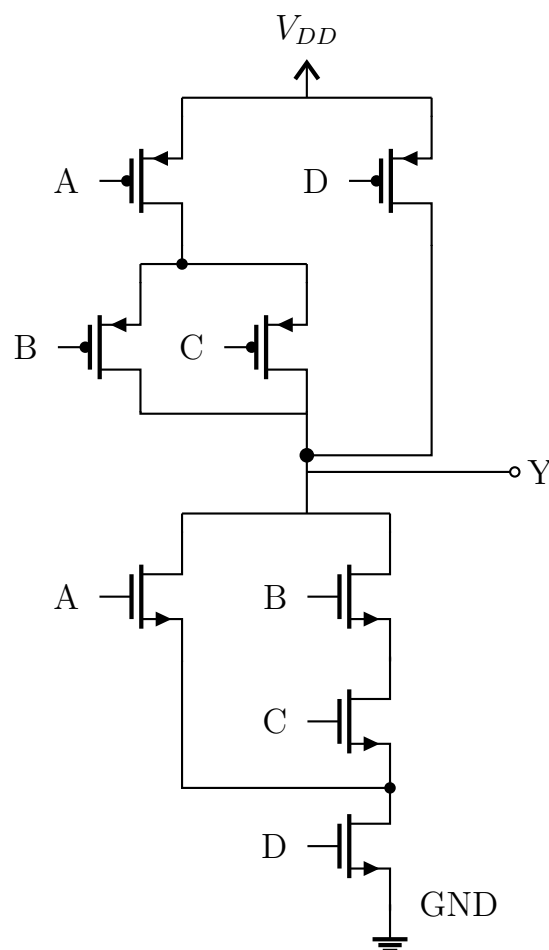


(20 marks)

[EEE 263 | L-2/T-1 | 2020–21]

## CMOS Sizing and Delay

**Q1.** For the logic circuit shown, choose transistor widths to achieve equal rise and fall resistance as a unit inverter. Assume  $\mu_n = 3\mu_p$ . What is the minimum and maximum propagation delay in the circuit?

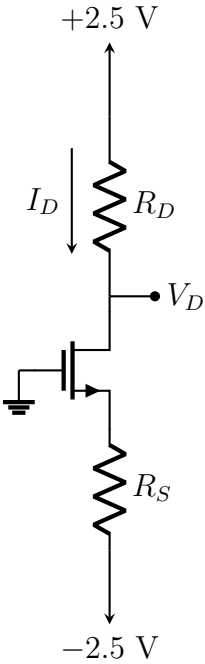


(26 marks)

[EEE 263 | L-2/T-1 | 2020–21]

## NMOS Circuit Design

**Q1.** Design the circuit shown so that the transistor operates at  $I_D = 0.3 \text{ mA}$  and  $V_D = 0.4 \text{ V}$ . The NMOS transistor has a threshold voltage of  $1 \text{ V}$ , aspect ratio of  $40$ , and a process transconductance parameter of  $60 \mu\text{A/V}^2$ . Neglect channel-length modulation.



(23 marks)

[EEE 263 | L-2/T-1 | 2020–21]

BUET · EEE 263 · Electronic Circuits

# Electronic Circuits

Part II — Wave Shaping Circuits

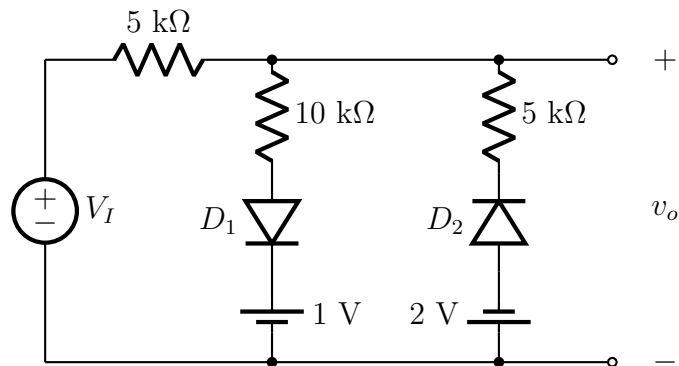
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Clipping, clamping, comparator, and switching circuits using diodes and BJTs

## S4 Clipping, Clamping & Switching Circuits

### Clipping Circuits — Waveform & Transfer Characteristics

- Q1.** For the circuit given, plot  $v_o$  vs time. The input  $v_1$  is a sine wave with 5 V amplitude and 1 kHz frequency. Assume 0.7 V voltage drop across diodes in forward bias.



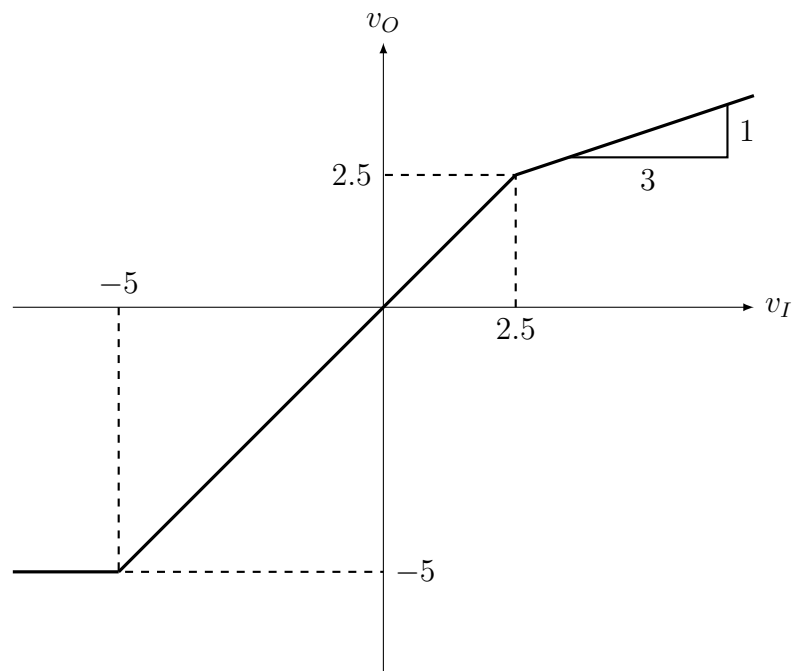
(14 marks)

[EEE 263 | L-2/T-1 | 2022–23]

- Q2.** Design a limiter circuit using only diodes and 10 kΩ resistors to provide an output signal limited to the range  $\pm 2.1$  V. Assume each diode has a 0.7 V drop when conducting. (15 marks)

[EEE 263 | L-2/T-1 | 2021–22]

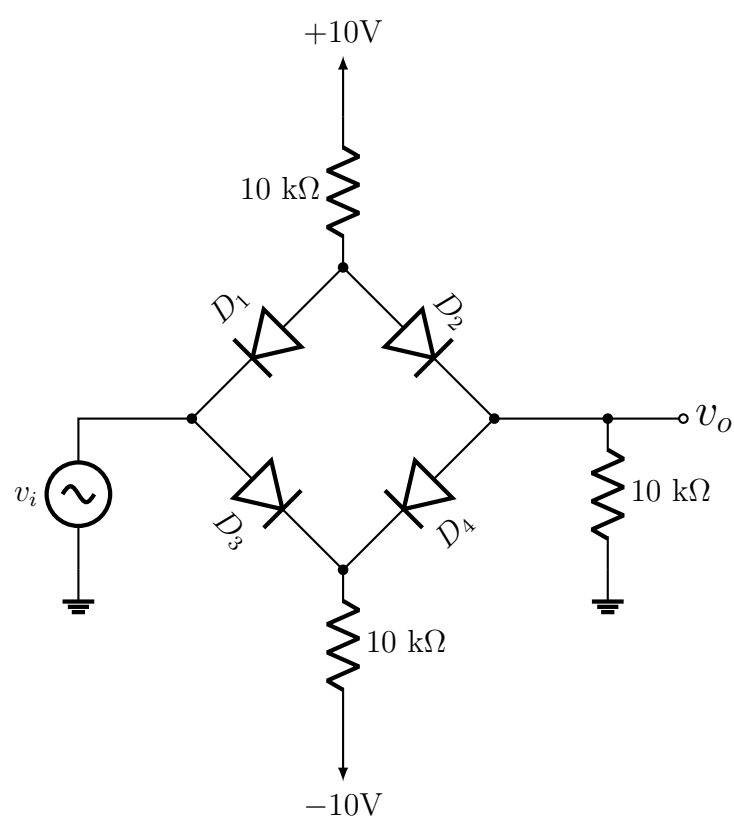
- Q3.** Design a parallel-based clipper that will yield the voltage transfer function shown in the figure. Assume 0.7 V as diode cut-in voltage.



(18 marks)

[EEE 263 | L-2/T-1 | 2017–18]

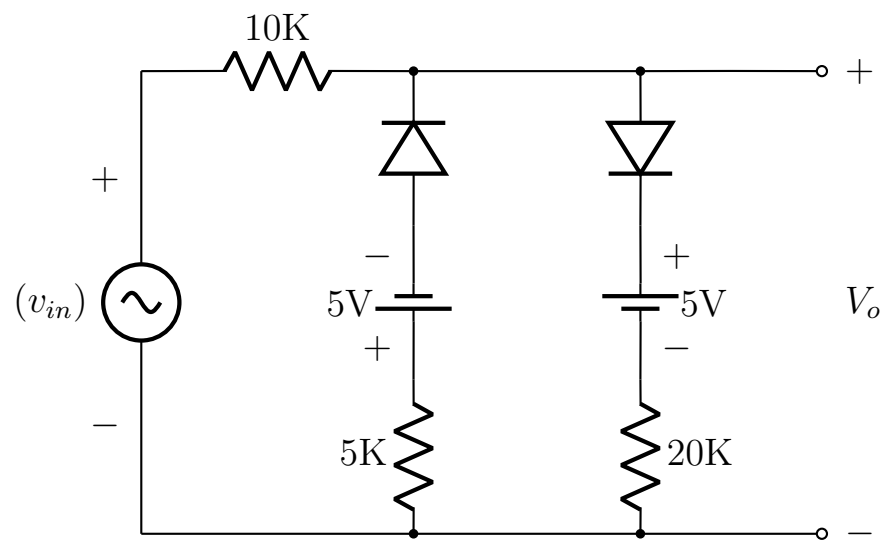
- Q4.** For the circuit shown in Figure , find and sketch the voltage transfer characteristics with proper labelling. Utilize the constant-voltage drop model (0.7 V) for each conduction diode.



(13 marks)

[EEE 263 | L-2/T-1 | 2016–17]

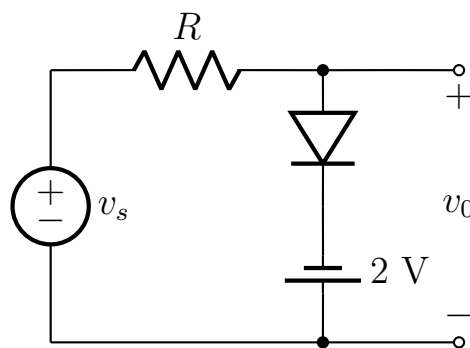
- Q5.** For the circuit shown, neatly draw the output ( $v_o$ ) waveform corresponding to the input ( $v_{in}$ ). Here,  $v_{in} = 10 \sin(200\pi t)$  and diodes are non-ideal.



(16 marks)

[EEE 263 | L-2/T-1 | 2015–16]

- Q6.** Sketch the output waveform and transfer characteristics for the circuit shown. Assume the diode is ideal and  $V_s$  is a sinusoid with 5 V peak amplitude and frequency 5 kHz.

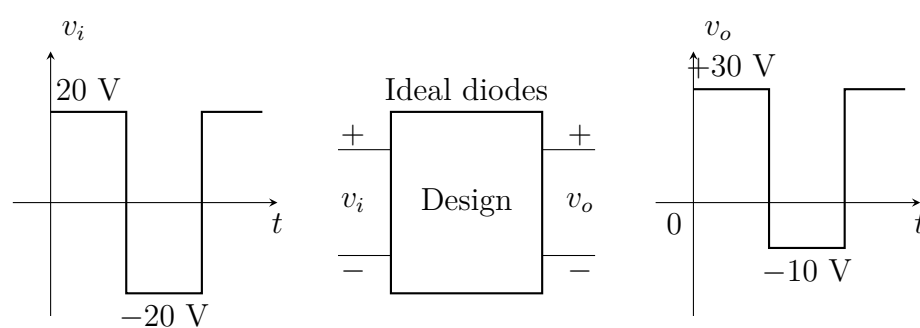


(15 marks)

[EEE 263 | L-2/T-1 | 2011–12]

### Clamping Circuits

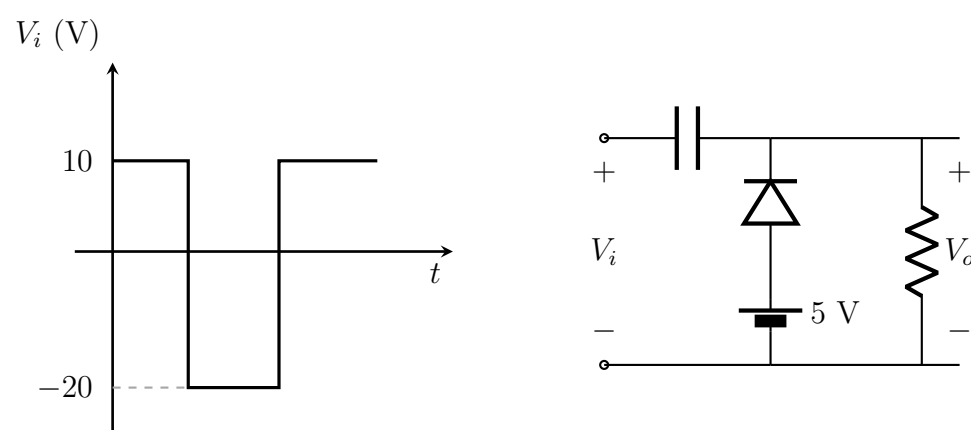
- Q1.** Design a clamper circuit to perform the function indicated in the figure (input  $v_i$ , output  $v_o$ ). Assume ideal diodes.



(12 marks)

[EEE 263 | L-2/T-1 | 2022–23]

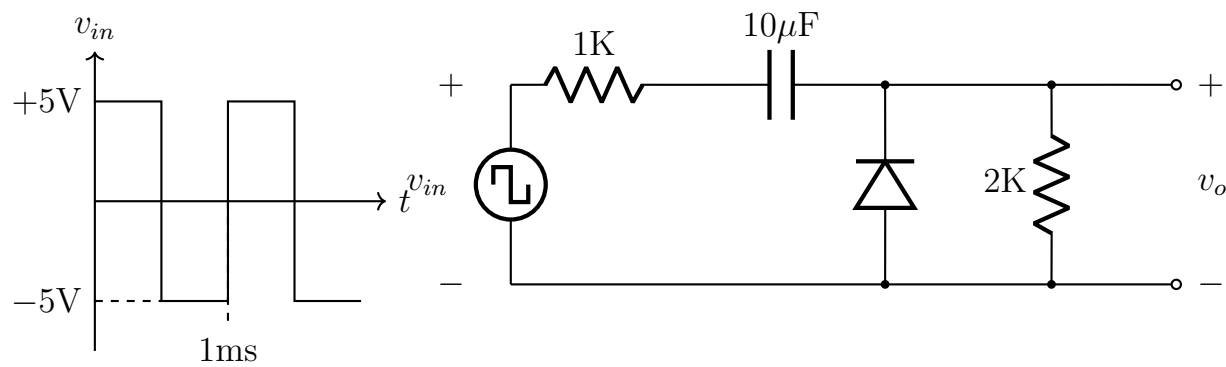
- Q2.** For the following circuit, draw the output waveshape for the given square input waveshape.



(16 marks)

[EEE 263 | L-2/T-1 | Jan 2021]

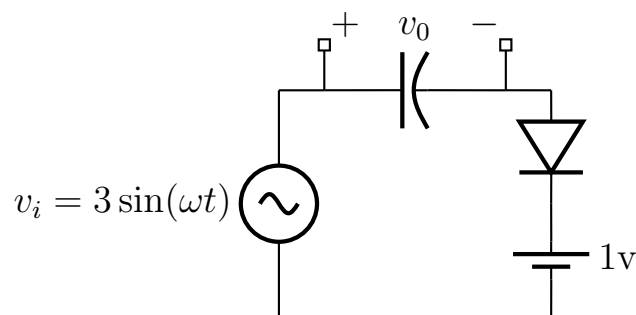
- Q3.** For the circuit shown, neatly draw the output waveform ( $v_o$ ) corresponding to the input waveform ( $v_{in}$ ) in steady state. Assume the diode is non-ideal.



(15 marks)

[EEE 263 | L-2/T-1 | 2015–16]

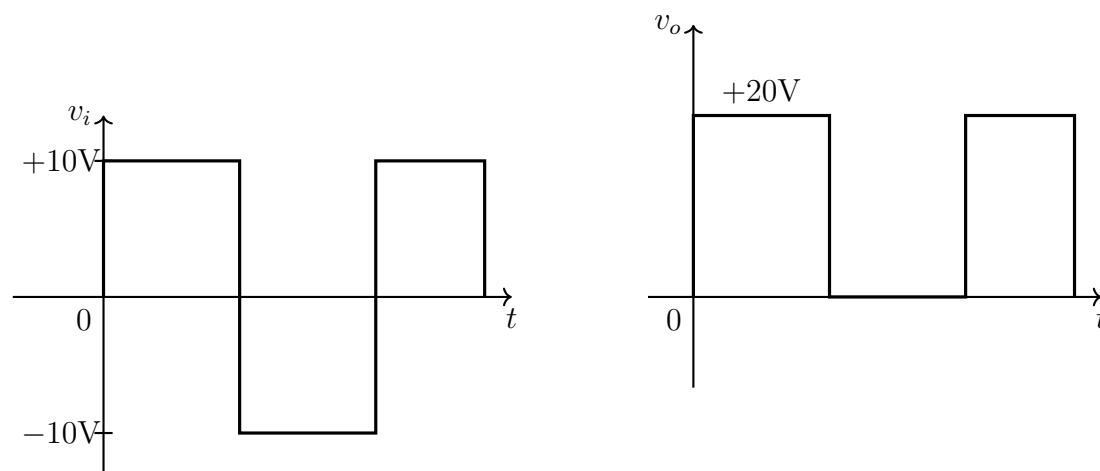
- Q4.** Draw the output wave shape ( $v_o$ ) and also the  $V_L$ - $V_o$  graph for the circuit shown. Assume a constant 0.7 V forward voltage drop across the diode.



(20 marks)

[EEE 263 | L-2/T-1 | 2013–14]

- Q5.** Design a clamper circuit to perform the function indicated in the figure using an ideal diode.



(12 marks)

[EEE 263 | L-2/T-1 | 2011–12]

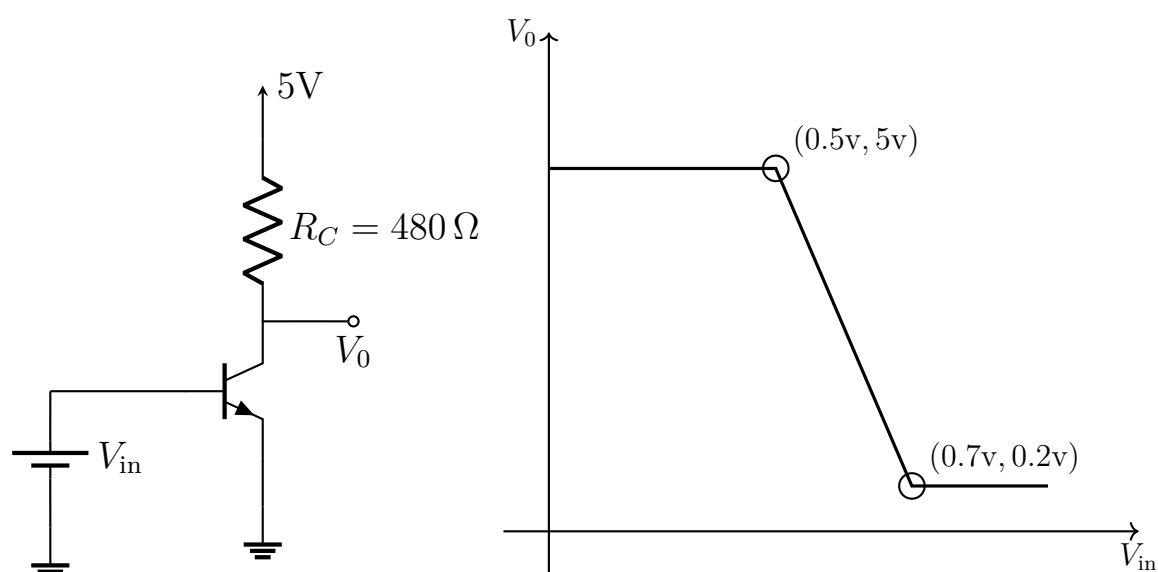
### Switching Circuits — Logic Gates using BJTs

- Q1.** Make a NAND and a NOR gate using BJTs. (10 marks)

[EEE 263 | L-2/T-1 | 2012–13]

### BJT Inverter Analysis

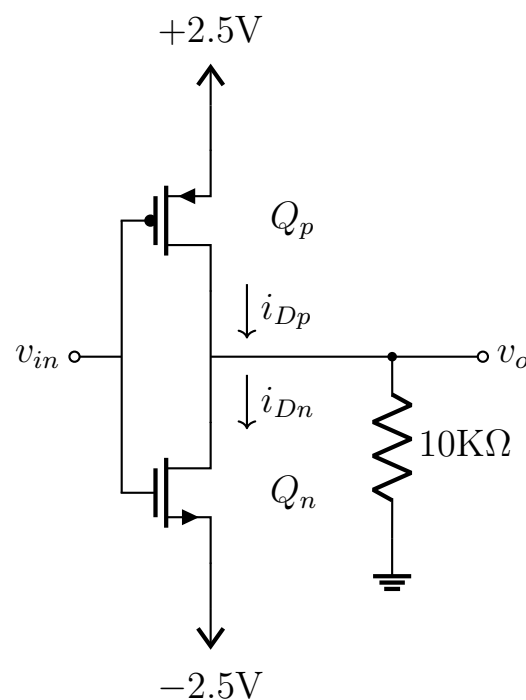
- Q1.** (a) Modify both the  $V_o$ - $V_{in}$  and  $I_c$ - $V_{in}$  graphs if supply voltage is decreased from 5 V to 2.5 V ( $R_c \approx 480 \Omega$ ). (10 marks) [EEE 263 | L-2/T-1 | 2013–14]



- (b) The figure shows an inverter gate built by an npn BJT and its  $V_o$ - $V_{in}$  characteristics. Copy the  $V_o$ - $V_{in}$  graph to your answer script exactly to scale and then draw the collector current ( $I_c$ ) vs.  $V_{in}$  over the same graph. Mention the co-ordinates of the bending points of the  $I_c$ - $V_{in}$  graph as shown for  $V_o$ - $V_{in}$ .  
 (16 marks) [EEE 263 | L-2/T-1 | 2012-13]
- (c) Modify both the  $V_o$ - $V_{in}$  and  $I_c$ - $V_{in}$  graphs if  $R_c$  is infinity. (10 marks) [EEE 263 | L-2/T-1 | 2012-13]
- (d) Modify both the  $V_o$ - $V_{in}$  and  $I_c$ - $V_{in}$  graphs if supply voltage is increased from 5 V to 10 V ( $R_c = 480 \Omega$ ). (10 marks) [EEE 263 | L-2/T-1 | 2012-13]
- (e) If the BJT in active mode shows a linear  $I_c$ - $V_{BE}$  behaviour instead of exponential, does voltage gain change with change in bias voltage  $V_{in}$ ? Does output voltage swing change with the change in  $V_{in}$ ? Explain why. (5 marks) [EEE 263 | L-2/T-1 | 2012-13]

### CMOS Inverter Analysis

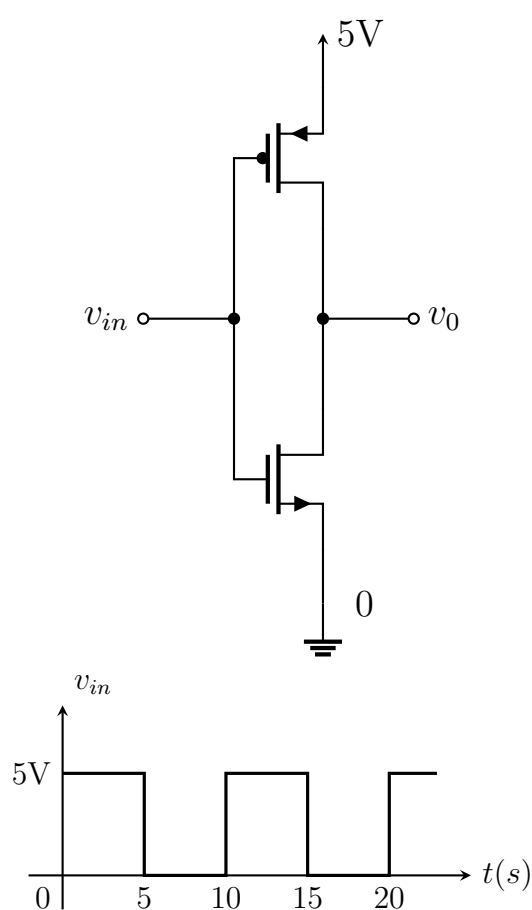
- Q1.** The inverter circuit with CMOS shown has  $k'_n(W_n/L_n) = k'_p(W_p/L_p) = 1 \text{ mA/V}^2$  and  $V'_{tn} = -V'_{tp} = 1 \text{ V}$ . Find (i)  $i_{DN}$ , (ii)  $i_{DP}$ , (iii)  $V_o$  for  $V_{in} = -2.5 \text{ V}$ .



(15 marks)

[EEE 263 | L-2/T-1 | 2015-16]

- Q2.** For the CMOS inverter, draw  $V_o$  for the given input  $V_{in}$  waveform.



(15 marks)

[EEE 263 | L-2/T-1 | 2013-14]

## Digital vs. Analog Systems

- Q1.** Why is digital processing of data advantageous over analog processing? **(6 marks)** [EEE 263 | L-2/T-1 | 2013–14]
- Q2.** Why do we need to miniaturize a transistor as much as possible? If you continue to miniaturize the transistor (switch), what problems may surface? **(5 marks)** [EEE 263 | L-2/T-1 | 2013–14]
- Q3.** Write a script in C language that takes a sinusoidal input and provides a full-wave rectified output. **(5 marks)** [EEE 263 | L-2/T-1 | 2013–14]
- Q4.** How does a computer program rectify the sinusoidal input mentioned above? Is there any diode inside the processor? If not, what makes that act of rectification possible? **(10 marks)** [EEE 263 | L-2/T-1 | 2013–14]

BUET · EEE 263 · Electronic Circuits

# Electronic Circuits

Part III — Amplifiers

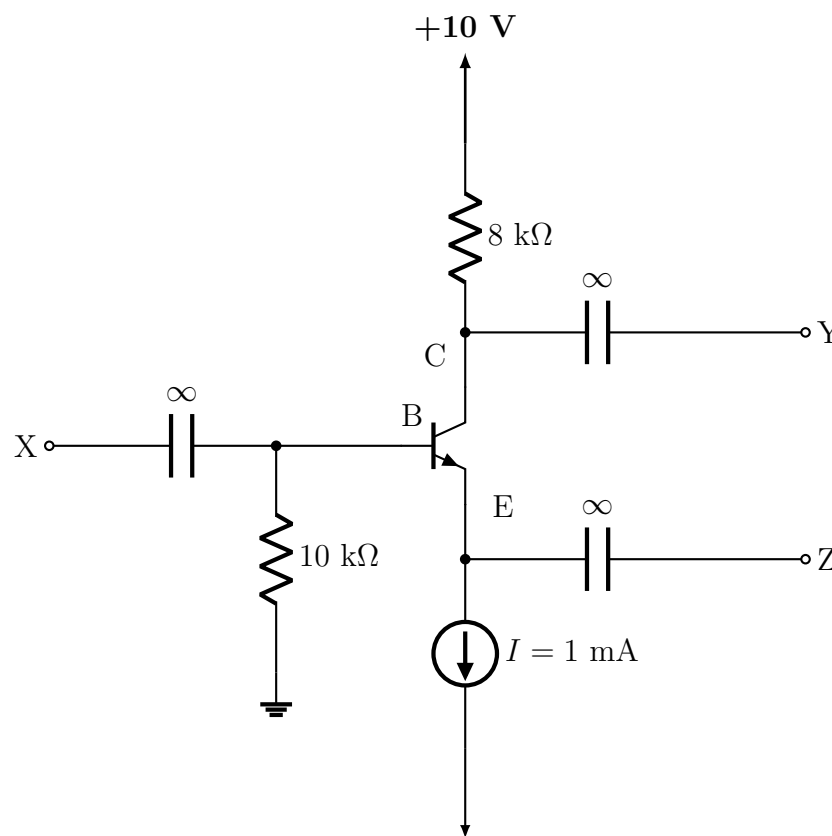
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BJT and MOSFET amplifier configurations, small-signal analysis, gain derivation

## S5 BJT Amplifiers

### Common Emitter / Trans-conductance Amplifier

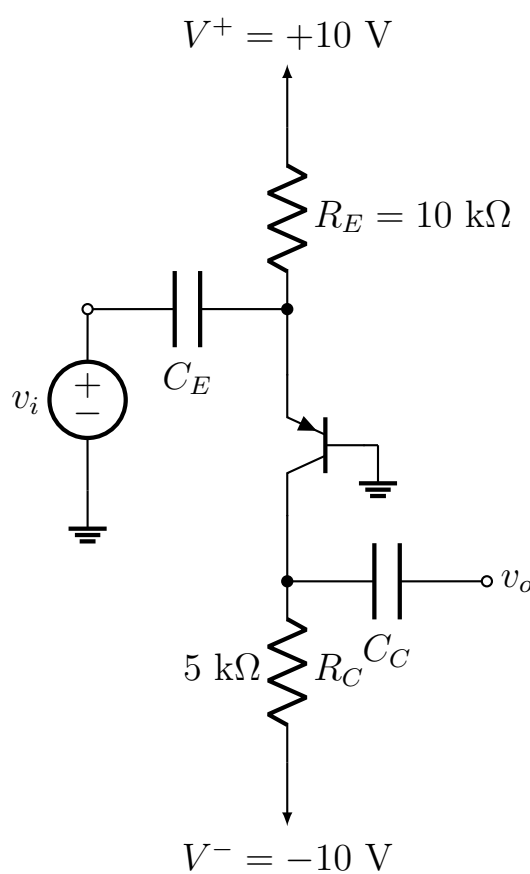
- Q1.** The transistor shown is biased with a constant current  $I = 1 \text{ mA}$  and has  $\beta = 100$  and  $V_A = 100 \text{ V}$ . (i) Find the DC voltages at the Base, Emitter, and Collector terminals. (ii) Find the small-signal equivalent circuit parameters  $r_o$ ,  $r_\pi$ ,  $g_m$  and draw the hybrid- $\pi$  model. (iii) If the Z terminal is connected to ground, X is connected to  $v_{sig}$  through  $R_{sig} = 2 \text{ k}\Omega$ , and Y is connected to  $R_L = 8 \text{ k}\Omega$ , find the overall voltage gain  $v_y/v_{sig}$ .



(20 marks)

[EEE 263 | L-2/T-1 | 2024–25]

- Q2.** Determine the voltage gain  $v_o/v_i$  of the transistor amplifier shown in the figure. Assume  $\beta = 100$ .



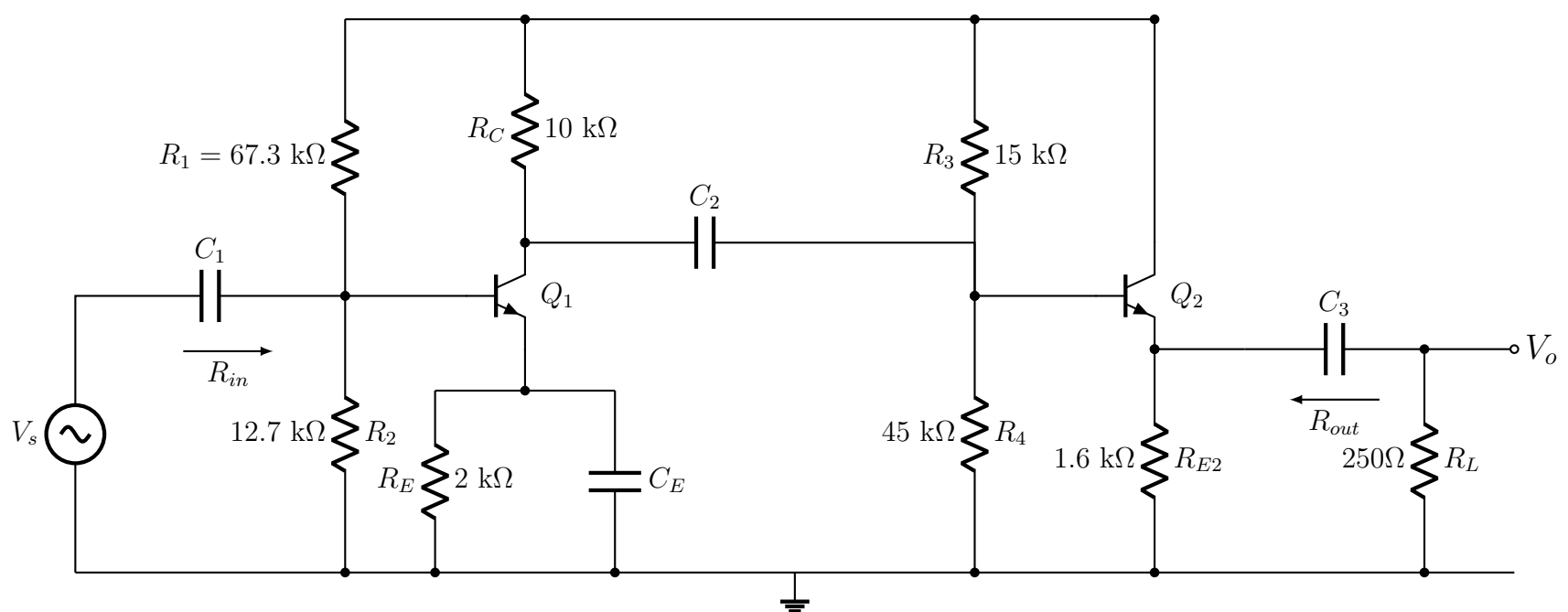
(26 marks)

[EEE 263 | L-2/T-1 | 2017–18]

- Q3.** A CE amplifier utilizes a BJT with  $\beta = 100$  and  $V_A = 100 \text{ V}$ , is biased at  $I_C = 1 \text{ mA}$  and has a collector resistance  $R_C = 5 \text{ k}\Omega$ . Find  $R_{in}$ ,  $R_o$ , and  $A_{vo}$ . If the amplifier is fed with a signal source having a resistance of  $5 \text{ k}\Omega$  and a load resistance  $R_L = 5 \text{ k}\Omega$  is connected to the output terminal, find the resulting  $A_v$  and  $G_v$ . If  $v_\pi$  is to be limited to  $5 \text{ mV}$ , what are the corresponding  $v_{sig}$  and  $v_o$  with the load connected? (20 marks)

[EEE 263 | L-2/T-1 | 2017–18]

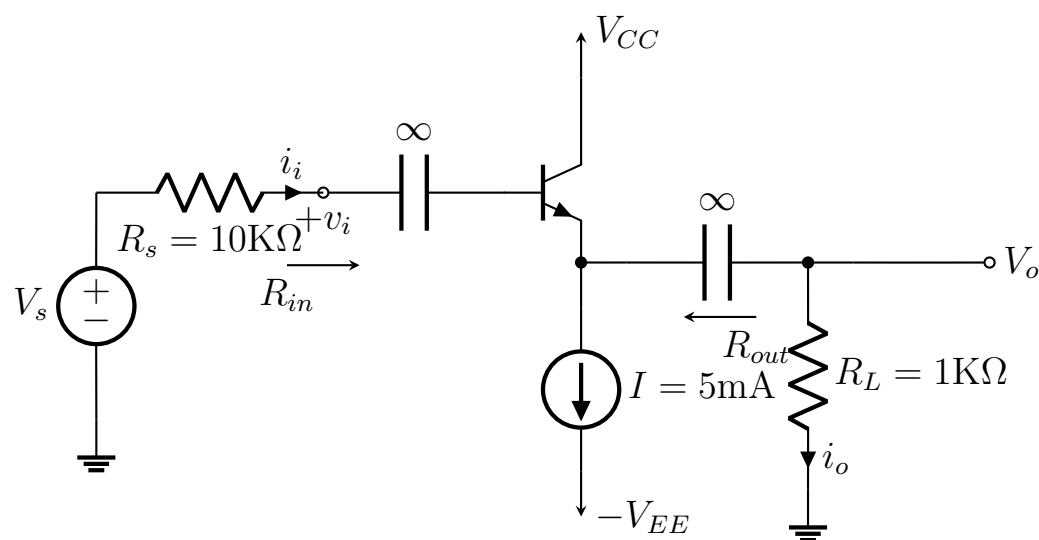
- Q4.** For the circuit shown, determine small-signal parameters  $g_m$ ,  $r_\pi$ , and  $r_e$  for both transistors, the overall small-signal voltage gain  $A_v = v_o/v_s$ , input resistance  $R_{in}$ , and output resistance  $R_{out}$ . Here  $\beta = 120$ .



(36 marks)

[EEE 263 | L-2/T-1 | 2016–17]

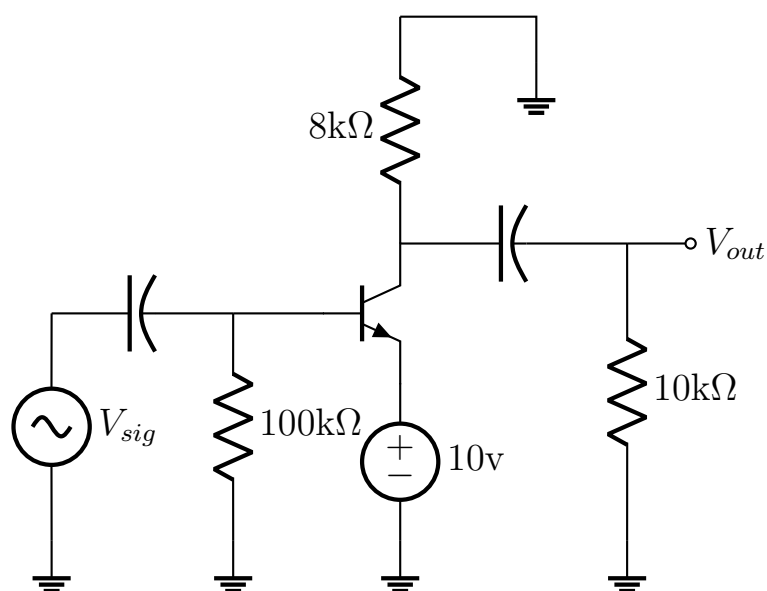
Q5. For the circuit shown, determine  $R_{in}$ ,  $R_{out}$ ,  $V_o/V_i$ , and  $i_o/i_i$ . The BJT is biased with  $I = 5 \text{ mA}$  and has  $\beta = 100$  and  $V_A = 100 \text{ V}$ .



(20 marks)

[EEE 263 | L-2/T-1 | 2014–15]

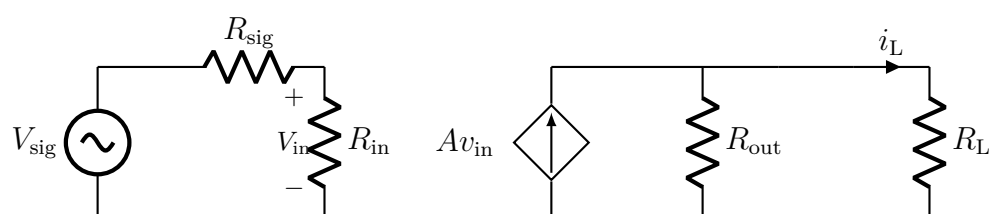
Q6. Draw the small signal circuit diagram of the CE circuit shown. Why is the circuit called a “Small Signal” circuit? Determine the values of input resistance  $R_{in}$ , output resistance  $R_{out}$ , and voltage gain  $A_v$ . If  $v_{sig} \approx 2 \text{ mV}$ , find the value of  $V_{out}$ .



(36 marks)

[EEE 263 | L-2/T-1 | 2013–14]

Q7. The figure shows the basic structure of a trans-conductance amplifier to be implemented by a BJT. Given  $AV_{in} = I_c R_{out}$  along with  $R_{sig} \approx 0$  and the load ( $R_L$ ) can be much greater than  $R_{out}$ ; which BJT configuration (common emitter, common base, or common collector) will you choose for best performance? Explain why. Comment on its gain and loading effect.

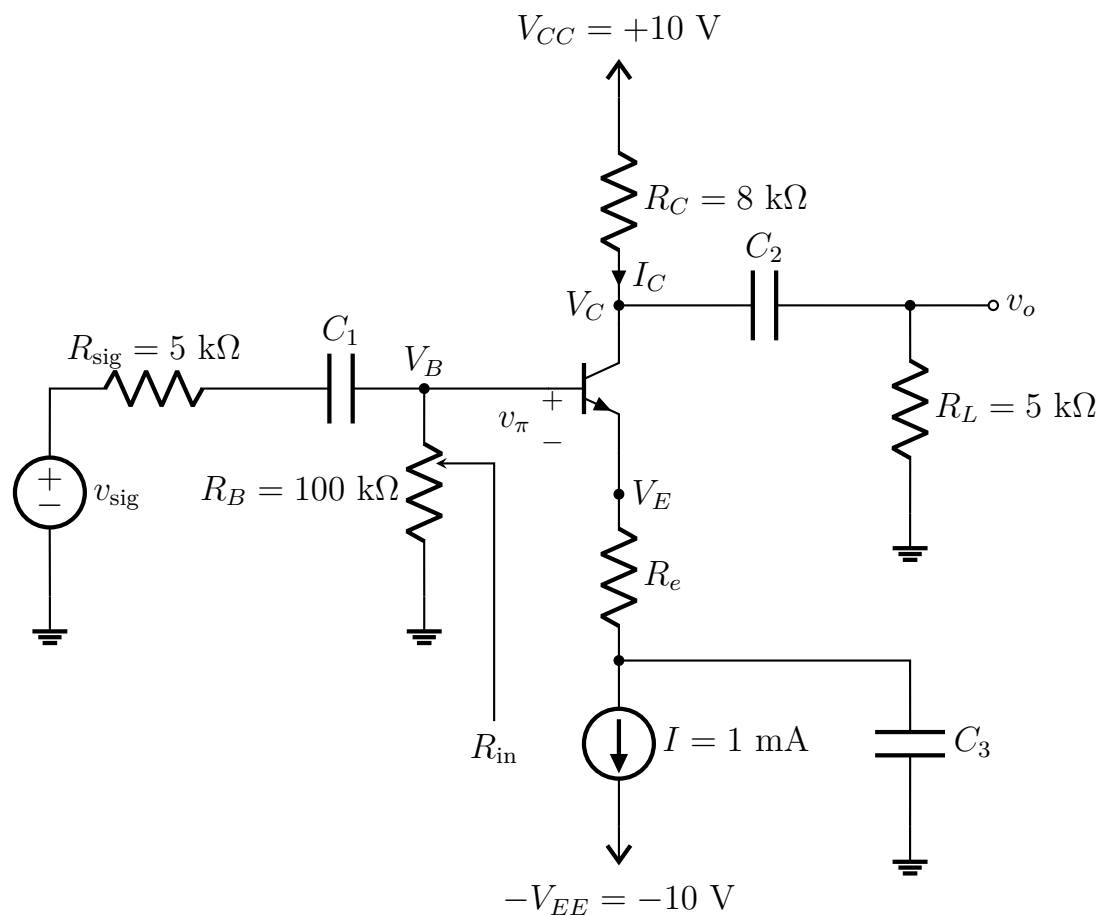


(20 marks)

[EEE 263 | L-2/T-1 | 2012–13]

Q8. Consider the emitter degenerated CE circuit of Fig. for Q. No. 8(b).

- Find the DC current  $I_C$  and DC voltage  $V_C$ ,  $V_B$  and  $V_E$ . Assume  $\beta = 100$ ,  $V_{BE} = 0.7$  V.
- What are the allowable signal swings at the collector in both directions?
- Determine the values of the BJT small signal model parameter:  $r_\pi$ ,  $r_e$ ,  $g_m$  at this bias point.
- Draw the complete small signal equivalent circuit of the amplifier using suitable model of BJT. Neglect  $r_o$ .
- Find the value of  $R_e$  that results in  $R_{in}$  equal to four times the source resistance  $R_{sig}$ .
- For this value of  $R_e$ , find  $A_{vo}$ ,  $R_{out}$ ,  $A_v$ ,  $G_v$  and  $A_{is}$  of the amplifier.

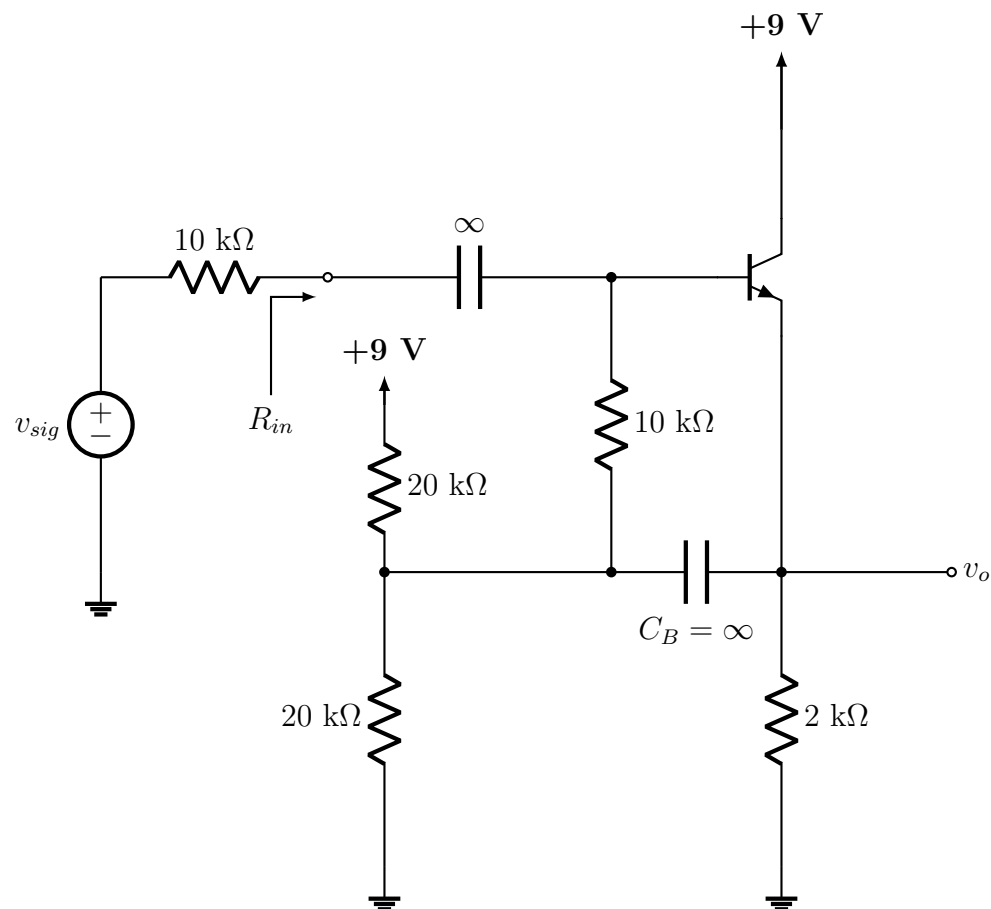


(36 marks)

[EEE 263 | L-2/T-1 | 2011–12]

### Common Base Amplifier

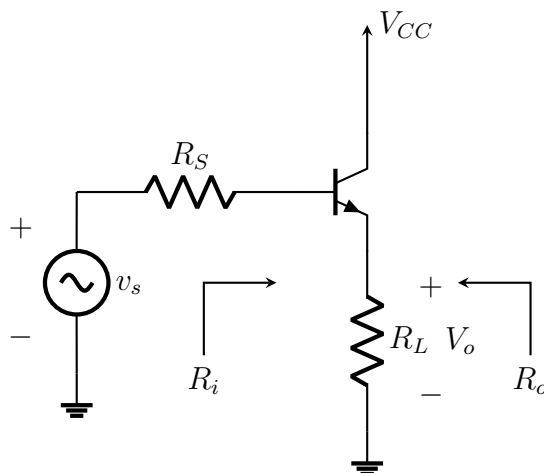
- Q1. For a Common Base Amplifier, using the appropriate small-signal equivalent model, derive the expressions of input resistance, output resistance, open-circuit voltage gain, and overall voltage gain. (14 marks) [EEE 263 | L-2/T-1 | 2022–23]
- Q1. For the bootstrapped follower circuit shown: (i) Find the DC emitter current and  $g_m$ ,  $r_e$ ,  $r_\pi$ . Assume  $\beta = 100$ . (ii) Replace the BJT with its T model (neglecting  $r_o$ ) and determine  $R_{in}$  and the voltage gain  $v_o/v_{sig}$ . (iii) Repeat (ii) for the case when capacitor  $C_B$  is open-circuited. Compare the results with (ii) and comment on the advantage of bootstrapping.



(15 marks)

[EEE 263 | L-2/T-1 | 2024–25]

- Q1.** For a common collector amplifier, derive the expressions for  $R_i$ ,  $R_o$ ,  $A_v$ , and  $A_i$  using the T model with early effect for the transistor. Where is the common collector amplifier used? (26 marks) [EEE 263 | L-2/T-1 | 2020–21]
- Q2.** For the common collector circuit shown, derive the expressions for  $A_v$ ,  $A_i$ ,  $R_i$ , and  $R_o$  using the T model for the transistor.



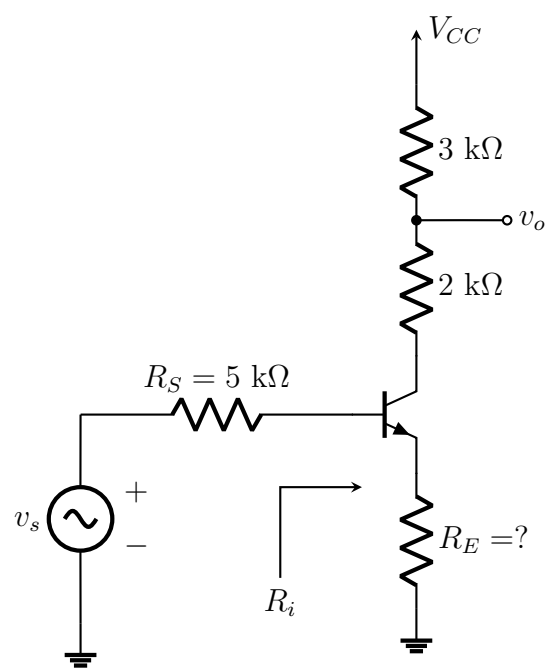
(23.67 marks)

[EEE 263 | L-2/T-1 | 2018–19]

- Q3.** The small-signal voltage gain  $A_v$  of a BJT CE amplifier is  $-I_C R_C / V_T$  without the Early effect. Show that if the Early effect is considered using  $i_C = I_S e^{v_{BE}/V_T} (1 + v_{CE}/V_A)$ , the gain becomes  $A_v = -\frac{I_C R_C / V_T}{1 + I_C R_C / (V_A + V_{CE})} = -\frac{(V_{CC} - V_{CE}) / V_T}{1 + (V_{CC} - V_{CE}) / (V_A + V_{CE})}$ . For  $V_{CC} = 5$  V,  $V_{CE} = 2.5$  V,  $V_A = 100$  V, calculate the gain with and without the Early effect. (20 marks) [EEE 263 | L-2/T-1 | 2024–25]
- Q4.** Derive the expression of  $R_i$ ,  $R_o$ ,  $A_v$ , and  $G_v$  for a common-emitter amplifier with emitter resistance  $R_e$  using the T model. (26 marks) [EEE 263 | L-2/T-1 | 2021–22]
- Q5.** For a BJT common-emitter amplifier with resistance in the emitter, derive the expressions for input impedance, output impedance, voltage gain, and current gain. What are the advantages and disadvantages of adding resistance in the emitter? (30 marks) [EEE 263 | L-2/T-1 | Jan–2021]

### Biasing and Input Impedance

- Q1.** For the circuit shown, find the value of  $R_E$  so that  $R_i$  is five times the value of  $R_s$ . Then using these values find  $R_i$ ,  $R_o$ ,  $A_v$ , and  $A_i$  for the circuit. Neglect early effect. Given:  $\beta = 100$  and  $r_e$  (T-model) =  $40 \Omega$ .

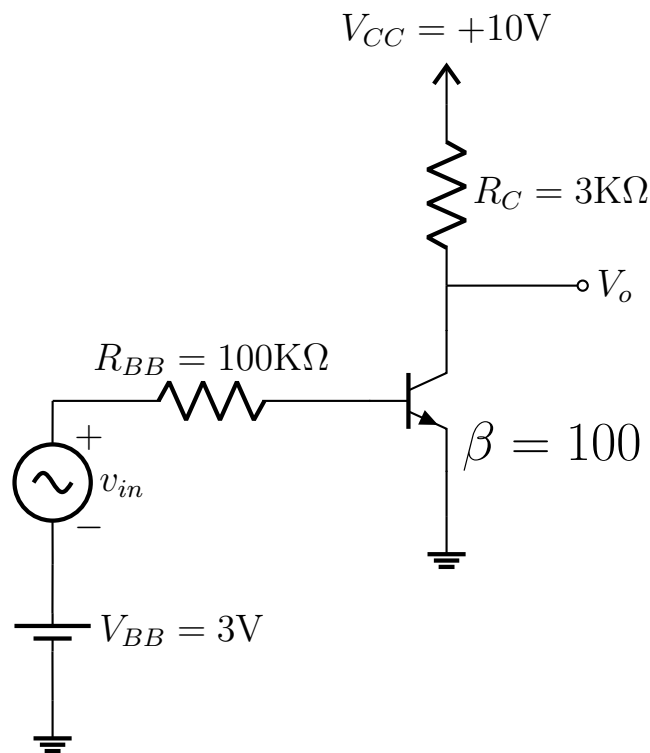


(26.67 marks)

[EEE 263 | L-2/T-1 | 2018–19]

BJT Input-Output Characteristics

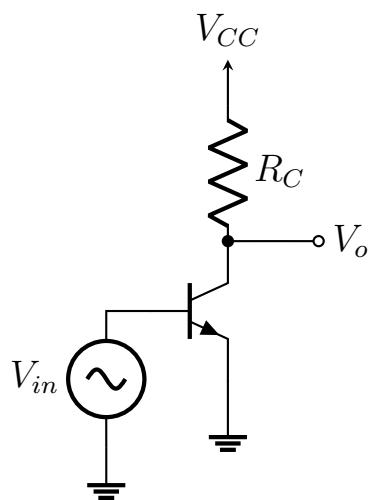
Q1. For the circuit shown, calculate (i) DC bias points, (ii) small-signal voltage gain, (iii) maximum allowable input AC voltage.

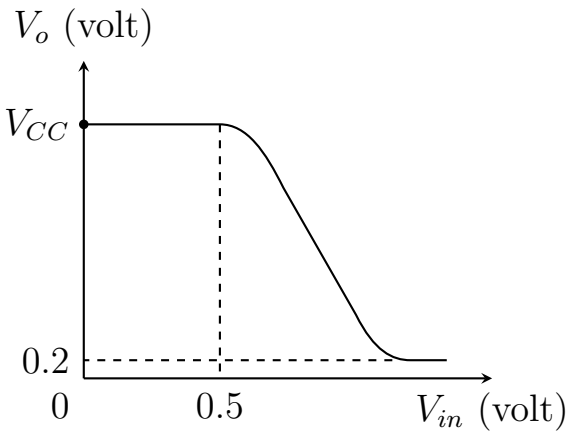


(26 marks)

[EEE 263 | L-2/T-1 | 2015–16]

Q2. The input-output characteristics of a BJT circuit are given. What will happen (i) if  $V_{cc}$  is increased, (ii) if  $R_c$  is increased? Explain with the characteristic curve.





(10 marks)

[EEE 263 | L-2/T-1 | 2014–15]

BUET · EEE 263 · Electronic Circuits

# Electronic Circuits

S6 — MOSFET Amplifiers

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Common-source configuration, small-signal model, gain derivation

## S6 MOSFET Amplifiers

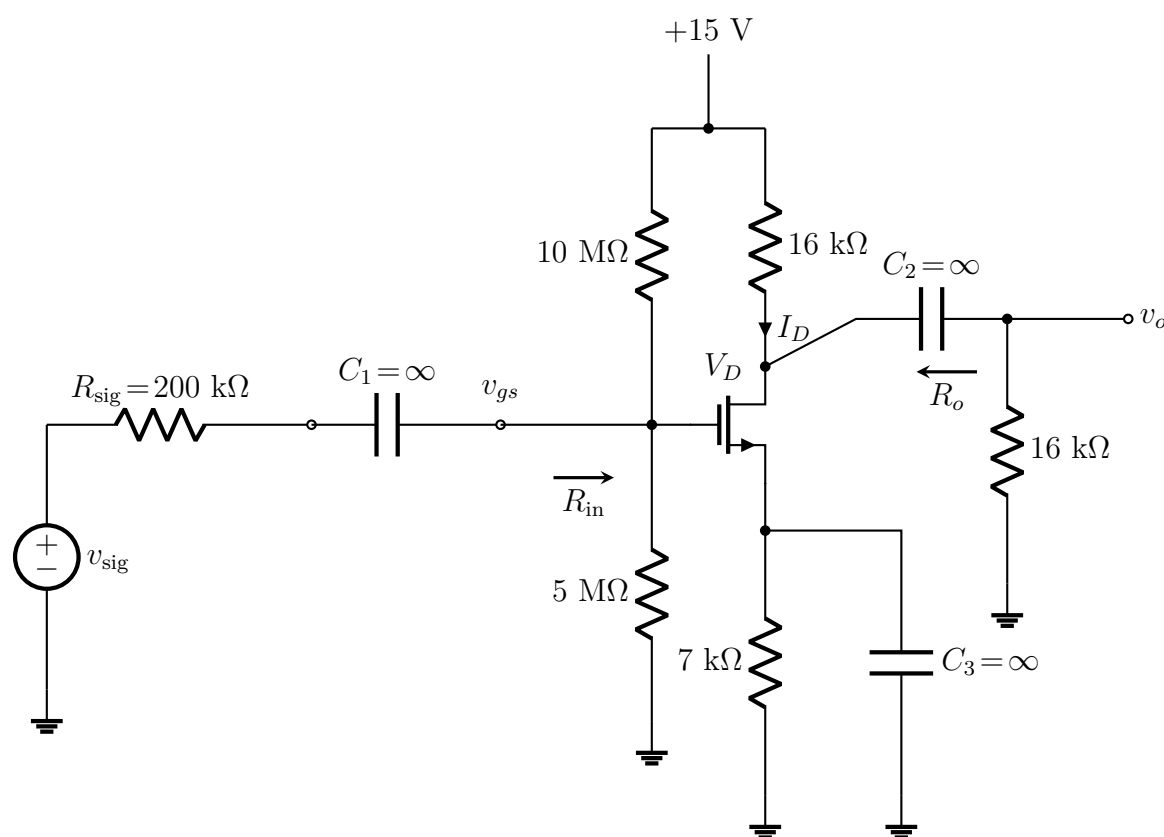
### Common Gate Amplifier

- Q1.** Derive the expression for input resistance, output resistance, and overall voltage gain of a common-gate (CG) amplifier. **(26 marks)**  
[EEE 263 | L-2/T-1 | 2013–14]

### Common Source Amplifier

- Q1.** For the Common Source amplifier circuit shown, the transistor has  $V_t = 1\text{ V}$ ,  $\mu_n C_{ox}(W/L) = 4\text{ mA/V}^2$ , and  $V_A = 100\text{ V}$ . All capacitors behave as open circuits under DC and short circuits under AC.

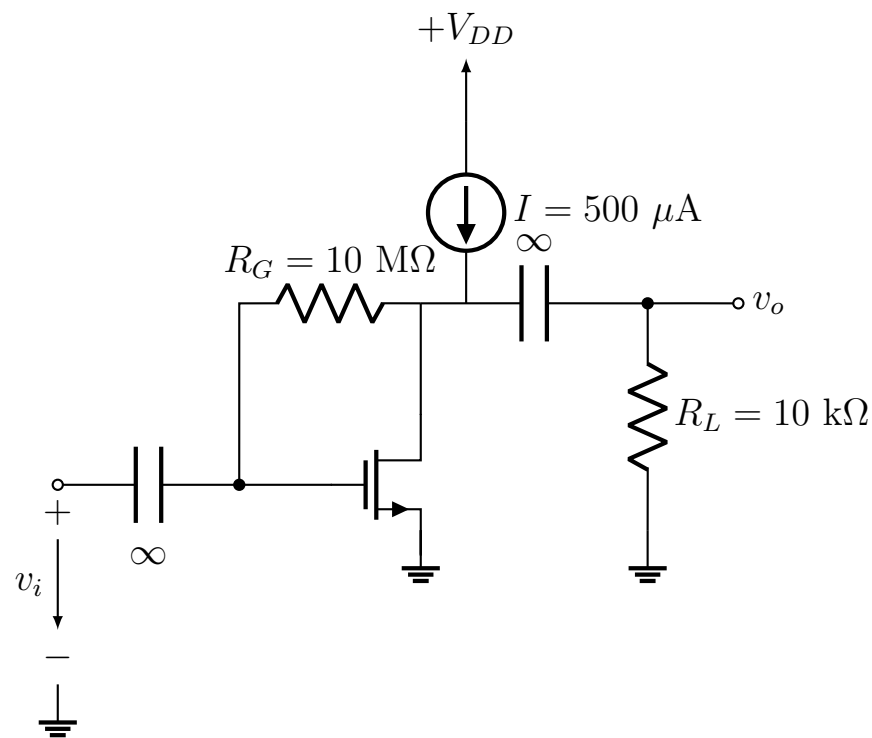
- Find the DC bias voltage  $V_D$  and drain current  $I_D$  (ignore Early effect for bias).
- Find  $g_m$ ,  $r_o$  and draw the small-signal equivalent circuit.
- Determine  $R_{in}$ ,  $R_o$ , and the overall voltage gain  $v_o/v_{sig}$ .
- Calculate the maximum peak value of  $v_{sig}$  such that  $v_{gs} \leq 0.1 V_{OV}$  (small-signal approximation valid).
- Suggest a circuit modification that will double the maximum  $v_{sig}$  found in (iv) without changing the DC bias point.



**(35 marks)**

[EEE 263 | L-2/T-1 | 2024–25]

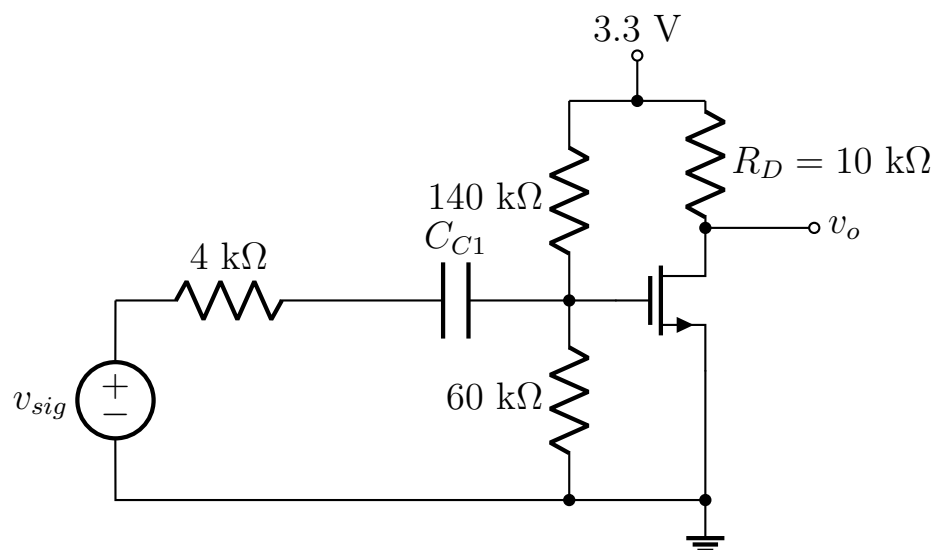
- Q2.** Explain why adding a source resistor  $R_S$  to a Common-Source MOSFET amplifier improves its bias stability compared to directly grounding the source terminal. Detail how  $R_S$  actively compensates for manufacturing variations in the threshold voltage  $V_t$  and transconductance parameter  $k_n$ . **(12 marks)**  
[EEE 263 | L-2/T-1 | 2024–25]
- Q3.** A CS amplifier is connected to a source  $v_{sig}$  with resistance  $R_{sig}$  and it utilizes a MOSFET with  $\mu_n C_{ox} = 400\mu\text{A/V}^2$ ,  $W/L = 10$ , and  $V_A = 10\text{ V}$ . It is biased at  $I_D = 0.2\text{ mA}$  and uses  $R_D = 6\text{ k}\Omega$ . Find  $R_{in}$ ,  $A_{vo}$ , and  $R_o$ . If a load resistance of  $10\text{ k}\Omega$  is connected to the output, what will be the overall voltage gain  $G_v$ ? If a  $0.2\text{-V}$  peak sine-wave signal is required at the output, what must the peak amplitude of  $v_{sig}$  be? **(15 marks)**  
[EEE 263 | L-2/T-1 | 2023–24]
- Q4.** In the circuit shown, the NMOS transistor has  $|V_t| = 0.9\text{ V}$  and  $V_A = 50\text{ V}$  and is operated with  $V_D = 2\text{ V}$ . What is the voltage gain  $v_o/v_i$ ? What do  $V_D$  and the gain become for  $I = 1\text{ mA}$ ?



(17 marks)

[EEE 263 | L-2/T-1 | 2022–23]

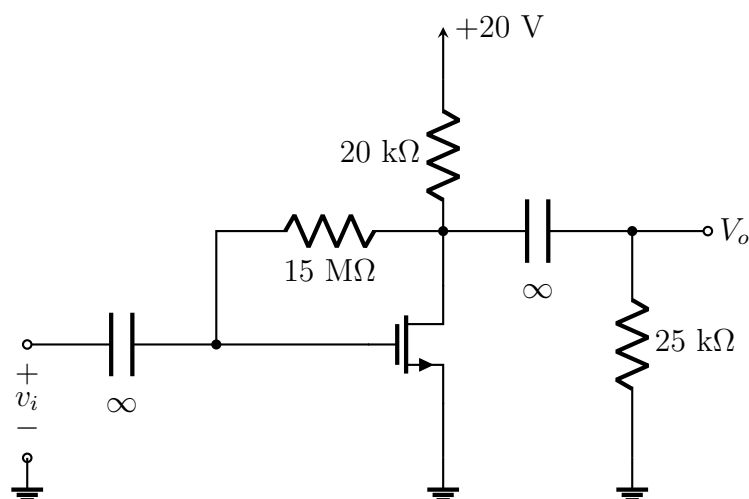
- Q5.** A CS amplifier utilizes a MOSFET biased at  $I_D = 0.25 \text{ mA}$  with  $V_{ov} = 0.25 \text{ V}$  and  $R_D = 20 \text{ k}\Omega$ . The amplifier is fed with a signal source having  $R_{sig} = 100 \text{ k}\Omega$ , and a  $20\text{-k}\Omega$  load is connected to the output. Find  $R_{in}$ ,  $A_{vo}$ ,  $R_o$ ,  $A_v$ , and  $G_v$ . (20 marks) [EEE 263 | L-2/T-1 | 2021–22]
- Q6.** Determine the overall small-signal voltage gain of the common source amplifier shown. The transistor parameters are:  $V_t = 0.4 \text{ V}$ ,  $k_n = 0.5 \text{ mA/V}^2$ ,  $\lambda = 0.02 \text{ V}^{-1}$ .



(26 marks)

[EEE 263 | L-2/T-1 | 2020–21]

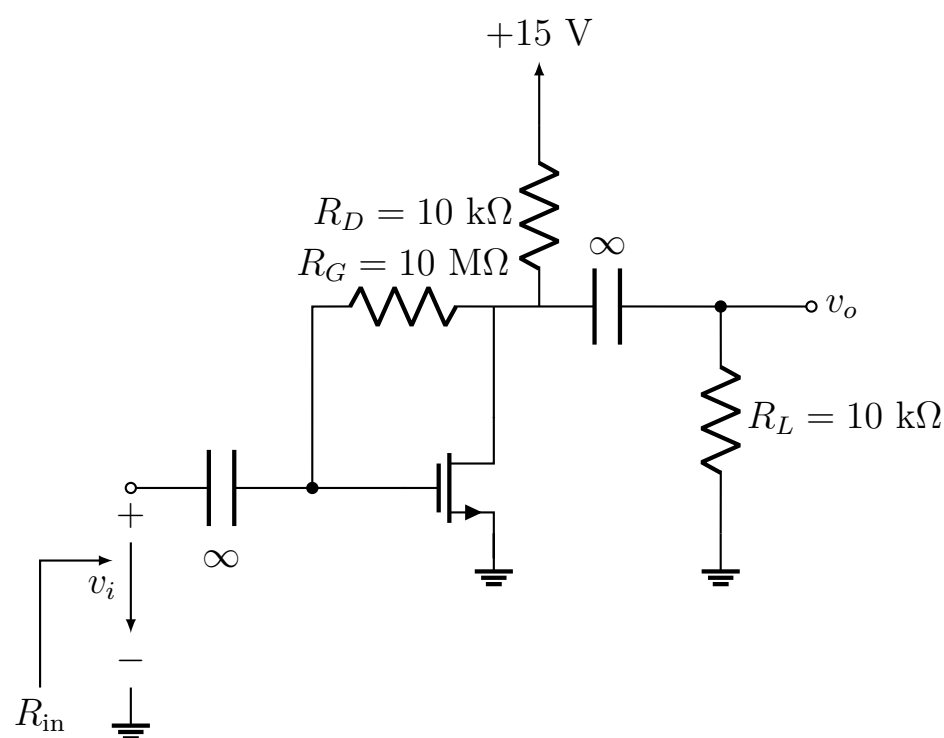
- Q7.** For the circuit shown, find the small-signal input impedance and voltage gain. Given:  $V_t = 1.2 \text{ V}$ ,  $k'_n(W/L) = 0.5 \text{ mA/V}^2$ , and  $V_A = 55 \text{ V}$ .



(30 marks)

[EEE 263 | L-2/T-1 | Jan–2021]

- Q8.** Determine the voltage gain  $v_o/v_i$  of the common-source MOSFET amplifier shown in the figure. The transistor has  $V_t = 1.5 \text{ V}$ ,  $k'_n(W/L) = 0.25 \text{ mA/V}^2$ , and  $V_A = 50 \text{ V}$ .



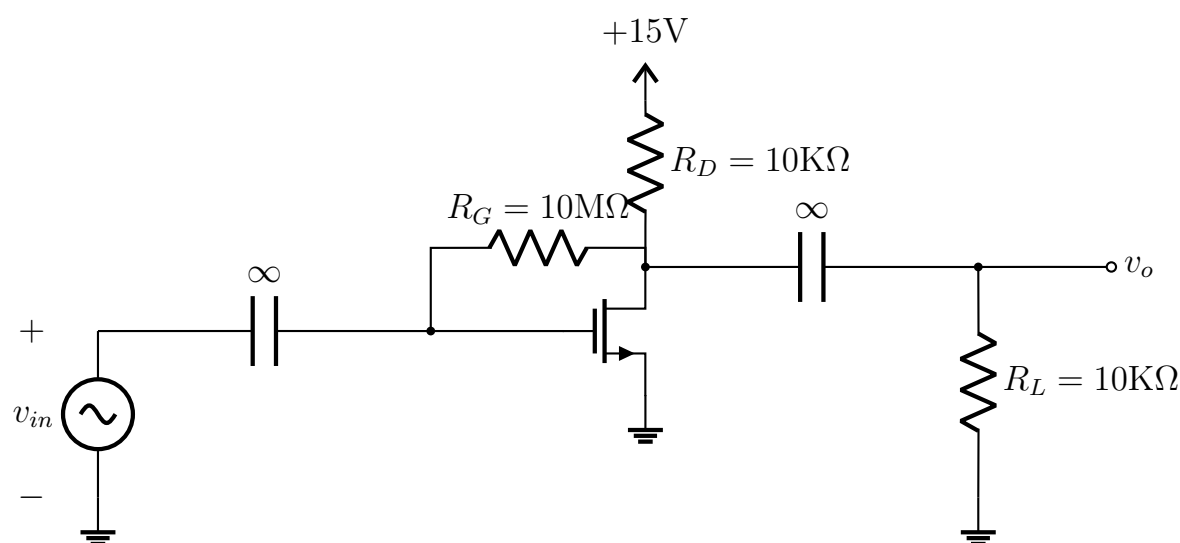
(20 marks)

[EEE 263 | L-2/T-1 | 2017–18]

Q9. For the circuit shown:

- Calculate DC bias point ( $I_D$  and  $V_D$ ).
- Draw the small-signal equivalent circuit.
- Calculate the small-signal voltage gain.
- Find the input resistance.
- Determine the largest allowable input signal voltage.

The transistor has  $V_t = 1.5$  V,  $k'_n(W/L) = 0.25$  mA/V<sup>2</sup>, and  $V_A = 50$  V.



(31 marks)

[EEE 263 | L-2/T-1 | 2015–16]

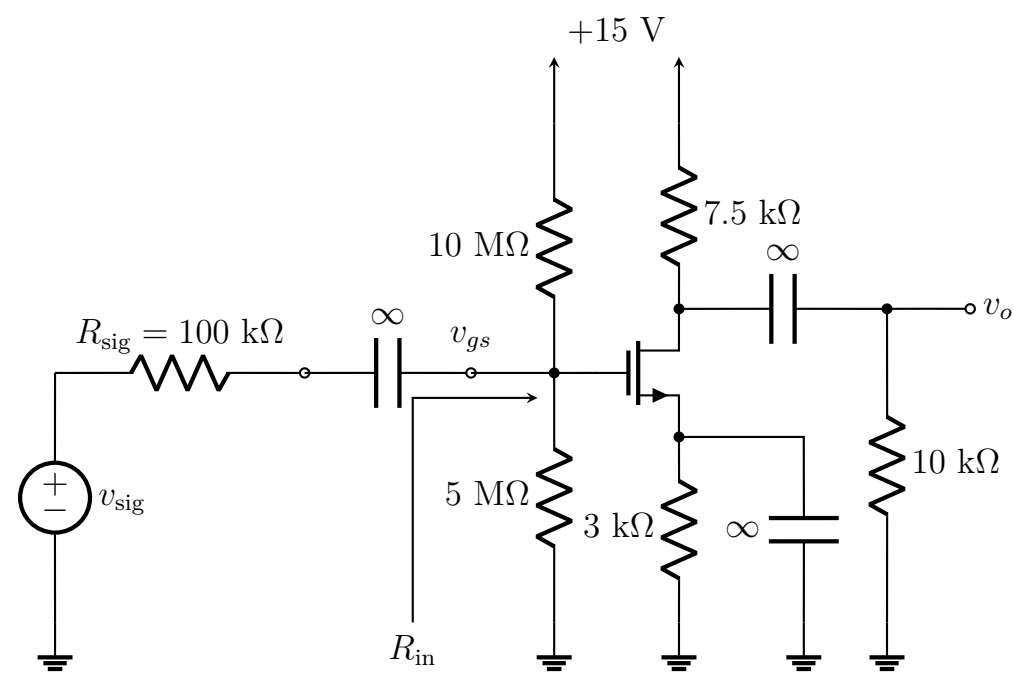
Q10. For the CS amplifier, derive the expressions for input resistance ( $R_{in}$ ), output resistance ( $R_{out}$ ), open-circuit voltage gain ( $A_{vo}$ ), and overall voltage gain ( $G_v$ ). (26 marks)

[EEE 263 | L-2/T-1 | 2012–13]

Q11. A common source amplifier circuit is shown in the figure.

- Prove that this circuit is operating in the saturation region.
- Draw the small-signal equivalent circuit.
- Find  $g_m$ ,  $R_{in}$ ,  $R_{out}$ ,  $A_v$ , and  $G_v$ .

Assume  $V_t = 1$  V,  $K'_n W/L = 2$  mA/V<sup>2</sup>,  $V_A = 100$  V.



(46 marks)

[EEE 263 | L-2/T-1 | 2011–12]

BUET · EEE 263 · Electronic Circuits

# Electronic Circuits

Part IV — Linear Integrated Circuits

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Op-amp ideal characteristics, mathematical operations, active filters, ADC

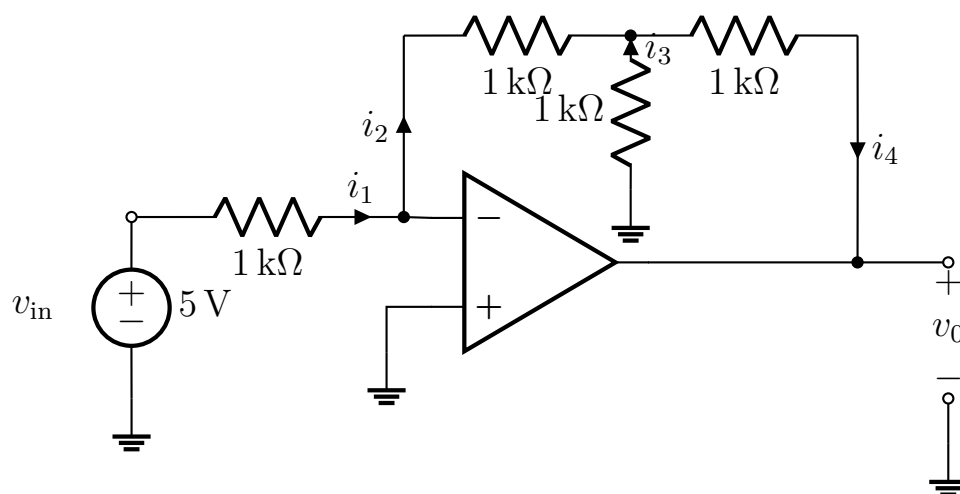
## S7 Operational Amplifiers (Op-Amps)

### Ideal Op-Amp Characteristics

- Q1.** What are the characteristics of an ideal op-amp? (5 marks) [EEE 263 | L-2/T-1 | 2013–14]
- Q2.** Explain graphically why in a closed-loop feedback circuit of an Op-Amp the differential voltage  $E_d$  is taken as zero. (12 marks) [EEE 263 | L-2/T-1 | 2012–13]
- Q3.** Derive the closed-loop gain of an inverting Op-Amp amplifier as a function of its open-loop gain, provided that the open-loop gain is not infinity. (13 marks) [EEE 263 | L-2/T-1 | 2012–13]
- Q4.** Write down the properties of an ideal Op-Amp. (10 marks) [EEE 263 | L-2/T-1 | 2011–12]

### Mathematical Operations

- Q1.** For the circuit shown in Figure below, find all the marked node voltages and branch currents:



(17 marks)

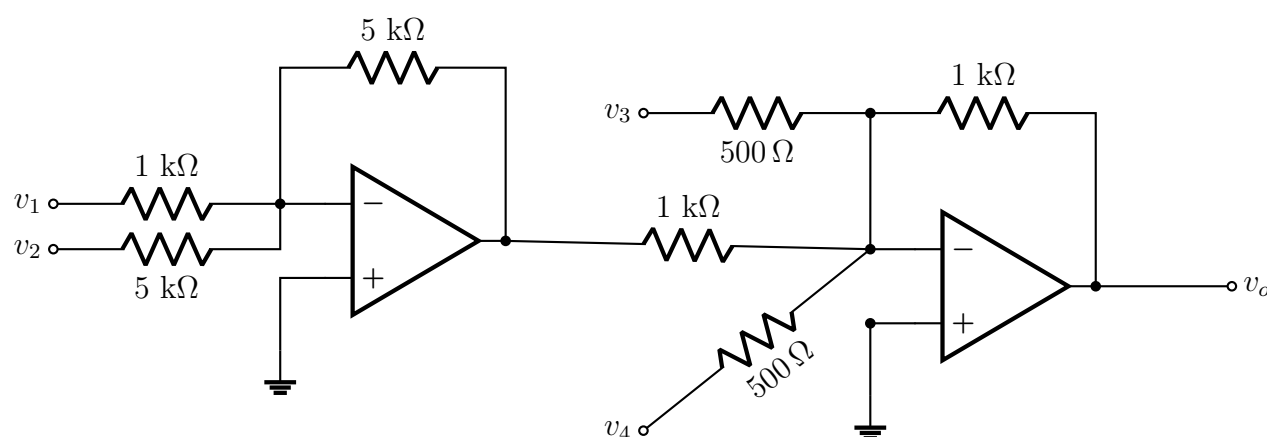
[EEE 263 | L-2/T-1 | 2023–24]

- Q2.** Suppose, you have three voltage sources:  $V_1 = \sin(40t)$ ,  $V_2 = \sin(100t)$ ,  $V_3 = 1V$ . Design a circuit using op-amps that will generate the following output:

$$V_{out} = 5 \sin(40t) - 4 \cos(100t) + 4$$

Draw the overall connection diagram with required values of elements. You may assume that the op-amps are biased using a 15V source. (18 marks) [EEE 263 | L-2/T-1 | 2023–24]

- Q3.** Find the expression of output voltage  $v_0$  in terms of input voltages and resistances in the circuit shown. Also calculate the value of output voltage at  $t = 1$  sec given that  $v_1 = 2V$ ,  $v_2 = 5V$ ,  $v_3 = 1V$ , and  $v_4 = 5 \sin(100\pi t)$ .



(20 marks)

[EEE 263 | L-2/T-1 | 2018–19]

- Q4.** For the circuit shown, find  $V_0$ . The MOSFET is in saturation with  $k'_n(W/L) = 1 \text{ mA/V}^2$  and  $V_t = 0.7V$ . The op-amp is ideal.

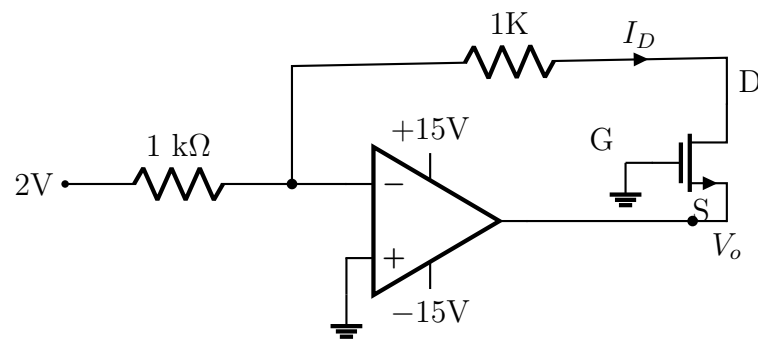
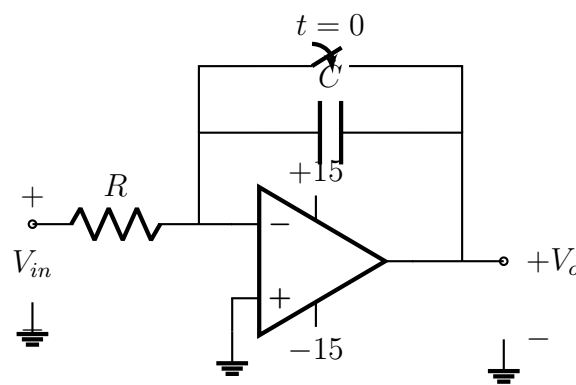
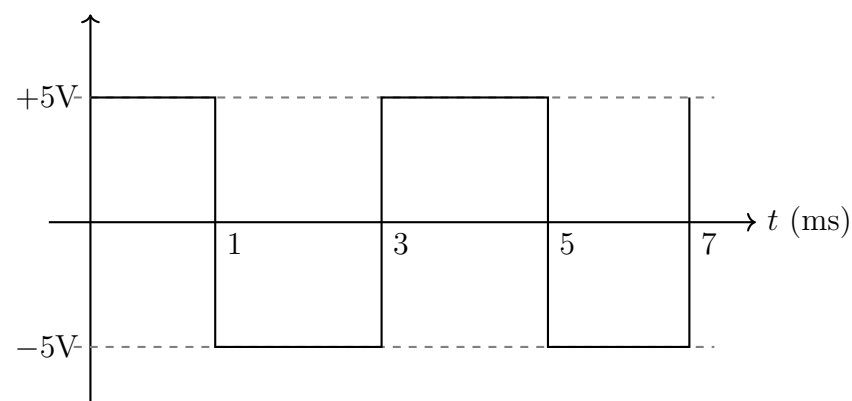


Fig for Q. 5(a)

(10 marks)

[EEE 263 | L-2/T-1 | 2017–18]

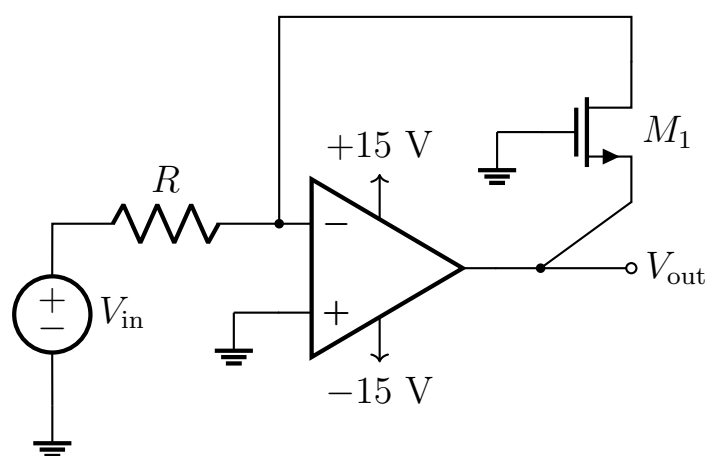
- Q5.** Consider the circuit and input voltage waveform shown in the figure. (i) If  $R = 10 \text{ k}\Omega$ ,  $C = 0.1 \mu\text{F}$ , and the Op-Amp is ideal, sketch the output voltage waveform to scale. (ii) If  $R = 10 \text{ k}\Omega$ , what value of  $C$  is required for the peak-to-peak output amplitude to be  $2 \text{ V}$ ?



(16 marks)

[EEE 263 | L-2/T-1 | 2017–18]

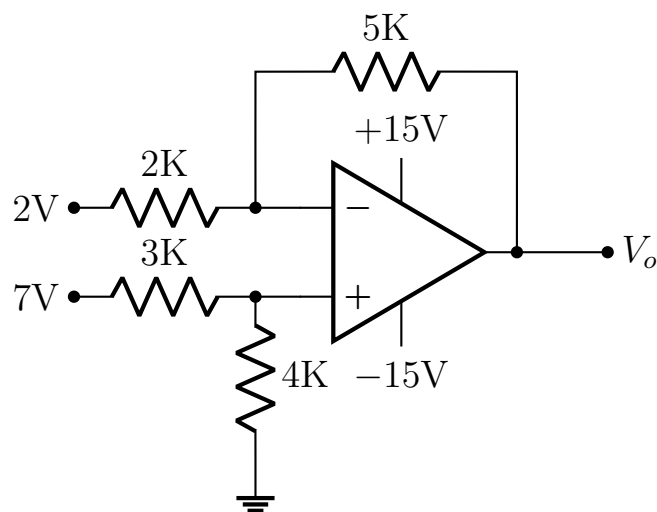
- Q6.** For the circuit shown, (i) determine output voltage  $V_{out}$  in terms of input voltage  $V_{in}$ ; (ii) using this circuit perform the following operation:  $V_{out} = 3V_{in}^2$ . Use additional Op-Amps and a DC voltage source if necessary. Specify values of all the parameters used. (Threshold voltage  $V_{TH} = 0.5 \text{ V}$  for the MOSFET M1.)



(20 marks)

[EEE 263 | L-2/T-1 | 2015–16]

- Q7.** What is the value of the output voltage  $V_o$  for the circuit shown?



(16.67 marks)

[EEE 263 | L-2/T-1 | 2014–15]

Q8. Design an Op-Amp circuit that performs the following mathematical operation:

$$v_{out} = 5v_1 - 2\frac{dv_2}{dt} + 6 \int v_3 dt$$

where  $v_1$ ,  $v_2$ , and  $v_3$  are three input signals and  $v_{out}$  is the overall output signal. (16 marks)

[EEE 263 | L-2/T-1 | 2012–13]

Q9. Propose a circuit using Op-Amp which will produce the following as output:

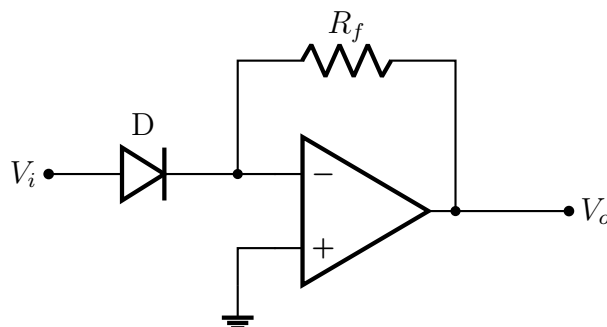
$$V_{out} = -4V_1 + 2V_2 + 7\frac{dV_1}{dt} - 8 \int V_2 dt$$

where  $V_1$  and  $V_2$  are two different input sources. (20 marks)

[EEE 263 | L-2/T-1 | 2011–12]

#### Logarithmic / Exponential Amplifier

Q1. For the circuit shown, deduce the equation for output voltage  $V_o$  and explain the function of this circuit.



(17 marks)

[EEE 263 | L-2/T-1 | Jan-2021]

Q2. Show that a proper connection of a diode, a resistor, and an op-amp can perform the exponential function. Mention the assumptions, if any. (Unspecified marks)

[EEE 263 | L-2/T-1 | 2013–14]

Q3. Draw the circuit diagram of an inverting logarithmic amplifier and derive the expression of its output voltage. (26 marks)

[EEE 263 | L-2/T-1 | 2011–12]

#### Difference Amplifier

Q1. Design a subtractor circuit using op-amps so that the output voltage is  $v_0 = v_1 - v_2$ . (15 marks)

[EEE 263 | L-2/T-1 | 2022–23]

Q2. Design a circuit using a single Op-Amp that performs the following operation:  $V_o = 3(V_1 - V_2)$  where  $V_1$  and  $V_2$  are input signals (AC or DC). Specify values of all resistors used in the design. (14 marks)

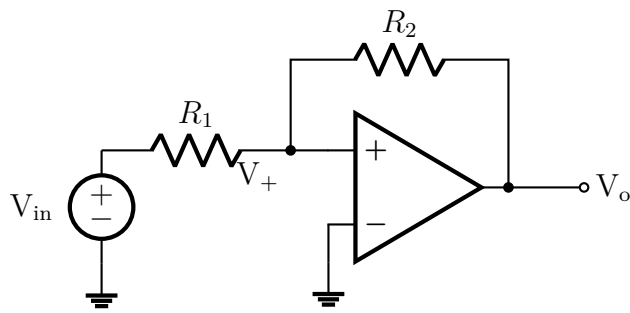
[EEE 263 | L-2/T-1 | 2015–16]

Q3. Implement  $z = 5V_1 - 3V_2$  using only one op-amp, where  $V_1$  and  $V_2$  are input signals. (15 marks)

[EEE 263 | L-2/T-1 | 2013–14]

#### Comparator and Schmitt Trigger

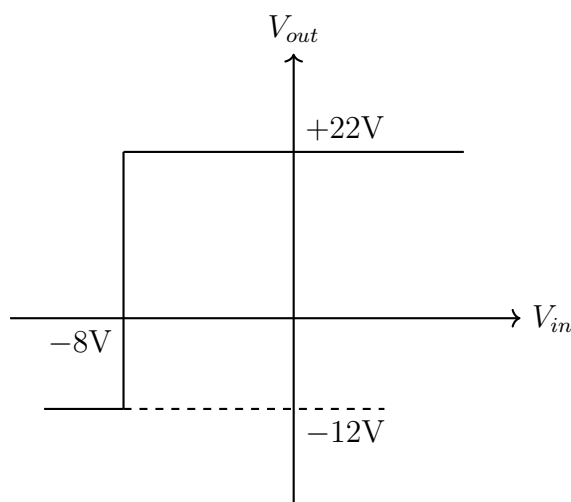
Q1. Write down the characteristics of an ideal op-amp. Draw the voltage transfer characteristics of the circuit shown in Figure below. Voltage transfer characteristics shows the variation in output voltage in response to variations in the input voltage.



(15 marks)

[EEE 263 | L-2/T-1 | 2023–24]

**Q2.** Determine the op-amp circuit that generates the transfer characteristics shown in the figure.



(11 marks)

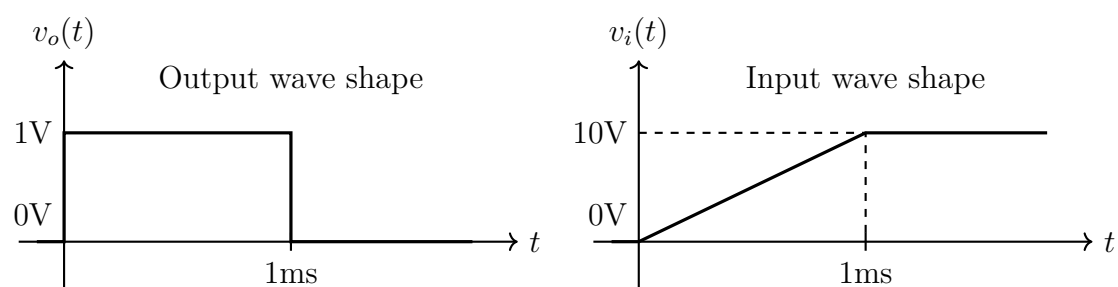
[EEE 263 | L-2/T-1 | 2016–17]

**Q3.** Design a Schmitt trigger that triggers from  $+16\text{ V}$  to  $-16\text{ V}$  at an input voltage of  $12\text{ V}$  and triggers from  $-16\text{ V}$  to  $+16\text{ V}$  at an input voltage of  $8\text{ V}$ . Specify all the values of resistors and reference DC voltage. (15 marks) [EEE 263 | L-2/T-1 | 2014–15]

**Q4.** Draw a zero-crossing detector circuit using Op-Amp and briefly describe its operation. (15 marks) [EEE 263 | L-2/T-1 | 2012–13]

### Waveform Generation

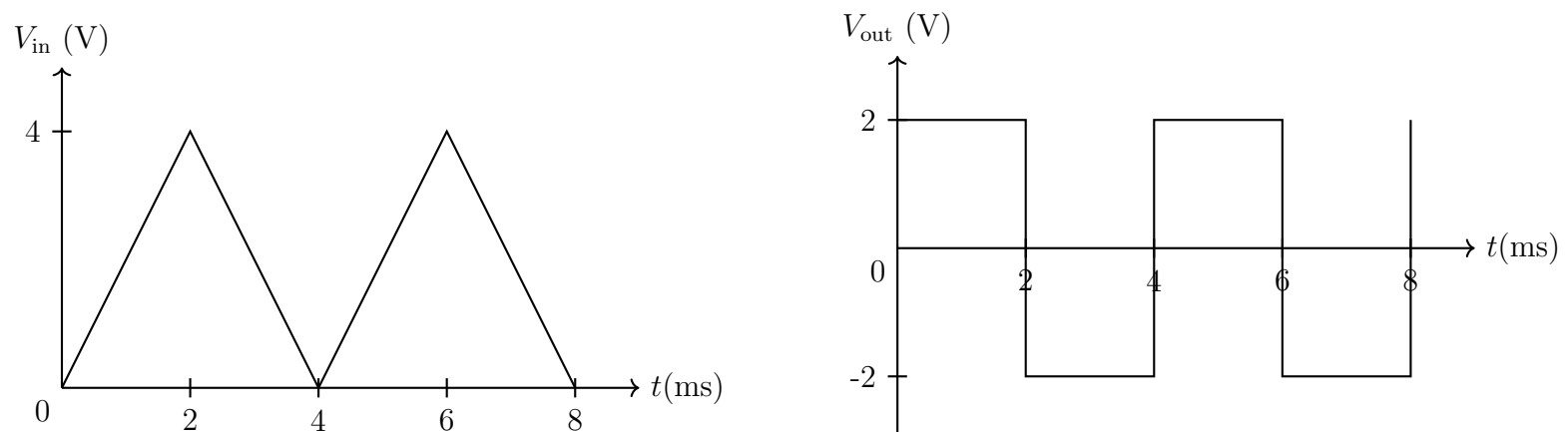
**Q1.** Design a circuit using op-amp(s) for given input  $v_i(t)$  and output  $v_o(t)$  waveforms shown in the figure.



(16 marks)

[EEE 263 | L-2/T-1 | 2018–19]

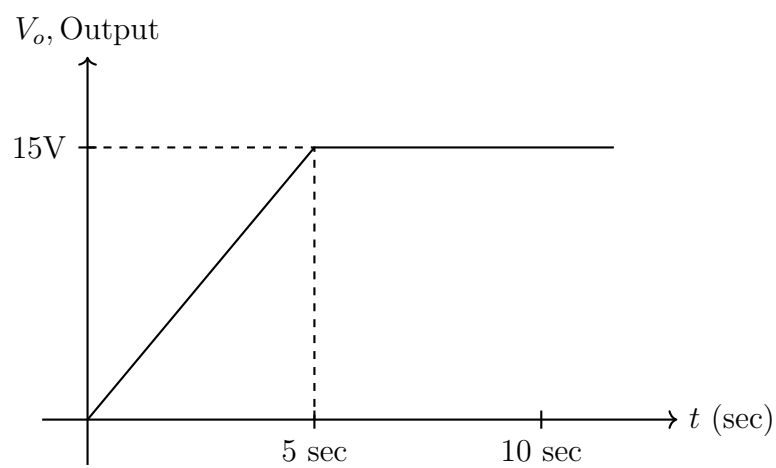
**Q2.** Input signal  $V_{in}$  and output signal  $V_{out}$  of a circuit are given in the figure. Design the circuit using Op-Amps. Show necessary calculations.



(12 marks)

[EEE 263 | L-2/T-1 | 2015–16]

**Q3.** Design a circuit using Op-Amps that gives a specified output waveform from a  $1\text{ V}$  DC input. Use  $+15\text{ V}$  and  $-15\text{ V}$  for DC biasing. Specify all the values of resistors and capacitors used.



(15 marks)

[EEE 263 | L-2/T-1 | 2014–15]

### Analog Computing (Differential Equations)

**Q1.** Design a circuit using op-amps and other passive elements to solve:

$$2\frac{d^2v}{dt^2} - 3\frac{dv}{dt} + 4v + 5 = 0$$

Draw the overall connection diagram and specify element values. Assume op-amps biased with  $\pm 15$  V; all initial conditions are zero.

(17 marks)

[EEE 263 | L-2/T-1 | 2024–25]

**Q2.** Using op-amps, design a circuit to solve the following differential equation where  $x$  is input and  $y$  is output:

$$\frac{d^2y}{dt^2} - 2\frac{dy}{dt} + 2y = 8x$$

(20 marks)

[EEE 263 | L-2/T-1 | 2022–23]

**Q3.** Design an electronic analog computer using op-amp to solve the following differential equation. Give a brief explanation of the operation of the designed computer.

$$\frac{d^2v}{dt^2} + 4\frac{dv}{dt} + 5v - 5 = 0$$

(26 marks)

[EEE 263 | L-2/T-1 | 2018–19]

**Q4.** Design an analog computer circuit to solve the following differential equation for  $t > 0$ :

$$\frac{d^2v}{dt^2} + 3\frac{dv}{dt} + 2v + 4\cos(10t) = 0$$

with initial conditions  $v(0) = 2$ ,  $v'(0) = 0$ . An oscillator is available to provide  $\sin(10t)$  signal. (20 marks) [EEE 263 | L-2/T-1 | 2015–16]

**Q5.** Design a circuit using Op-Amps that can solve the following differential equation:

$$3\frac{d^2x(t)}{dt^2} + 4x(t) = 0$$

Point out the node that represents the solution  $x(t)$ . Specify all the values of resistors and capacitors used. (20 marks) [EEE 263 | L-2/T-1 | 2014–15]

**Q6.** Design a circuit using op-amp that can provide solution to the following equation:

$$3\frac{d^2v}{dt^2} + K\frac{dv}{dt} - 3v + 8 = 0$$

where  $K$  = last digit of your student ID. Briefly discuss the operation of your designed circuit. (29 marks) [EEE 263 | L-2/T-1 | Jan–2021]

### Multiplier Circuit Design

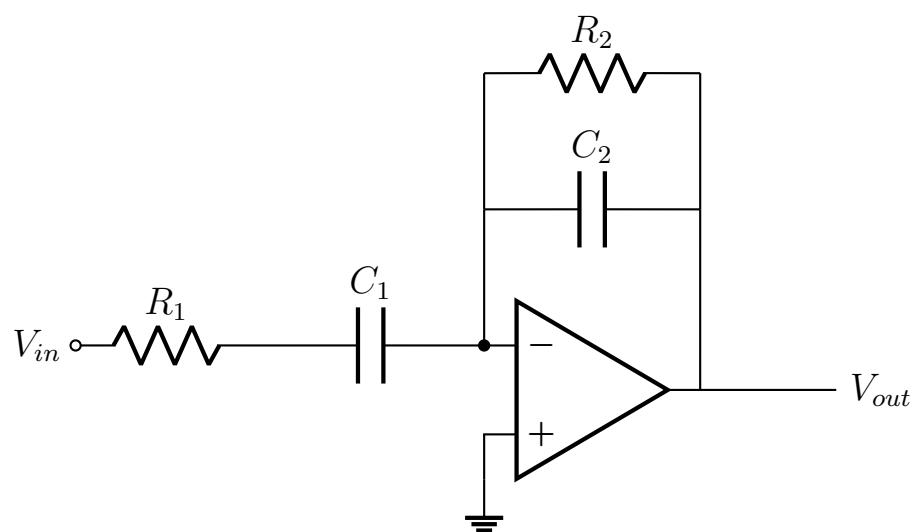
**Q1.** Suppose you have two input voltages  $X_1$  and  $X_2$ . Design a circuit using Op-Amps that gives output  $V_o = 10X_1X_2$ . The diodes available maintain the characteristic equation  $I_D = I_s e^{V_D/nV_T}$  where  $I_s = 10^{-6}$  A. Specify all the values of resistors used. (26 marks) [EEE 263 | L-2/T-1 | 2014–15]

## Frequency Multiplier

- Q1.** You have two supply voltage sources,  $v_1 = \sin(100t)$  V and  $v_2 = 1$  V. Design a circuit using op-amps that will produce an output voltage  $v_o = (\sin(50t))^2$  V. Also show that this circuit produces this output. **(20 marks)** [EEE 263 | L-2/T-1 | 2021–22]
- Q2.** Suppose you have a sinusoidal input voltage  $V_{in}(t) = \sin(4000\pi t)$  volt. Design a frequency multiplier circuit using Op-Amps that multiplies the frequency by a factor of 2 to produce  $V_{out}(t) = \sin(8000\pi t)$  volt. Use resistors, capacitors, inductors, and diodes (having reverse saturation current  $10^{-6}$  A) if necessary. Show necessary calculations and specify all design parameter values. **(26 marks)** [EEE 263 | L-2/T-1 | 2015–16]

## Active Filters

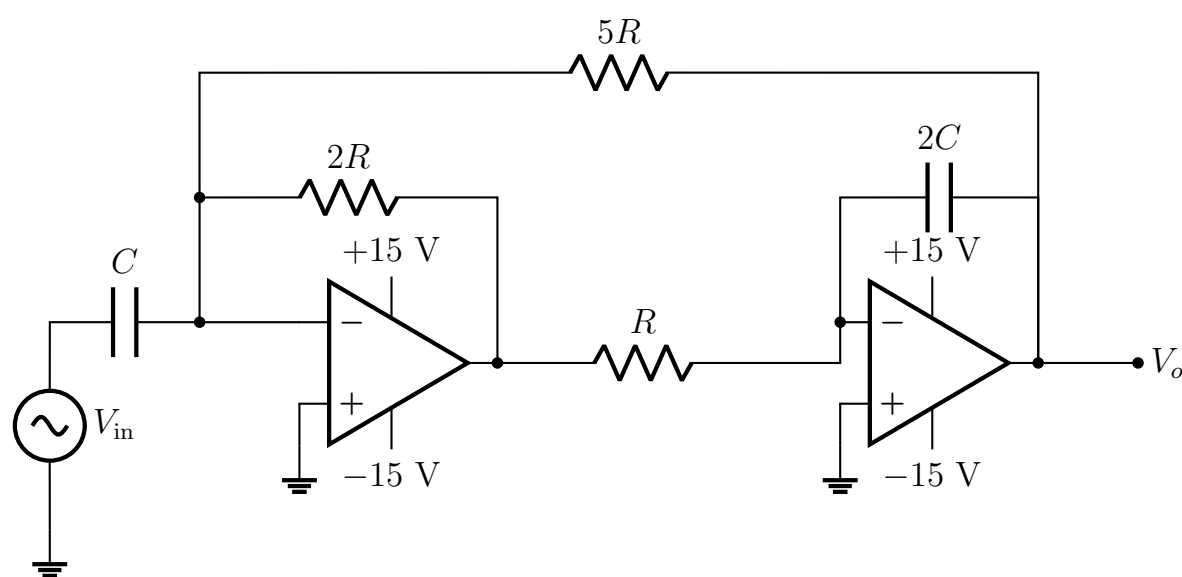
- Q1.** Design a 2nd order bandpass filter with lower cutoff frequency  $f_{CL} = 20$  kHz and higher cutoff frequency  $f_{CH} = 200$  kHz. **(10 marks)** [EEE 263 | L-2/T-1 | 2017–18]
- Q2.** Determine the type of filter shown in the circuit by computing the transfer function in the frequency domain.



(20 marks)

[EEE 263 | L-2/T-1 | 2016–17, 2017–18]

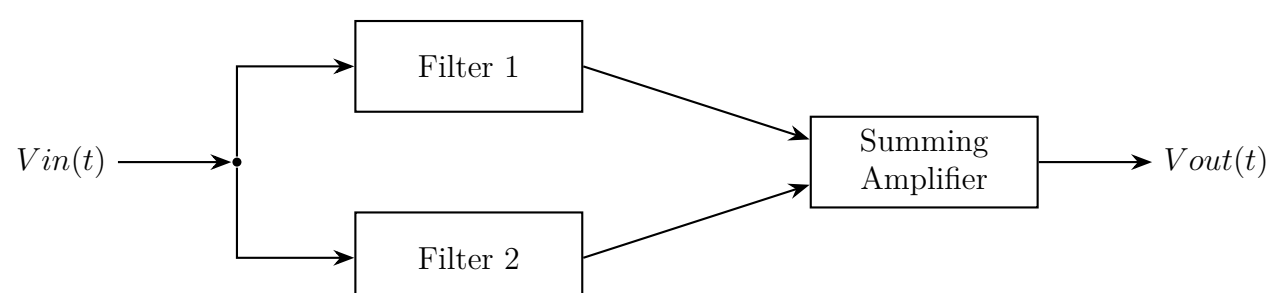
- Q3.** For the filter circuit given, determine (i) its type and comment on its order; derive the transfer function and find the cutoff frequency. (ii) If  $R = 10$  k $\Omega$  and  $C = 0.1$   $\mu$ F, would it be possible to pass a 100 Hz signal through this filter without distortion? Justify your answer. Note that the first Op-Amp block performs summation of input and output signals.



(23 marks)

[EEE 263 | L-2/T-1 | 2015–16]

- Q4.** In the circuit shown, input voltage is  $v_{in}(t) = 5 + 3 \cos(300\pi t - 30) + 8 \sin(1000\pi t) + 7 \sin(1600\pi t + 45)$  volt. Design the active filter blocks and summing amplifier to obtain output  $V_{out}(t) = 7 + 12 \sin(1000\pi t)$  volt. Sketch the frequency response and specify the cutoff frequency for each filter. Specify all values of resistors and capacitors/inductors.



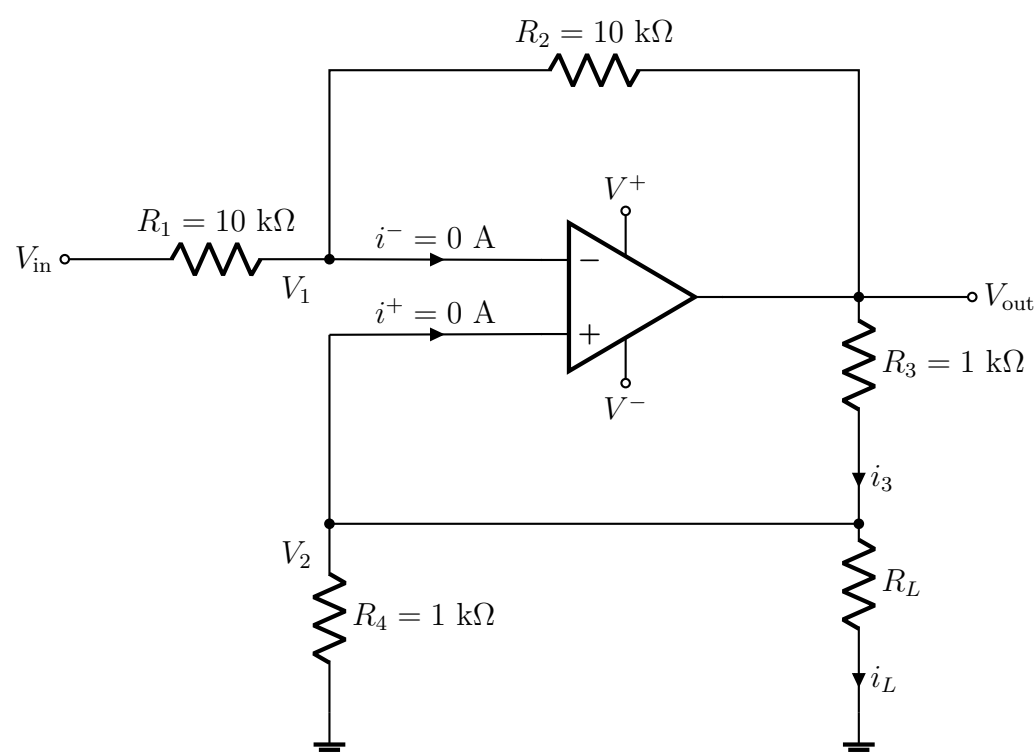
(23 marks)

[EEE 263 | L-2/T-1 | 2015–16]

- Q5.** Write the equation of magnitude of gain of a  $-20$  dB/decade low pass filter. Why is it called a  $-20$  dB/decade filter? Can it also be called a  $-6$  dB/octave filter? (20 marks) [EEE 263 | L-2/T-1 | 2014–15]
- Q6.** Design a notch filter that eliminates any signal between  $8$  kHz and  $12$  kHz using op-amps. Specify all the values of resistors and capacitors used. (20 marks) [EEE 263 | L-2/T-1 | 2014–15]
- Q7.** A system blocks  $20$  Hz and  $20$  kHz but passes  $10$  kHz. (i) Draw the frequency response of the system. (ii) Draw a block diagram of the system. (15 marks) [EEE 263 | L-2/T-1 | 2013–14]
- Q8.** Design a  $40$  dB low pass filter (LPF) with a cutoff frequency of  $1$  kHz. You have only one op-amp. (15 marks) [EEE 263 | L-2/T-1 | 2013–14]
- Q9.** Draw the circuit diagram of a  $-20$  dB/dec low pass filter. Also write down the design procedure. Design a  $-20$  dB/dec low pass filter with a cut-off frequency of  $100$  kHz. (16 marks) [EEE 263 | L-2/T-1 | 2011–12]

### Voltage-to-Current Converter

- Q1.** The circuit shown in Figure functions as a voltage-to-current converter. The op-amp used in this circuit is biased using  $\pm 20$  V sources. The overall circuit operates in the linear region as long as  $V_{out}$  stays within the bias voltage range, and the concept of virtual short (i.e.  $V_1 = V_2$ ) can be applied under such circumstances. If  $V_{out}$  attempts to exceed these limits, the amplifier enters saturation, and the virtual short assumption becomes invalid. Given the circuit parameters  $R_1 = R_2 = 10$  k $\Omega$ ,  $R_3 = R_4 = 1$  k $\Omega$  and  $R_L = 100$   $\Omega$ , calculate the load current  $i_L$  and output voltage  $V_{out}$  for an input voltage of  $V_{in} = -5$  V.

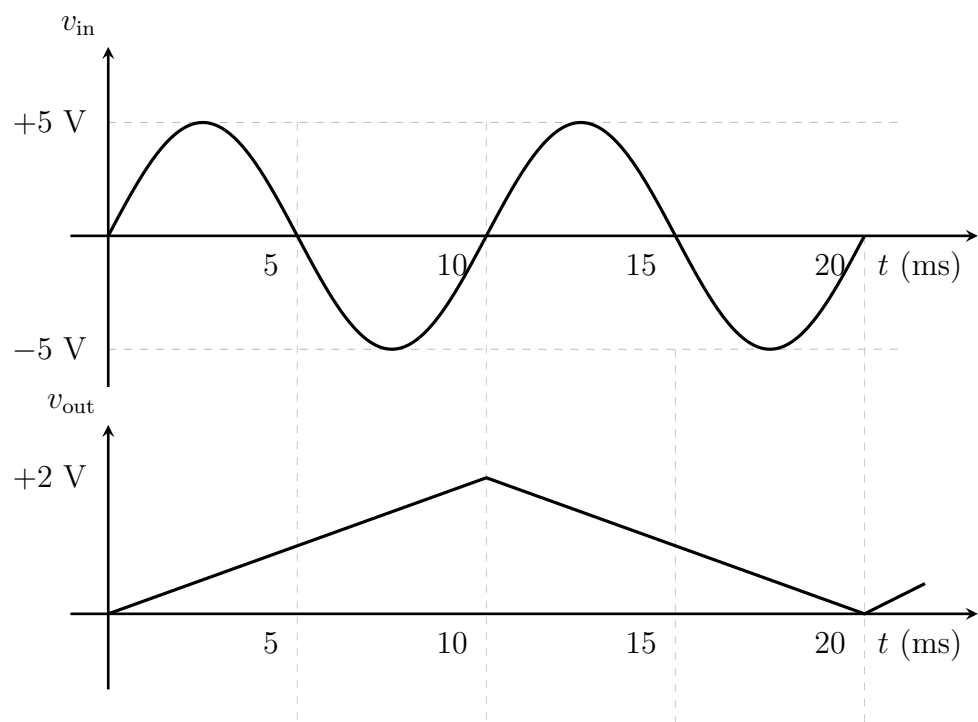


(18 marks)

[EEE 263 | L-2/T-1 | 2024–25]

### System Design (Analog to Digital Interface)

- Q1.** Design a circuit using Op-Amp and Digital Circuits for given input wave shape  $v_{in}$  and output wave shape  $v_{out}$  in Figure . Assume a  $5$  V level to be digital 1 and  $-5$  V level to be digital 0.



(26.67 marks)

[EEE 263 | L-2/T-1 | 2017–18]

**Q2.** A circuit is to be designed where depending on the range of input voltage a digital sequence is passed to a computer. The ranges and corresponding digital sequences are:

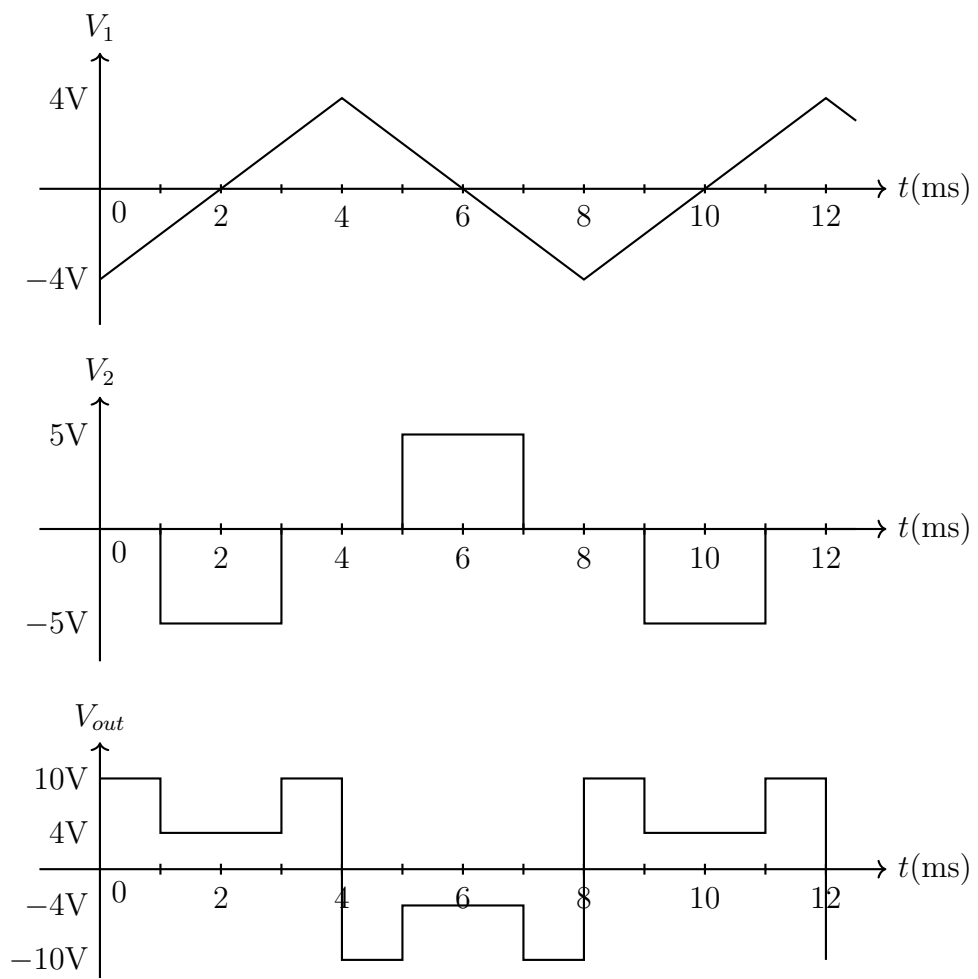
| Range       | Digital Sequence |
|-------------|------------------|
| 0 – 1.2 V   | 010              |
| 1.2 – 3.2 V | 110              |
| 3.2 – 3.5 V | 001              |
| 3.5 – 5 V   | 111              |

Design a circuit using op-amps and digital circuits (gates, multiplexers) to perform the task. Assume 5 V = digital 1 and 0 V = digital 0. (25 marks)

[EEE 263 | L-2/T-1 | 2016–17]

Waveform Realization

**Q1.** Determine an op-amp circuit realization for given input wave shapes  $V_1$  and  $V_2$  and output wave shape  $V_{out}$ .



(15 marks)

[EEE 263 | L-2/T-1 | 2016–17]

Integrator

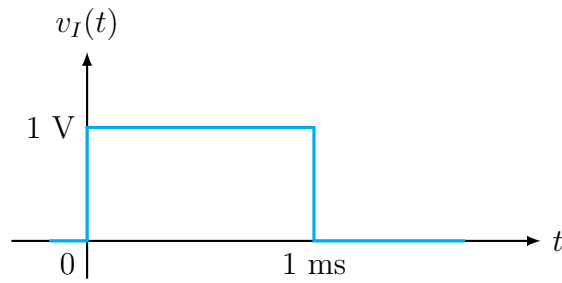
**Q1.** Design a circuit using ideal operational amplifiers that implements the following equation:

$$v_0 \propto \int (v_1 + v_2) dt$$

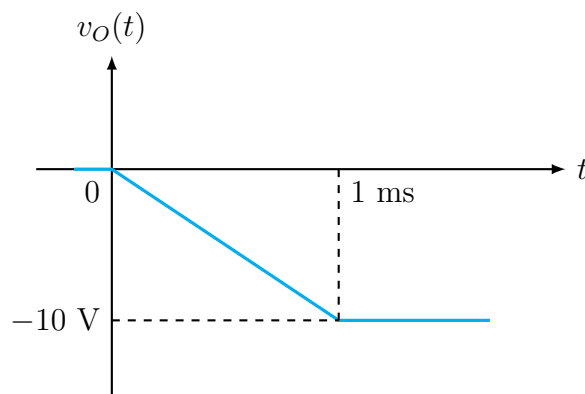
Write the final output expression in terms of the circuit components used in your design. **(20 marks)**

[EEE 263 | L-2/T-1 | 2020–21]

**Q2.** The input voltage  $v_I$  and output voltage  $v_O$  of an op-amp based circuit are given. Design the circuit.



(a)



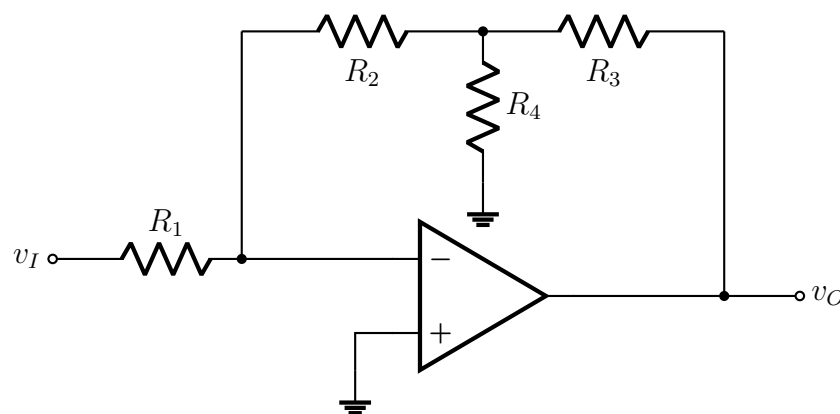
(b)

**(17 marks)**

[EEE 263 | L-2/T-1 | Jan–2021]

### Op-Amp Network Analysis

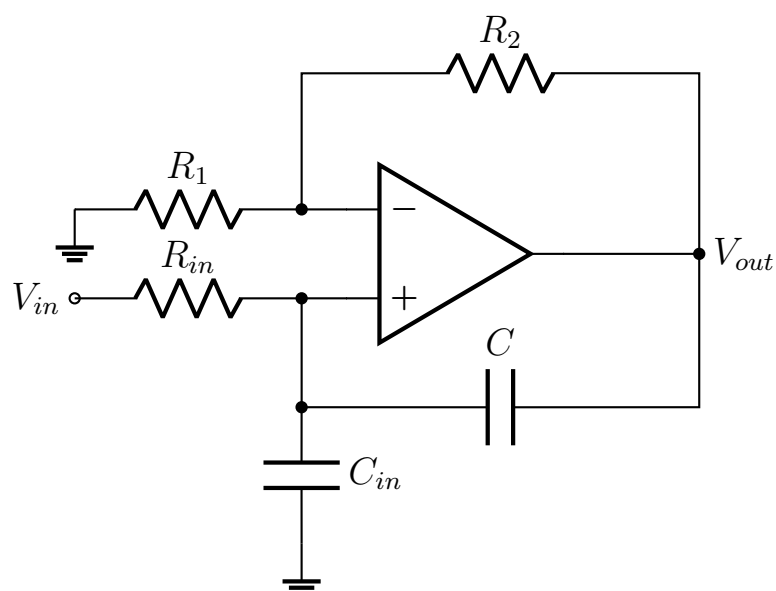
**Q1.** Determine the closed-loop voltage gain and current gain of the ideal op-amp network shown in the figure.



**(23 marks)**

[EEE 263 | L-2/T-1 | 2020–21]

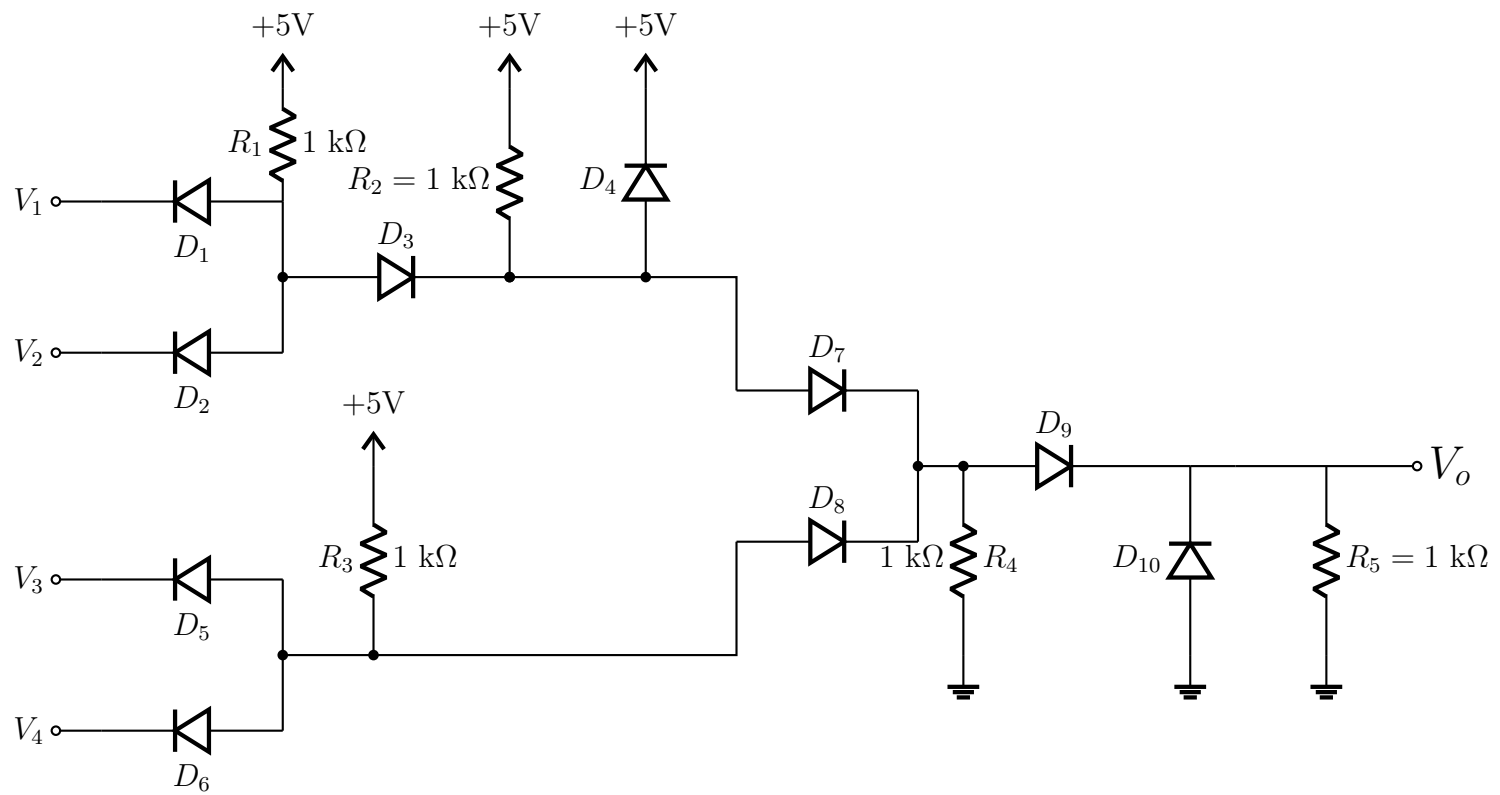
**Q2.** Determine the differential equation solved by the circuit shown: (i) when the capacitor  $C$  is absent; (ii) when the capacitor  $C$  is present. At what value of  $C$  does the circuit become a non-inverting amplifier?



(21 marks)

[EEE 263 | L-2/T-1 | 2016–17]

Q3. Determine the Boolean expression for  $V_o$  in terms of the four input voltages for the circuit shown (diode logic).



(10 marks)

[EEE 263 | L-2/T-1 | 2016–17]

#### Op-Amp Practical Imperfections

Q1. What are the AC imperfections of a practical op-amp? How can we find out the value of input offset voltage and input offset current? (15 marks) [EEE 263 | L-2/T-1 | 2013–14]

#### Dual Slope A/D Converter

Q1. By drawing appropriate circuits, explain the operation of a dual slope A/D converter. (20 marks) [EEE 263 | L-2/T-1 | 2020–21]

BUET · EEE 263 · Electronic Circuits

# Electronic Circuits

Part V — Oscillators & Timers

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Colpitts oscillator, 555 Timer astable design, triangular wave generators

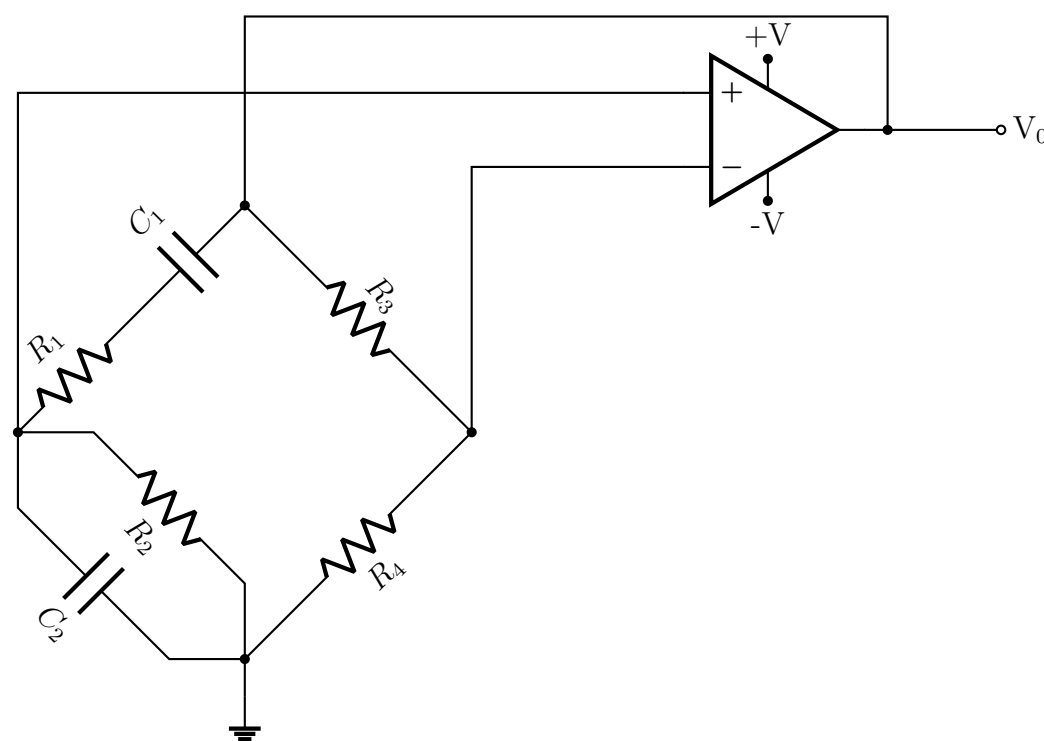
## S8 Oscillators, Timers & Function Generators

### Barkhausen Stability Criterion

- Q1.** Write down the Barkhausen criterion. (10 marks) [EEE 263 | L-2/T-1 | 2018–19]
- Q2.** Derive expression of Desensitivity. Explain positive and negative feedback from desensitivity. (10+5 marks) [EEE 263 | L-2/T-1 | 2014–15]
- Q3.** Write down the Barkhausen stability criterion for an oscillation in a linear electronic circuit to sustain. (6 marks) [EEE 263 | L-2/T-1 | 2014–15]

### Wien Bridge Oscillator

- Q1.** Define Barkhausen criterion. Describe the operating principle of an op-amp based Wein bridge oscillator. Derive the conditions for Barkhausen criteria to be satisfied in this oscillator and the expression of oscillation frequency. (17 marks) [EEE 263 | L-2/T-1 | 2023–24]
- Q2.** For the Wien bridge circuit connected to an op-amp shown:
- state conditions for sinusoidal oscillation;
  - derive the equation for oscillation frequency;
  - show that the voltage gain  $A_V$  must be  $\geq \left(1 + \frac{R_1}{R_2} + \frac{C_1}{C_2}\right)$  for oscillation to sustain;
  - if  $R_1 = R_2 = 10 \text{ k}\Omega$  and  $C_1 = C_2 = 0.1 \mu\text{F}$ , design the values of  $R_3$  and  $R_4$  so that oscillation sustains.



(28 marks)

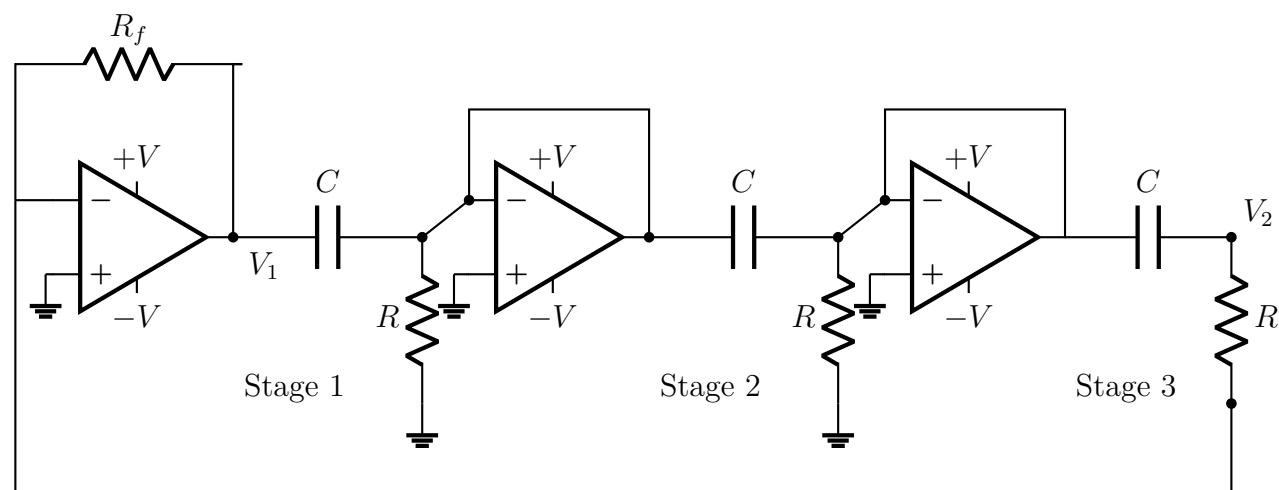
[EEE 263 | L-2/T-1 | Jan–2021]

- Q3.** Design a Wien Bridge Oscillator with an oscillation frequency of 1 kHz. What is the gain of the feedback network of your design? Specify all the values of resistors and capacitors used. (20 marks) [EEE 263 | L-2/T-1 | 2014–15]
- Q4.** Design a Wien Bridge Oscillator with an oscillation frequency of 1000 rad/sec. Available capacitors are 0.1  $\mu\text{F}$ , 1  $\mu\text{F}$ , and 100 pF. (Unspecified marks) [EEE 263 | L-2/T-1 | 2013–14]
- Q5.** Draw a Wien Bridge oscillator circuit and describe its operating principle. (17 marks) [EEE 263 | L-2/T-1 | 2012–13]

### RC Phase Shift Oscillator

- Q1.** For an RC phase shift oscillator, derive the expression for oscillation frequency. Also derive the conditions for the Barkhausen criterion to be satisfied. Assume all resistors in the feedback network are of the same value and all capacitors are of the same value. (26 marks) [EEE 263 | L-2/T-1 | 2021–22]

**Q2.** For the oscillator circuit shown below, find oscillation frequency ( $\omega$ ),  $\beta$ -network gain ( $\beta$ ), and amplifier gain ( $A$ ). Also design this oscillator for an oscillation frequency of 1 kHz. Note that, each stage is separated by a source follower circuit. Hence each stage will give a phase shift of  $60^\circ$  and 3 stages will give in total  $180^\circ$  phase shift. Here,  $\beta = \frac{v_2}{v_1}$ .

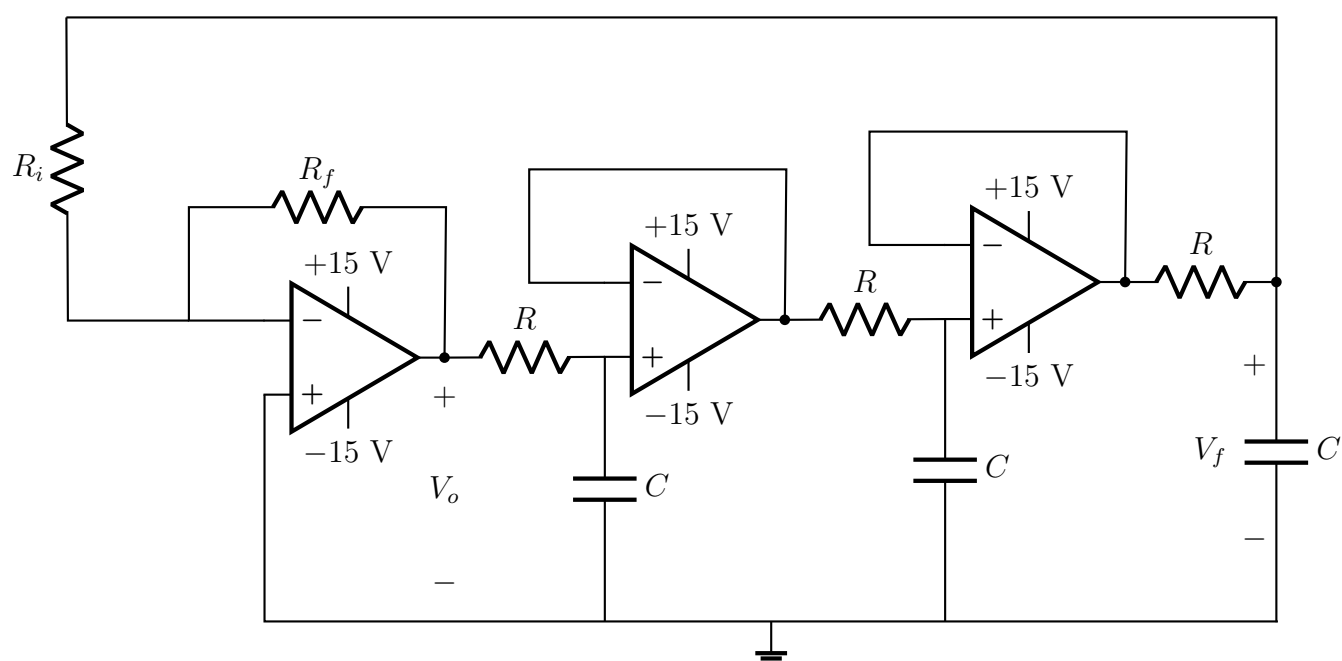


(18 marks)

[EEE 263 | L-2/T-1 | 2017–18]

**Q3.** For the RC phase shift oscillator shown:

- Determine the oscillation frequency  $f_0$  and obtain the relationship between  $R_i$  and  $R_f$  for sustainable oscillation.
- How can you generate a 100 kHz signal from this oscillator?



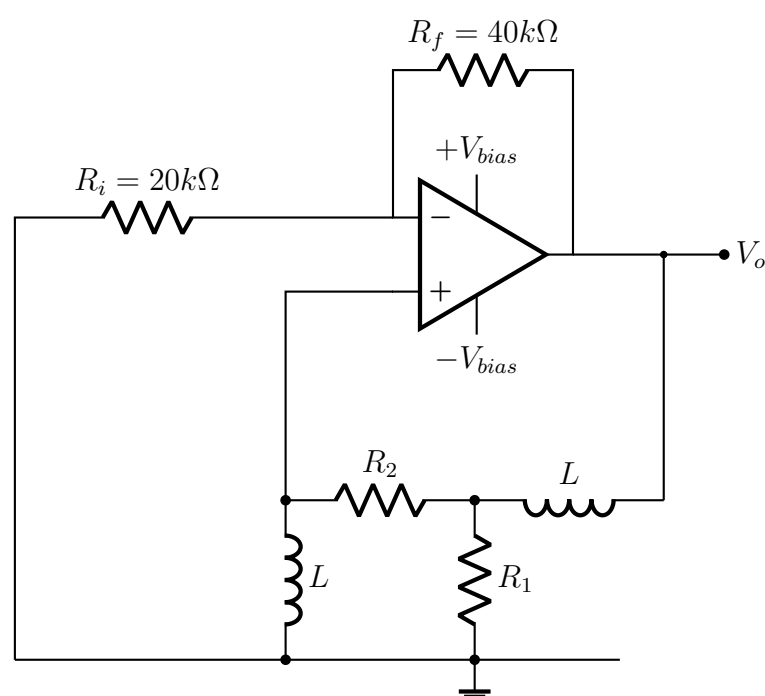
(20 marks)

[EEE 263 | L-2/T-1 | 2015–16]

### LC Oscillator

**Q1.** For the oscillator circuit given:

- Derive the expression for oscillation frequency  $f_0$ .
- Obtain the relationship between  $R_1$  and  $R_2$  for sustainable oscillation.
- Specify the bias conditions and values of  $L$ ,  $R_1$ ,  $R_2$  to obtain a 20 V peak-to-peak, 500 kHz sinusoidal signal from this oscillator.



(26 marks)

[EEE 263 | L-2/T-1 | 2015–16]

### Colpitts Oscillator

- Q1.** With an appropriate circuit diagram, describe the operation of an op-amp based Colpitts oscillator. Derive the frequency of oscillation and explain how this circuit satisfies the conditions for sustained oscillation. (18 marks) [EEE 263 | L-2/T-1 | 2024–25]
- Q2.** Derive the expression of oscillation frequency for a Colpitts oscillator. Then derive the conditions for the Barkhausen criterion to be satisfied. (20 marks) [EEE 263 | L-2/T-1 | 2022–23]
- Q3.** Draw the circuit of the Colpitts oscillator. Then, using the small-signal model of the MOSFET, show that the frequency of oscillation is:

$$\omega_0 = \frac{1}{\sqrt{L \cdot \frac{C_1 C_2}{C_1 + C_2}}}$$

and that  $\frac{C_1}{C_2} = g_m R$ . (26 marks)

[EEE 263 | L-2/T-1 | 2018–19, 2020–21]

### Crystal Oscillator

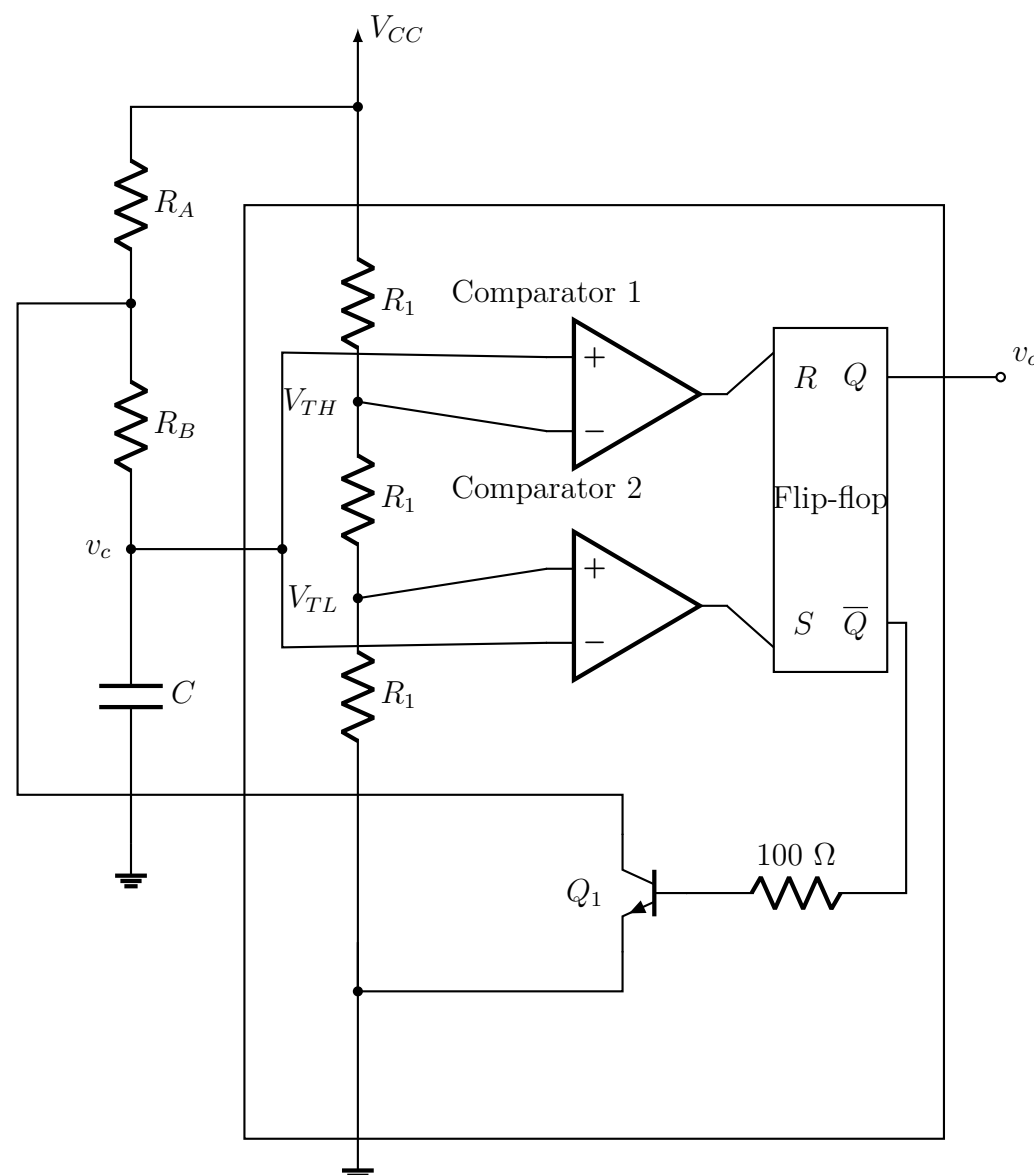
- Q1.** Draw the equivalent circuit of a piezoelectric crystal. Qualitatively draw the crystal reactance versus frequency and identify the inductive and capacitive reactance regions. (10 marks) [EEE 263 | L-2/T-1 | 2018–19]

### One-Shot (Monostable) Multivibrator

- Q1.** Draw a one-shot multivibrator circuit using an op-amp. Explain the operation of the circuit. (26 marks) [EEE 263 | L-2/T-1 | 2021–22]

### 555 Timer—Oscillator Design

- Q1.** A 555 timer is connected in the configuration shown in Figure below. The capacitor shown in the circuit is 1 nF. Find the values of  $R_A$  and  $R_B$  required to generate an oscillation frequency of 50kHz and duty cycle of 60%. What measures can be taken if we want to achieve duty cycle below 50%?



(12 marks)

[EEE 263 | L-2/T-1 | 2023–24]

**Q2.** With proper connection diagram, explain the monostable mode of operation of a 555 timer. Design a monostable multivibrator using a 555 timer that can generate an output pulse with duration  $T = 100\text{ms}$ . Draw the waveforms of output voltage and trigger voltage of your designed circuit. (18 marks)

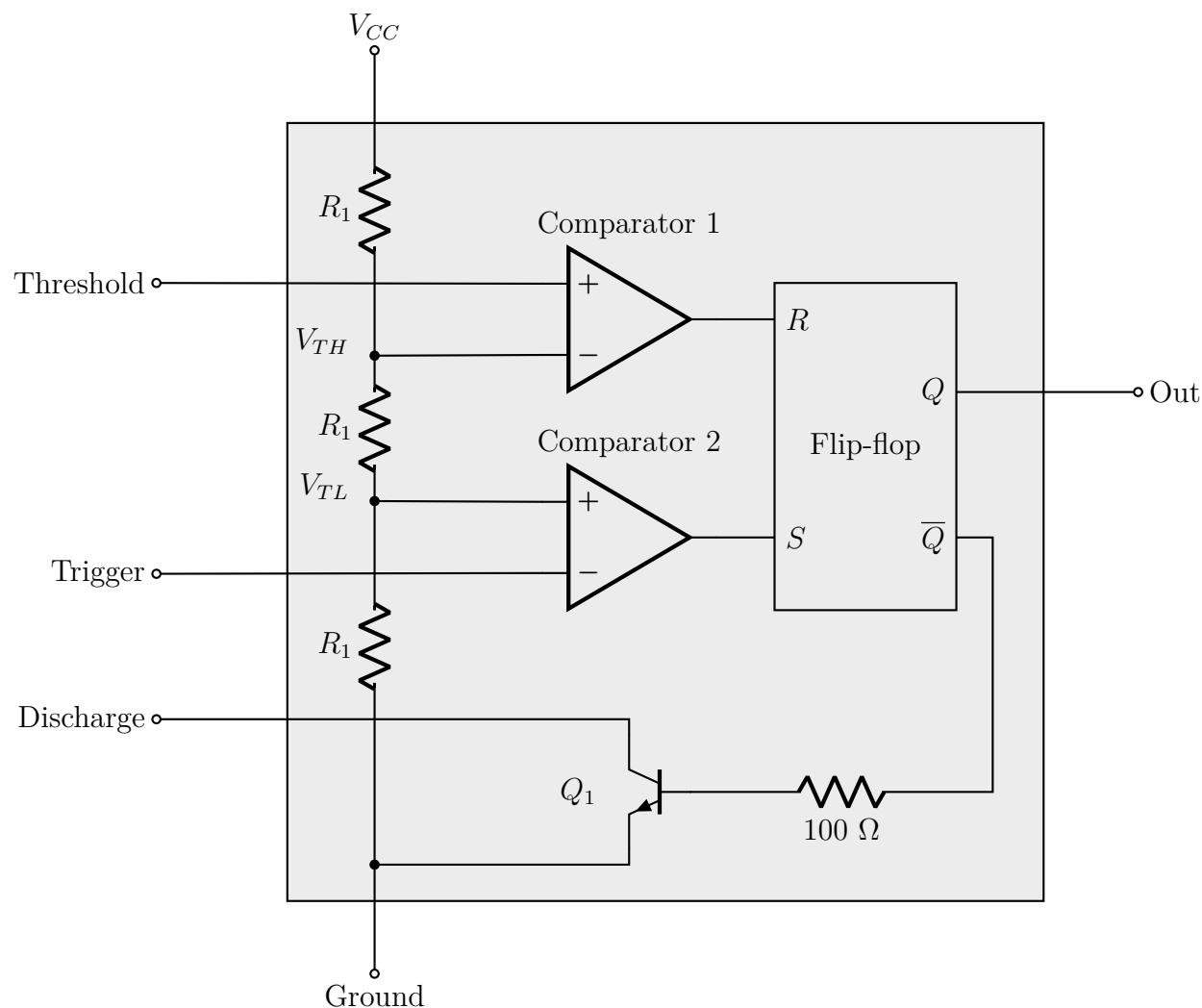
[EEE 263 | L-2/T-1 | 2023–24]

**Q3.** Design an oscillator circuit using a 555 Timer IC which has a square wave output with oscillation frequency of 20 kHz and duty cycle of 25%. Provide a necessary illustration of the designed circuit and draw the wave shape.

(20 marks)

[EEE 263 | L-2/T-1 | 2020–21]

**Q4.** Design a circuit using a 555 timer to generate a 100 kHz square wave with a duty cycle of 40%. Draw the final connection diagram with proper component values.



(15 marks)

[EEE 263 | L-2/T-1 | 2024–25]

**Q5.** Design a square wave generator with 10 ms period and 75% duty cycle using a 555 timer. (15 marks)

[EEE 263 | L-2/T-1 | 2022–23]

**Q6.** To get an output pulse of 10 ms width when a trigger is applied, an external resistor and a  $1\ \mu\text{F}$  capacitor are connected with a 555 timer. What should be the value of the resistor? (10 marks)

[EEE 263 | L-2/T-1 | 2021–22]

**Q7.** Design an astable multivibrator using a 555 timer with a period of 10 ms and 75% duty cycle. (10 marks)

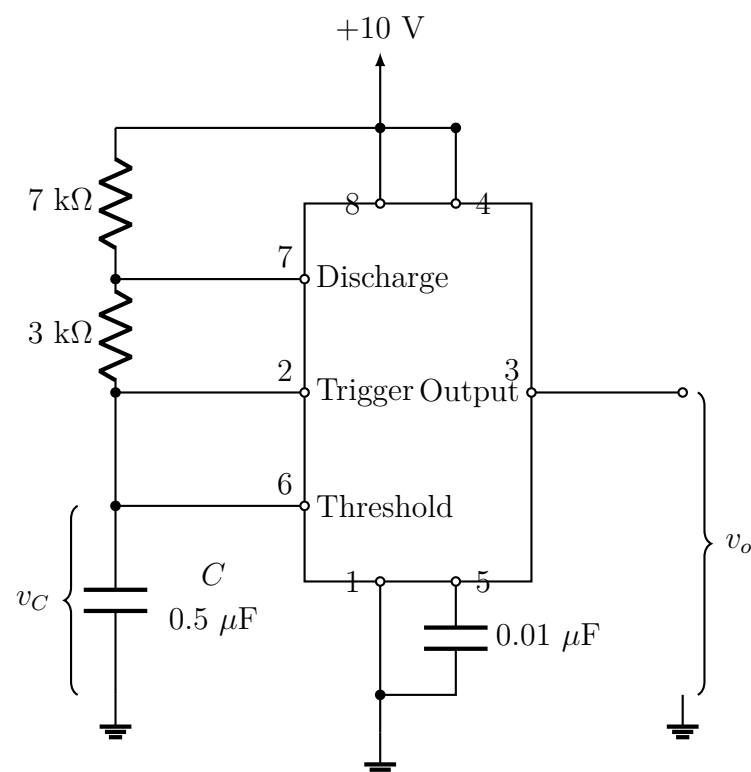
[EEE 263 | L-2/T-1 | 2021–22]

**Q8.** Design a circuit using a 555 IC which results in a square wave with oscillation frequency of 100 kHz and 75% duty cycle. Draw your designed circuit and output wave shape. (20 marks)

[EEE 263 | L-2/T-1 | 2018–19]

**Q9.** For the 555 IC based astable multivibrator circuit shown with  $R_1 = 7\text{ k}\Omega$ ,  $R_2 = 3\text{ k}\Omega$ ,  $C = 0.5\ \mu\text{F}$ :

- draw waveshapes of  $V_C$  and  $V_O$ ;
- determine  $t_{low}$  and  $t_{high}$ ;
- find duty cycle and free-running frequency;
- discuss how frequency depends on  $C$  with a graph;
- design resistors and capacitors so that duty cycle becomes 0.40.

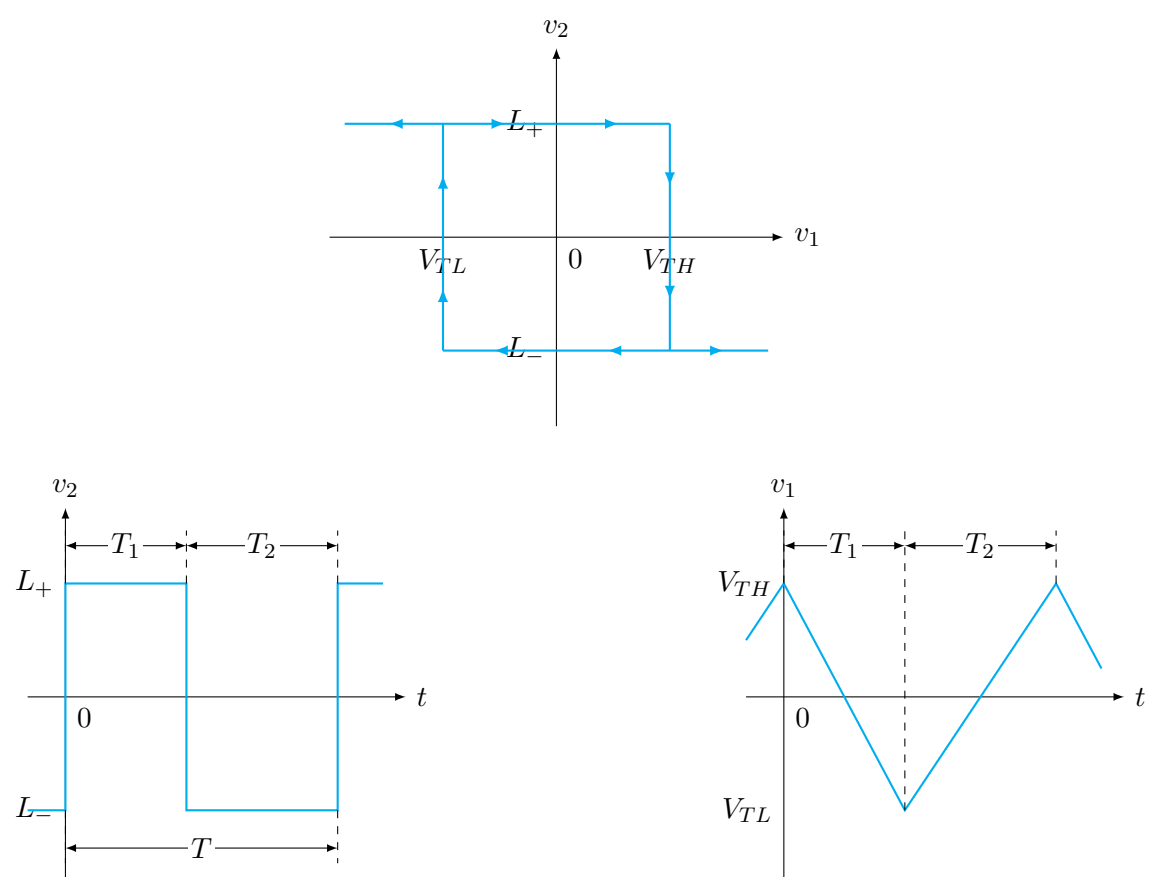


(29 marks)

[EEE 263 | L-2/T-1 | Jan-2021]

### Triangular Wave Generator

- Q1.** Draw and explain the operation of a triangular wave generator circuit using op-amps and a Schmitt trigger. Derive the frequency of oscillation for the circuit. **(20 marks)** [EEE 263 | L-2/T-1 | 2022-23]
- Q2.** Draw a sawtooth wave generator circuit using an op-amp. Explain the operation of the circuit. **(23 marks)** [EEE 263 | L-2/T-1 | 2021-22]
- Q3.** Drawing appropriate circuits, show how you can generate triangular waves using a bistable multivibrator and appropriate Op-amp circuits. **(26 marks)** [EEE 263 | L-2/T-1 | 2020-21]
- Q4.** Design a circuit using a bistable multivibrator which generates a triangular waveform with oscillation frequency of 2 kHz and 10 V peak-to-peak amplitude. **(16 marks)** [EEE 263 | L-2/T-1 | 2018-19]
- Q5.** A bistable multivibrator with a given hysteresis transfer characteristic is provided. Design a signal generator based on the bistable multivibrator that generates square and triangular waveforms. Choose values of your designed circuit components for which the frequency of oscillation is 1 MHz.



(18 marks)

[EEE 263 | L-2/T-1 | Jan-2021]

BUET · EEE 263 · Electronic Circuits

# Electronic Circuits

Part VI — Digital Logic and Sequential Circuits

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CMOS logic design, flip-flops, counters, and digital circuit implementation

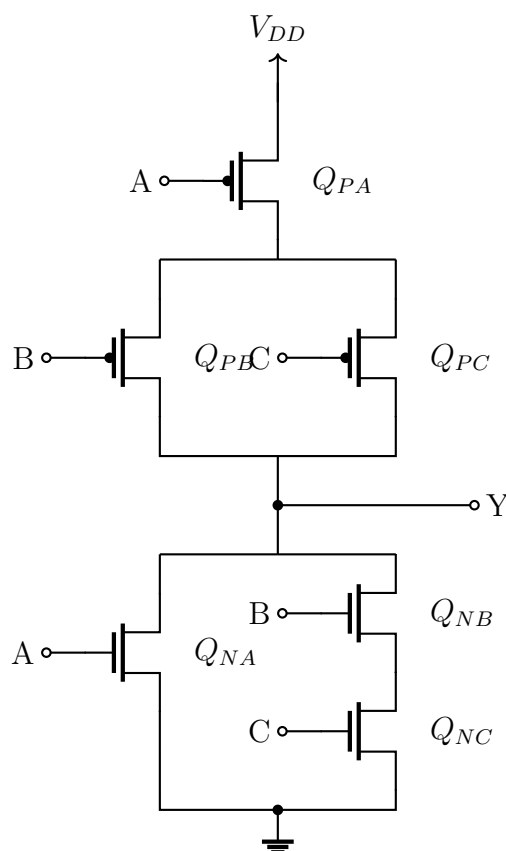
## S9 Digital Logic & Sequential Circuits

### CMOS Logic Design and Implementation

- Q1.** Discuss basic operational characteristics and parameters of CMOS and TTL logic families. **(23 marks)** [EEE 263 | L-2/T-1 | 2021–22]
- Q2.** Implement the function  $f(A, B, C, D) = \sum m(0, 4, 8, 9, 10, 11, 12)$  using CMOS. **(13 marks)** [EEE 263 | L-2/T-1 | 2018–19]
- Q3.** Draw a CMOS complex gate for the logic function  $f(a, b, c, d) = \sum m(0, 1, 2, 4, 6, 8, 10, 12, 14)$ . Use as few transistors as possible. **(15 marks)** [EEE 263 | L-2/T-1 | 2017–18]
- Q4.** Derive complex gates for the following logic functions using as few transistors as possible:
- (a)  $f(X_1, X_2, X_3, X_4) = \sum m(0, 4, 6, 8, 9, 15) + D(3, 7, 11, 13)$
- (b)  $\bar{f} = (A + B)(C + D)(\bar{A} + B + D)$
- (21 marks)** [EEE 263 | L-2/T-1 | 2016–17]
- Q5.** Implement the function  $f(A, B, C, D) = \sum m(1, 2, 4, 7, 14, 15)$  using (i) CMOS circuit and (ii) pseudo NMOS circuit. Which circuit would you prefer between your two designed circuits and why? **(25 marks)** [EEE 263 | L-2/T-1 | Jan–2021]

### CMOS Transistor Sizing

- Q1.** Calculate transistor  $W/L$  ratios for the logic circuit shown. Assume that for the basic inverter  $n = 1.5$  and  $P = 5$  and the channel length is  $0.25 \mu\text{m}$ .



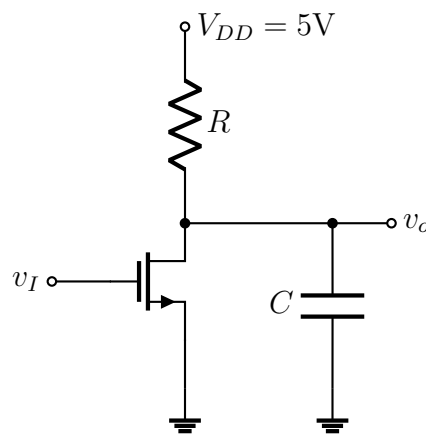
**(16 marks)**

[EEE 263 | L-2/T-1 | 2018–19]

- Q2.** Design transistor  $W/L$  ratios for the logic  $Y = \overline{(AB + D)}C$  implemented using CMOS circuit. Assume that for the basic inverter  $(W/L)_n = n = 2$  and  $(W/L)_p = p = 7$  and the channel length is  $0.15 \mu\text{m}$ . **(21 marks)** [EEE 263 | L-2/T-1 | Jan–2021]

### Inverter Propagation Delay

- Q1.** For the inverter shown with a capacitor  $C$  connected between the output node and ground: if at  $t = 0$  the input goes low to high, find the equation of time for  $v_0$  to reach  $\frac{1}{2}(V_{OL} + V_{OH})$ . Also calculate the high-to-low propagation time for  $R = 25 \text{ k}\Omega$  and  $C = 20 \text{ pF}$ .



(10.67 marks)

[EEE 263 | L-2/T-1 | 2018–19]

### Transmission Gate Implementation

Q1. Implement the following function  $Y$  using CMOS transmission gate configuration:

$$Y = AB + \bar{A}C$$

(7 marks)

[EEE 263 | L-2/T-1 | 2018–19]

### CMOS Flip-Flop Implementation

Q1. Show a CMOS implementation of edge-triggered D flip-flops with asynchronous reset. (25 marks) [EEE 263 | L-2/T-1 | 2016–17]

### Flip-Flop Construction

Q1. Show how D flip-flop and T flip-flop can be constructed from J-K flip-flops. (15 marks) [EEE 263 | L-2/T-1 | 2016–17]

### Counter Design

Q1. Design a MOD-3 counter using JK flip-flops and verify the design showing the truth table. (15 marks) [EEE 263 | L-2/T-1 | 2021–22]

Q2. Design a modulo-12 up counter with synchronous reset. (15 marks) [EEE 263 | L-2/T-1 | 2017–18]

Q3. Design a 4-bit up/down counter using T flip-flops. It should include a control pin  $\overline{UP}/DOWN$ . When  $\overline{UP}/DOWN = 0$  the circuit behaves as an up counter; when  $\overline{UP}/DOWN = 1$  it behaves as a down counter. The circuit should also have a synchronous active-low  $\overline{RESET}$  pin. (20 marks) [EEE 263 | L-2/T-1 | 2016–17]

### Combinational Logic Design

Q1. Draw a PLA programmed to implement the following functions:

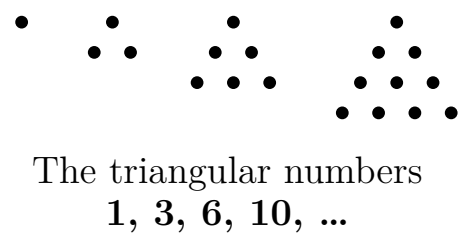
$$\begin{aligned} Y_1 &= A\bar{B}C + \bar{A}C + B \\ Y_2 &= \bar{B}AC + AB\bar{C} + \bar{B} \\ Y_3 &= \bar{A}C + B\bar{C}A \end{aligned}$$

(15 marks)

[EEE 263 | L-2/T-1 | 2017–18]

Q2. Implement the function  $f(x_1, x_2, x_3) = \sum m(2, 3, 4, 6, 7)$  using two-input LUTs (Look Up Tables). You can use as many LUTs as needed. (16 marks) [EEE 263 | L-2/T-1 | 2017–18]

Q3. Triangular numbers are sequences formed by continuous summation of natural numbers (1, 3, 6, 10, ...). Design the simplest possible logic circuit that outputs 1 whenever it detects a 4-bit triangular number.



(11 marks)

[EEE 263 | L-2/T-1 | 2016–17]

Priority Encoder

**Q1.** Design a four-line to two-line priority encoder circuit with active high inputs and outputs, with priority assigned to the higher-order data input line ( $I_4 > I_3 > I_1$ ). Use NOR gates only for the circuit implementation. **(16 marks)** [EEE 263 | L-2/T-1 | 2017–18]

Shift Register Design

**Q1.** Design a 4-bit shift register using D flip-flops and a four-to-one multiplexer with mode selection inputs  $S_1$  and  $S_0$ . The register operates according to the following function table:

| $S_1$ | $S_0$ | Register Operation                           |
|-------|-------|----------------------------------------------|
| 0     | 0     | No change                                    |
| 0     | 1     | Complement the four outputs                  |
| 1     | 0     | Clear register to 0 (synchronous with clock) |
| 1     | 1     | Shift Right                                  |

(15 marks)

[EEE 263 | L-2/T-1 | 2017–18]

BUET · EEE 263 · Electronic Circuits

# Electronic Circuits

Part VII — Data Converters, PLL & Memory

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DAC, ADC, Phase-Locked Loop, DRAM, Flash Memory

S10 Data Converters, PLL & Memory

Digital-to-Analog Converter (DAC)

Q1. What are DAC circuits? How does an R-2R ladder DAC circuit operate? Explain using a circuit diagram and mathematical analysis. (17 marks) [EEE 263 | L-2/T-1 | 2021–22]

Analog-to-Digital Converter (ADC)

Q1. A 6-bit A/D converter has reference voltage  $V_{REF} = 8\text{ V}$  and reference count  $n_{REF} = 2^6$ . Values of some consecutive analog sampled voltages  $|V_A|$  are: 0 V, 3 V, 5.5 V, 7.125 V. If these voltages are given as inputs to the converter, find the corresponding digital outputs. (12 marks) [EEE 263 | L-2/T-1 | 2021–22]

Phase-Locked Loop (PLL)

Q1. Draw a simplified block diagram of a PLL and briefly explain its operation. (17 marks) [EEE 263 | L-2/T-1 | 2021–22]

Memory: DRAM and Flash

Q1. Explain the read and write operation in a DRAM cell. (16 marks) [EEE 263 | L-2/T-1 | 2021–22]

Q2. Explain briefly the programming and erase operation in a Flash Memory. (15 marks) [EEE 263 | L-2/T-1 | 2021–22]